5 Noise Impact

5.1 Legislation, Standards and Guidelines

5.1.1.1 The relevant legislation and associated guidance applicable to the present study for the assessment of noise impacts include:

- Noise Control Ordinance (NCO) (Cap.400);
- Environmental Impact Assessment Ordinance (EIA Ordinance) (Cap. 499);
- Technical Memoranda (TM) on Noise from Construction Work other than Percussive Piling (TM-GW);
- TM on Noise from Percussive Piling (TM-PP);
- TM on Noise from Construction Work in Designated Areas (TM-DA);
- TM for the Assessment of Noise from Places other than Domestic Premises, Public Places or Construction Sites (TM-Places); and
- Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM).

5.1.2 Construction Noise

Construction Noise during Non-restricted Hours

5.1.2.1 The NCO provides the statutory framework for noise control in Hong Kong. Assessment procedures and standards are set out in the respective TM promulgated under the NCO.

5.1.2.2 To ensure a better environment, the EIAO-TM promulgated under the EIA Ordinance has imposed more stringent criteria. For construction, there is no statutory limit on daytime construction noise under the NCO and related TMs. Nevertheless, the EIAO-TM stipulates noise standards of 65 – 75 dB(A) for daytime construction activities, as shown in **Table 5.1.1**.

Table 5.1.1 Noise standards for daytime construction activities

Uses	Noise Standards [1], $L_{eq\ (30\ mins)}$ dB(A)
	0700 to 1900 hours on any day not being a Sunday or general holiday
All domestic premises including temporary housing accommodation	75
Hotels and hostels	75
Educational institutions including kindergartens, nurseries and all others where unaided voice communication is required	70 65 (During examinations)

Note:
[1] The above standards apply to uses that rely on opened windows for ventilation.

Construction Noise during Restricted Hours

5.1.2.3 The NCO also provides statutory control on general construction works during restricted hours (i.e. 1900 to 0700 hours (of the next day) from Monday to Saturday and at any time on Sundays or public holidays).

The use of Powered Mechanical Equipment (PME) for construction works during restricted hours would require a Construction Noise Permit (CNP). The TM-GW details the procedures adopted by Environmental Protection Department (EPD) for assessing such application. The granting of a CNP is subject to conditions stated in the CNP and it may be revoked at any time for failure to comply with the permit conditions.

5.1.2.4 In addition to the general controls on the use of PME during restricted hours, the use of Specified Powered Mechanical Equipment (SPME) and the undertaking of Prescribed Construction Work (PCW) during the restricted hours in a designated area are controlled by the TM-DA. Construction plant or equipment classified as SPME under the TM-DA includes hand-held breakers, bulldozers, concrete lorry mixers, dump trucks and vibratory pokers. The PCW includes the erection or dismantling of formwork or scaffolding, hammering, loading, unloading or handling of rubble, wooden boards, steel bars, wood or scaffolding material, and the disposal of rubble through plastic chutes.

5.1.2.5 The TM-DA details the procedures that should generally be adopted by the Noise Control Authority for assessing the use of SPME during restricted hours and for determining whether a CNP would be issued.

5.1.2.6 Maximum noise levels from construction activities during restricted hours at affected Noise Sensitive Receivers (NSRs) are controlled under the TMs and shall not exceed the specified Acceptable Noise Levels (ANLs). These ANLs are stipulated in accordance with the Area Sensitivity Ratings (ASRs) established for the NSRs. The ANLs for construction works in Designated Areas are more stringent than those given in the TM-GW and summarized in **Table 5.1.2**.

Table 5.1.2 ANLs for construction during restricted hours

Time Period	ANL, dB(A)		
	ASR A	ASR B	ASR C
All weekdays during the evening (1900 to 2300 hours), and general holidays (including Sundays) during the day-time and evening (0700 to 2300 hours)	60 (45)	65 (50)	70 (55)
All days during the night-time (2300 to 0700 hours)	45 (30)	50 (35)	55 (40)

Note: Figures in brackets are ANLs for SPME construction work in designated areas

5.1.2.7 As defined in Noise Control Designated Area Plan No. EPD/AN/NT-01, Yuen Long Areas such as Tin Liu Tsuen, Sham Chung Tsuen, Muk Kiu Tau Tsuen, etc. are within the Designated Area.

5.1.2.8 Despite any description made in this EIA, there is no guarantee that a CNP will be issued for the project construction. The Noise Control Authority will consider a well-justified CNP application, once filed, for construction works within restricted hours as guided by the relevant TMs issued under the NCO. The Noise Control Authority will take into

account contemporary conditions / situations of adjoining land uses and any previous complaints against construction activities at the site before making a decision in granting a CNP. Nothing in this EIA report shall bind the Noise Control Authority in making a decision. If a CNP is to be issued, the Noise Control Authority shall include in it any conditions demand. Failure to comply with any such conditions will lead to cancellation of the CNP and prosecution under the NCO. According to the proposed construction methodology for the Project, night-time construction is not anticipated.

Percussive Piling

5.1.2.9 Under the TM-PP, CNPs are also required for percussive piling involving the use of diesel, pneumatic and / or steam hammer. This TM specifies the permitted hours and other conditions for percussive piling. **Table 5.1.3** lists the acceptable percussive piling noise levels for various types of NSR.

Table 5.1.3 ANLs for percussive piling

NSR Window Type or Means of Ventilation	ANL (dB(A))
(i) NSR (or part of NSR) with no window or other opening	100
(ii) NSR with central air conditioning system	90
(iii) NSR with windows or other openings but without central air conditioning system	85

5.1.2.10 Depending on the numbers and types of piling machines and the separation from NSRs, percussive piling may be restricted to 12, 5 or 3 hours per day. For NSRs that are particularly sensitive to noise, such as hospitals, medical clinics, educational institutions and courts of law, a further reduction of 10 dB(A) shall be applied to the above ANLs. According to the proposed construction methodology for the Project, percussive piling is not anticipated.

5.1.3 Operational Noise

5.1.3.1 The EIAO-TM (Annex 5 of TM) has stipulated the noise standards for various noise sources as shown in **Table 5.1.4**. It should, however, be noted that the following noise criteria are only applicable to uses that rely on opened windows for ventilation.

Table 5.1.4 Noise standards for operational phase

Common Uses	Noise Standards[1]				
	Aircraft Noise (Noise Exposure Forecast: NEF)	Helicopter Noise L_{max} dB(A) 0700-1900 hours	Road Traffic Noise $L_{10\ (1hour)}$ dB(A)	Rail Noise[2]	Fixed Noise Sources
All domestic premises including temporary housing accommodation	25	85	70	(a) The appropriate ANLs shown in Table 2 of the Technical Memorandum for the	(a) 5dB(A) below the appropriate ANLs shown in Table 2 of the Technical Memorandum
Hotel and hostels	25	85	70		
Offices	30	90	70		

Common Uses	Noise Standards[1]				
	Aircraft Noise (Noise Exposure Forecast: NEF)	**Helicopter Noise L_{max} dB(A) 0700-1900 hours**	**Road Traffic Noise $L_{10\ (1hour)}$ dB(A)**	**Rail Noise[2]**	**Fixed Noise Sources**
Educational institutions including kindergartens, nurseries and all others where unaided voice communication is required	25	85	65	Assessment of Noise from Places Other than Domestic Premises, Public Places or Construction Sites and (b) L_{max} (2300-0700 hours) = 85dB(A)	for the Assessment of Noise from Places Other than Domestic Premises, Public Places or Construction Sites, or (b) the prevailing background noise levels (For quiet areas with level 5 dB(A) below the ANL)
Places of public Worship and courts of law	25	85	65		
Hospitals, clinics, convalescences and homes for the aged, diagnostic rooms, wards	25	85	55		

Notes:
[1] The above standards apply to uses that rely on opened windows for ventilation.
[2] The above standards should be viewed as the maximum permissible noise levels assessed at 1m from the external facade.

Road Traffic Noise

5.1.3.2 The criteria for assessing road traffic noise is given in the EIAO-TM and tabulated in **Table 5.1.4**. For domestic premises, hotels, hostels and offices, the criterion is 70dB(A). For educational institutions and places of public worship, the criterion is 65dB(A). For hospitals, clinics etc, a more stringent criterion of 55dB(A) is stipulated. It should be noted that all these criteria only apply to NSRs that rely on opened windows for ventilation.

Fixed Noise Sources

5.1.3.3 Operational noise from fixed noise sources is controlled under the TM-Places. To plan for a better environment, the EIAO-TM has specified the following requirements for the planned fixed noise sources, whichever is more stringent.

- 5dB(A) below the appropriate ANLs in the TM-Places; or
- the prevailing background noise levels.

5.1.3.4 Besides, ANL stipulated in the TM-Places has been adopted in the evaluation of potential cumulative fixed noise sources impact assessment from existing and planned fixed noise sources.

5.1.3.5 The ANLs for different ASRs during different periods are summarised in **Table 5.1.5**.

Table 5.1.5 ANLs for fixed noise sources

Time Period	ANL, dB(A)		
	ASR A	ASR B	ASR C
Day (0700 to 1900 hours)	60	65	70
Evening (1900 to 2300 hours)	60	65	70
Night (2300 to 0700 hours)	50	55	60

5.1.3.6 For assessing fixed noise sources, the ASRs at the NSRs are defined in accordance with the relevant TMs. More details are discussed in **Section 5.4**.

Rail Noise

5.1.3.7 The noise criterion for assessing rail noise is given in **Table 5.1.4**. In addition, noise level during the period between 11:00pm and 7:00am of the following day should not exceed L_{max} 85dB(A). Similar to road traffic noise assessment, all these criteria only apply to NSRs relying on opened windows for ventilation.

5.2 Description of the Environment

5.2.1 Existing Environment

5.2.1.1 As discussed in **Section 1**, the Potential Development Area (PDA) (As shown in **Figure 1.1**) is bounded by Yuen Long Highway (YLH) to the north, domestic premises such as Tin Liu Tsuen, Sham Chung Tsuen, Muk Kiu Tau Tsuen, etc. and some rural industrial operations to the east, Tai Lam Country Park to the south, hillside to the west. The current landuse is a mixed zoning of Residential, Government, Institution or Community (G/IC), Village Type Development, Industrial and Undetermined. The existing noise climate in the north of the PDA is dominated by road traffic noise from YLH, Tong Yan San Tsuen (TYST) Interchange, Tin Shui Wai West (TSWW) Interchange and industrial operations in the northern part. There are scattered rural industrial operations in the eastern and southern parts of the PDA.

5.2.2 Prevailing Noise Levels

5.2.2.1 Prevailing noise levels have been measured in the vicinity of the PDA boundary from 2013 to 2017 during daytime, evening and night-time periods. The selected measurement locations cover the accessible areas within and in its immediate vicinity of the PDA (PNM-01 to PNM-08). **Appendix 5.1** shows the locations of prevailing noise measurement.

5.2.2.2 Given the future development would comprise of high-rise residential and commercial developments, multi-storey storage compound and open storage, whereas the existing environment is comprised of low-rise residential development, scattered rural industrial operations and open storage, the prevailing noise measurements conducted in the vicinity of the existing site could not reflect the future noise climate. Therefore, prevailing noise measurements were also selected at alternative locations for reference with a noise climate similar to the future development in these areas. PNM-R1 is located at Long Ping Estate which is a high rise residential area with public housing in Yuen Long, and is therefore considered representative of the future prevailing noise level of the Area 2 and Area 3 development areas. Another reference prevailing noise measurement location PNM-R2 is located in Wang Chau, which is close to low-rise residential and close to industrial estate in Yuen Long, and is therefore representative of the future prevailing noise level of the Area 1 within the PDA. **Appendix 5.1** shows the locations of these measurement points. A summary of the prevailing noise measurement results is given in **Table 5.2.1**.

Table 5.2.1 Prevailing noise measurements

Monitoring Location[1]	Measurement Location	Prevailing Noise Levels[2], dB(A)			Dominant Noise Sources
		Day[3]	Evening[3]	Night[3]	
PNM-01	Kung Um Road near Muk Kiu Tau Tsuen	72 – 74	69 – 70	67 – 69	Traffic noise from Kung Um Road and noise from industrial activities
PNM-02	Lutheran Academy	62 – 65	59 – 60	56 – 60	Traffic noise from Lam Hau Tsuen Road and noise from industrial activities
PNM-03	Shan Ha Tsuen	57 – 58	53 – 55	53 – 54	Minor noise from industrial activities
PNM-04	Fui Sha Wai	60 – 62	60 – 61	59 – 60	Traffic noise from local road near Fui Sha Wai and noise from industrial activities
PNM-05	Park Villa	60 – 62	48 – 50	46 – 48	Minor noise from industrial activities
PNM-06	South of Shan Ha Tsuen	58 – 60	53 – 54	53 – 54	Minor noise from industrial activities
PNM-07	Pak Sha Shan Road	66 – 67	57 – 60	51 – 54	Traffic noise from Kung Um Road and Pak Sha Shan Road and minor noise from industrial activities
PNM-08	Tong Yan San Tsuen	65 – 66	59 – 60	57 – 60	Traffic noise from TYST Road and minor noise from industrial activities
PNM-R1	Long Ping Estate	71 – 72	65 – 67	58 – 60	Traffic noise from Long Tin Road and community noise
PNM-R2	Wang Chau	73 – 74	67 – 68	61 – 62	Traffic noise from Fuk Hi Street and

Monitoring Location[1]	Measurement Location	Prevailing Noise Levels[2], dB(A)			Dominant Noise Sources
		Day[3]	Evening[3]	Night[3]	
					minor noise from industrial activities

Note:
[1] PNM represents prevailing noise measurement locations which are indicated in **Appendix 5.1**.
[2] The noise measurement descriptor is A-weighted equivalent continuous sound pressure level (L_{eq}) measured using Type 1 sound level meter. Measurements were conducted in January 2013, February 2013, May 2013, January 2014, June 2015, June 2016, January 2017 and February 2017, covering typical daytime, evening and night-time periods (unless specified otherwise). Depending on site conditions at the measurement location, a series of measurements with duration of 10-20 minutes were taken during non-peak traffic hours to represent the prevailing background noise. Facade correction was adopted.
[3] Day: 0700 to 1900 hours, Evening: 1900 to 2300 hours, Night: 2300 – 0700 hours.

5.2.3 Previous Studies and Current Studies

Previous Studies

Hung Shui Kiu New Development Area (HSK NDA)Planning and Engineering Study

5.2.3.1 The HSK NDA EIA report has been approved in December 2016 (AEIAR-203/2016). The HSK NDA is located at the north west of the YLS PDA. Their 300m noise assessment area has covered a small area near Long Tin Road. The relevant NSRs and assessment results have been reviewed and considered as necessary. Cumulative impacts during both construction and operational phases are anticipated.

Engineering Works at Lin Cheung Road Site, Sham Shui Po and Wang Chau, Yuen Long

5.2.3.2 To support the proposed public housing development, CEDD commissioned the study for the provision of essential infrastructure. The first phase of building works of the housing flats at Wang Chau is targeted to commence in mid 2018.

5.2.3.3 As the works boundary of the proposed Wang Chau site is located more than 1km away from the YLS PDA, cumulative impact during construction phase is not anticipated. However, road traffic noise impact during operational phase is anticipated from the induced traffic.

Improvement to Pok Oi Interchange

5.2.3.4 To improve the traffic capacity of Pok Oi Interchange to relieve the existing traffic pressure and meet the future traffic demand. The Project includes provision of an additional lane of YLH (Kowloon bound) at Pok Oi Interchange, construction of a slip road connecting the elevated and ground level sections of YLH (Tuen Mun bound), and construction of a segregated left-turn lane at the approach road of Pok Oi Roundabout near Pok Oi Hospital.

5.2.3.5 The scheme commenced in late 2012 and is completed in 2016. Cumulative impact during construction phase is not anticipated. Road traffic noise impact during operational phase is anticipated from the induced traffic.

Current Studies

Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long, Feasibility Study

5.2.3.6 A District Council (DC) Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10" has been discussed at the DC meeting on 21 February 2017. As mentioned in the DC Discussion Paper, the project area would be approximate 10ha for developing 11 blocks of public housing for accommodation of a population of 31,200. The construction period and population intake under this project would be divided into 2 stages, in which the population intake years of stages 1 and 2 are anticipated to be Year 2024/25 and Year 2027/28 respectively.

5.2.3.7 The project area of the development at Long Bin is located approximately 150m away from the YLS PDA opposite to YLH. Available information relating to Long Bin development has also been reviewed. However, since the detailed construction programme, plant inventory, etc. are yet to be developed by the respective project proponent, only committed and planned NSR has been added within the Long Bin development to account for the impacts from the development of YLS PDA.

5.2.3.8 In addition, road traffic noise impact during operational phase is anticipated from the induced traffic. Cumulative impact on its planned receivers have also been assessed under this project.

Site Formation and Infrastructural Works for the Development near Tan Kwai Tsuen, Yuen Long – Feasibility Study

5.2.3.9 A DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10" has been discussed at the DC meeting on 21 Feb 2017. As mentioned in the DC Discussion Paper, the project area would be approximate 10.6ha for developing 7 blocks of public housing for accommodation of a population of 20,600, and the population intake year is anticipated to be Year 2027/28.

5.2.3.10 The project area of the development at Tan Kwai Tsuen is located approximately 300m away from the YLS PDA next to the Tan Kwai Tsuen Fresh Water Service Reservoir along YLH. Road traffic noise impact during operational phase are anticipated from the induced traffic.

Property Development of Kam Sheung Road Station and its Adjoining Areas

5.2.3.11 Land Use Review has been conducted to examine the development potential of the Kam Tin South and Pat Heung Areas. The Land Use Review has demonstrated that it would be technically feasible to transform Kam Tin South and Pat Heung into a sustainable suburban township with enhancement on infrastructure and utility services.

5.2.3.12 As the area is located more than 1km away from the YLS PDA, cumulative impact during construction phase is not anticipated. However, road traffic noise impact during operational phase is anticipated from the induced traffic.

5.3 Inventory of Noise Sources

5.3.1 Construction Phase

5.3.1.1 As discussed in **Section 3.7**, the PDA is divided into three areas and will be implemented in four stages. The tentative construction period for these four stages are summarized as below:

- Stage 1 between Year 2020 and Year 2029;
- Stage 2 between Year 2022 and Year 2033;
- Stage 3 between Year 2031 and Year 2038; and
- Stage 4 between Year 2032 and Year 2038.

5.3.1.2 Potential construction noise impacts would be arising from the use of PME during the following major construction works:

- Site clearance;
- Site formation;
- Superstructure of residential, G/IC, commercial, industrial development and open space areas;
- Construction of internal and external roads;
- Road widening of existing roads;
- Construction of sewage treatment works, sewage pumping stations; and
- Construction of utilities.

5.3.1.3 Other minor works including ground investigation (GI), temporary traffic diversion, etc. would unlikely cause adverse environmental impacts.

5.3.1.4 The plant inventory as provided by the Project Engineer and the details of PME adopted for the assessment are presented in **Appendix 5.2** and **Appendix 5.3** according to the development programme and work site locations as given in **Appendix 5.4** and **Appendix 5.5** respectively. The construction plant inventory has also been confirmed with relevant government departments.

5.3.2 Operational Phase

5.3.2.1 During operational phase, the identified noise sources include the following:

- Existing and planned roads within 300m from the PDA boundary and the associated infrastructure works;
- Existing and planned fixed noise sources; and
- Rail noise from the proposed Environmental Friendly Transport System (EFTS) within the PDA boundary.

Road Traffic Noise

Existing and Planned Roads

Existing Roads Comprises existing roads surrounding the PDA including:

- YLH
- Long Tin Road
- Hung Tin Road
- Section of Kung Um Road
- Section of TYST Road
- Sha Tseng Road
- Shan Ha Road
- Lam Yu Road
- Castle Peak Road
- Shap Pat Heung Road
- Tai Tong Road
- Lam Hau Tsuen Road
- Lam Hi Road
- Tai Shu Ha Road East
- Fung Ki Road
- Fung Cheung Road
- Siu Sheung Road
- Sham Chung Road
- Town Park Road South

	• San Hi Tseun Street • Ping Tong Street East • Ping Tong Street South • Ping Tong Street West • San Fui Street • Hung Shui Kiu Tin Sam Road Kung Um Road and TYST Road will have road improvements to increase the carriageway width to standard width 6.75m. It will remain same configuration (i.e. two lane local distributor). Low noise road surfacing (LNRS) will be provided together with the road improvement works.
Planned Designated Project (DP) Roads (within PDA)	Comprises: • TSWW Interchange (trunk roads and underpass) • TYST Interchange (primary distributor roads) • Road D1 and D2 (district distributor roads) (As shown in **Figure 5.1**)
Planned Non – DP Roads (within PDA)	Comprises local roads within the PDA including: • Local Roads (L1 – L27) • Lam Tai East Road • Lam Tai West Road • Section of Kung Um Road • Tan Kwai Tsuen Road • Section of TYST Road
Planned DP Roads (outside PDA)	Comprises: • TSWW Interchange (trunk roads) • TYST Interchange (primary distributor roads) (As shown in **Figure 5.1**).
Planned Non – DP Roads (outside PDA)	Comprises local roads including: • Long Ho Road • Shung Ching Road

- Tai Kei Leng Road
- Shan Ha Road
- Yuen Ching Road
- Slip roads to YLH from Kung Um Road eastward

5.3.2.2 All roads mentioned above will have certain road traffic noise impacts on both existing and committed and planned NSRs, and have been suitably considered in the current assessment. Key existing and planned road traffic noise sources are illustrated in **Figure 5.1**.

Fixed Noise Sources

5.3.2.3 The area for fixed noise sources impact assessment includes i) 300m assessment area for committed and planned NSRs affected by planned and existing fixed noise sources; ii) 300m assessment area for planned fixed noise sources. Key existing and planned fixed noise sources are illustrated in **Figure 5.1** and are summarised in **Table 5.3.1**.

Table 5.3.1 List of key existing and planned fixed noise sources

Types	Noise Sources	Reference Locations[1]
Existing Noise Sources	Industrial Zone (OZP No.: S/YL-TYST/10)	Outside PDA near Fui Sha Wai
	Livestock Farm[1]	AGR 3.1
	Lung Tin Sewage Pumping Station[1]	To the northwest of Tin Liu Tsuen (OU2.2)
Planned Noise Sources	Electricity Substation	OU2.1
	Refuse Collection Point and Sewage Pumping Station	OU1.12, OU2.2
	Sewage Treatment Works	OU3.3
	Divisional Fire Station and Ambulance Depot	G2.3
	Public Transport Interchange	OU1.1, OU2.3, OU3.2a, in the vicinity of Yoho Midtown
	Sports Centre and Market	G2.1
	District Police Station	G3.2
	Open Storage	OU1.6
	Storage Use	OU1.5, OU1.7, OU1.9, OU1.10
	Storage and Workshop	OU1.8
	Retention Tank	OU1.11
	Government Reserve (intended for Government Depot)	G3.1, G3.3

Note:
[1] Existing use retained in the PDA.

5.3.2.4 The nature and operation of these noise sources are described as follows.

Existing Industrial Zone (Outside PDA near Fui Sha Wai)

5.3.2.5 An industrial zone ("I" zone) is located outside the PDA to the north of YLH and to the east of Fui Sha Wai. The nearest existing NSR is within "I" zone and the nearest planned NSR is located at about 330m away from the "I" zone. Fixed noise sources impacts have been addressed in this noise assessment.

Existing Livestock Farm (AGR3.1)

5.3.2.6 A livestock farm (AGR3.1) is located in Area 3 within the PDA and will be retained under the Recommended Outline Development Plan (RODP). The major noise sources of the livestock farm would mainly consist of ventilation system with louver and generator etc. The livestock farm (AGR3.1) is located at 90m from the nearest existing NSR and 185m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Existing Lung Tin Sewage Pumping Station (OU2.2)

5.3.2.7 The existing Lung Tin sewage pumping station (LTSPS) is located in Area 2 within PDA at about 20m to the south of YLH and about 20m to the west of Kung Um Road. The major noise sources of the LTSPS (OU2.2) would mainly consist of ventilation system with louver. The LTSPS (OU2.2) is located at 75m from the nearest existing NSR and 190m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Electricity Substation (OU2.1)

5.3.2.8 An electricity substation (ESS) (OU2.1) is planned to be located in Area 2 at the middle of the PDA at about 3m to the west of Kung Um Road. The major noise sources of the ESS are from the operation of transformers and ventilation systems. The ESS (OU2.1) is located at 40m from the nearest existing NSR and 95m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Refuse Collection Point & Sewage Pumping Station (OU1.12, OU2.2)

5.3.2.9 Two refuse collection points (RCP) and two sewage pumping stations (SPS) (OU1.12, OU2.2) are planned within the PDA. One RCP and one SPS (OU1.12) will be located Area 1 at the northern part of the PDA to south of YLH and east of TYST Interchange. They will be located at about 200m from the planned PTI (OU1.1) and retention tank (OU1.11). The nearest existing NSR is located at about 70m from the RCP and SPS (OU1.12) and the nearest planned NSR is located at about 100m from the RCP and SPS (OU1.12).

5.3.2.10 Another RCP and SPS (OU2.2) will be located Area 2 at the northern part of the PDA to south of YLH and in the vicinity of the LTSPS. The nearest existing NSR is located at about 70m from the RCP and SPS

(OU2.2) and the nearest planned NSR is located at about 165m from the RCP and SPS (OU2.2).

5.3.2.11 The major noise sources of RCP and SPS are from the operation of pumps, mechanically racked screens, transformer and ventilation fan of deodorization unit. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Sewage Treatment Works (OU3.3)

5.3.2.12 A sewage treatment works (STW) (OU3.3) is planned to be located in Area 3 at the southern end of the PDA to the southwest of the existing Wong Nai Tun Tsuen. The major noise sources of the STW are from the operation of sewage facilities and ventilation system. The STW (OU3.3) is located at 260m from the nearest existing NSR and 180m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Divisional Fire Station and Ambulance Depot (G2.3)

5.3.2.13 A divisional fire station and ambulance depot (G2.3) is planned to be located in Area 2 at the middle of the PDA to the south of the existing Shan Ha Tsuen. The planned divisional fire station and ambulance depot (G2.3) is located at 270m from the nearest existing NSR and 60m from the nearest planned NSR.

5.3.2.14 The major noise sources from the planned divisional fire station and ambulance depot operations include loudspeakers, siren, vehicle sirens, etc. In addition, the operation of a fire station also require certain fixed noise sources such as condenser, transformer, etc. These noise may be potential nuisance to the nearby NSRs, however their occurrences are on as-needed basis and last for short duration only. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Public Transport Interchange (OU1.1, OU2.3, OU3.2a, in the vicinity of Yoho Midtown)

5.3.2.15 Three Public Transport Interchange (PTIs) (OU1.1, OU2.3, OU3.2a) are planned within the PDA and one PTI is planned near Yoho Midtown outside the PDA. One of the PTIs (OU1.1) will be located in Area 1 to south of TYST Interchange adjacent to a planned retention tank (OU1.11) and about 200m from a RCP and SPS (OU1.12). The nearest existing NSR is located at about 110m away from this PTI and the worst affected planned NSR is located at about 62m away from this PTI.

5.3.2.16 Another PTI (OU2.3) will be located in Area 2 at the middle of the PDA at about 35m to west of Kung Um Road adjacent to a planned sports centre and market (G2.1). The nearest existing NSR is located at about 80m away from this PTI and the worst affected planned NSR is located at about 50m away from this PTI.

5.3.2.17 Another PTI (OU3.2a) will be located in Area 3 to the west of Pak Sha Tsuen and at about 25m to the west of Kung Um Road adjacent to a planned police station (G3.2). The nearest existing NSR is located at

about 145m away from this PTI and the worst affected planned NSR is located at about 29m away from this PTI.

5.3.2.18 The fourth PTI will be located outside the PDA to the east of Yoho Midtown and north of Kwong Ming Ying Loi School. The nearest existing NSR is located at about 10m away from this PTI. There are no planned NSR in the vicinity of this PTI.

5.3.2.19 The major fixed noise sources of PTIs include the operation of ventilation fans, bus idling, etc.. All the four PTIs would be decked under proposed building structures and designed with no line-of-sight at the noise sensitive use. In addition, the PTI would be oriented with ingress and egress facing away from NSRs in close vicinity. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Sports Centre and Market (G2.1)

5.3.2.20 A sports centre and a market (G2.1) are planned to be co-located in Area 2 at the middle of the PDA at about 25m to west of Kung Um Road. The major fixed noise sources of the sports centre and market include chiller, ventilation louver, etc.. The planned sports centre and a market (G2.1) is located at 85m from the nearest NSR and 10m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned District Police Station (G3.2)

5.3.2.21 A district police station (G3.2) is planned to be located in Area 3 to the west of Pak Sha Tsuen and at about 150m to the west of Kung Um Road. The major fixed noise sources of the district police station include chiller, ventilation louver, etc.. The planned district police station (G3.2) is located at 240m from the nearest NSR and 8m from planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Sites for Storage Use, Open Storage as well as Storage and Workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10)

5.3.2.22 Several sites in Area 1 at the north western part of the PDA to the south of YLH are planned for open storage, storage use as well as storage and workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10). The major fixed noise sources of the sites for storage use, open storage as well as storage and workshop include loading and unloading, forklifting, chiller, ventilation louver, etc.. The nearest existing NSR to these OU sites is located at about 45m from the storage use site (OU1.7) and the nearest planned NSR to these OU sites is located at about 35m from the storage use site (OU1.10). Fixed noise sources impacts have been addressed in this noise assessment.

Planned Retention Tank (OU1.11)

5.3.2.23 A retention tank (OU1.11) is planned to be located in a district open space in Area 1 near TYST Interchange will be installed underground with pumps and ventilation fans operating above ground. The planned

retention tank (OU1.11) is located at 105m from the nearest existing NSR and 32m from the planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Planned Government Reserve (Intended for Government Depot) (G3.1, G3.3)

5.3.2.24 Two sites in Area 3 at the southern end of the PDA to the southwest of the existing Wong Nai Tun Tsuen are planned for government depot use (G3.1 and G3.3). The major noise sources of the government depot include the operation of chiller, ventilation louver, maintenance activities, etc.. The planned government depot (G3.1, G3.3) is located at 155m from the nearest existing NSR and 280m from the nearest planned NSR. Fixed noise sources impacts have been addressed in this noise assessment.

Rail Noise

5.3.2.25 An EFTS is proposed to serve the community within the PDA and the tentative alignment is given in **Figure 5.1**. This proposed alignment of EFTS connects to MTR Tin Shiu Wai Station via the proposed EFTS of the HSK NDA. Therefore, the operational characteristics of EFTS such as the configuration, operation details, operating speed, etc. will be similar to those in the approved HSK NDA EIA Report (AEIAR-203/2016). Rail noise impacts have been addressed in this noise assessment.

5.4 Noise Sensitive Receivers

5.4.1.1 With reference to Annex 13 of the EIAO-TM, NSRs shall include residential uses (all domestic premises including temporary housing), institutional uses (educational institutions including kindergarten and nurseries), hospitals, medical clinics, homes for the aged, convalescent homes, places of public worship, libraries, court of law, performing arts centres, auditoria and amphitheatres, country park and others.

5.4.1.2 The assessment area for noise impact should generally be defined by a distance of 300m from the PDA boundary and the associated infrastructure works. Representative NSRs within a distance of 300m from the PDA boundary and the associated infrastructure works have been identified with the first layer of NSRs selected for assessment. These NSRs include both the existing and committed and planned sensitive developments during both construction and operational phase. The planned sensitive development within the RODP and the indicative layout has been confirmed by Lands Department and Planning Department.

5.4.1.3 The representative existing and committed and planned NSRs are identified by means of reviewing topographic maps, aerial photos, land status plans, relevant Outline Zoning Plans (OZP), Outline Development Plans, Layout Plans and other relevant published land use plans, including plans and drawings published by Lands Department

and land use and development applications approved by the Town Planning Board, and supplemented by site inspections. Photos of the representative existing NSRs are shown in **Appendix 5.6**.

5.4.1.4 For the planned clinic and social welfare building, they are anticipated to be provided with central air-conditionings and would not rely on openable window for ventilation and therefore the standards in **Table 5.1.4** do not apply. The representative NSRs identified are summarized in **Table 5.4.1** and their locations are shown in **Figure 5.2**.

Table 5.4.1 Existing, committed and planned NSRs

NSR[1]	NSR Description[2]	Use[3]	No. of Storey[5]
Existing NSRs (Within PDA)			
E1_CLMY	Chuk Lam Ming Yuen Monastery	W	2
E1_ELD	Eldorado	R	4
E1_FLG	Fu Lai Garden	R	3
E1_GR	Greenville Residence	R	4
E1_KLG	Kam Lai Garden	R	3
E1_KLK	Kam Lan Koon	W	1
E1_KV	Kisland Villa	R	4
E1_LHT	Lam Hau Tsuen	R	3
E1_MG	Marbella Gardens	R	4
E1_PH	The Parkhill	R	4
E1_PV	Park Villa	R	3
E1_RLS	Recours La Serre	R	4
E1_SHT	Shan Ha Tsuen	R	3
E1_TYST	Tong Yan San Tsuen	R	1-4
E1_WG	Winsor Garden	R	4
E1_YHT	Yeung Hau Temple	W	1
E2_MKTT	Muk Kiu Tau Tsuen	R	1
E2_TLT	Tin Liu Tsuen	R	3
E2_PST	Pak Sha Tsuen	R	2
E2_TLY	Tin Lung Yuen	R	3
Existing NSRs (Outside PDA)			
E4_CFG	Chun Fai Garden	R	3
E4_CPG	Chung Pak Garden	R	3
E4_FSW	Fui Sha Wai	R	1-3

NSR[1]	NSR Description[2]	Use[3]	No. of Storey[5]
E4_FWM	Fu Wa Mansion	R	3
E4_GM	Garden Maple	R	3
E4_HCG	Hang Cheong Garden	R	3
E4_HG	Hoover Garden	R	3
E4_HYTT	Ha Yau Tin Tsuen	R	3
E4_HTS	Ha Tsuen Shi	R	3
E4_JC	Jasper Court	R	4
E4_KMYLS	Kwong Ming Ying Loi School	E	7
E4_KTST	Kong Tau San Tsuen	R	3
E4_LA	Lutheran Academy	E	8
E4_LFM	Lung Fai Mansion	R	3
E4_LG	La Grove	R	19
E4_LHT	Lam Hau Tsuen	R	1-3
E4_LHWT	Lam Hau Wai Tsuen	R	1-2
E4_LTG	Lung Teng Garden	R	3
E4_LTT	Lung Tin Tsuen	R	1-3
E4_MKT	Muk Kiu Tau Tsuen	R	3
E4_OHP	One Hyde Park	R	2
E4_PCG	Pui Chak Garden	R	2-3
E4_PHY	Pitt Hwa Yuen	R	3
E4_PS	Park Signature	R	22
E4_PST	Pak Sha Tsuen	R	2-3
E4_PSV	Parkside Villa	R	10
E4_RP	Regent's park	R	2-3
E4_SCST	Sham Chung San Tsuen	R	3
E4_SCT	Sham Chung Tsuen	R	3
E4_SG	Scenic Gardens	R	14
E4_SHT	Shan Ha Tsuen	R	2-3
E4_SMG	Sun Mei Garden	R	3
E4_STG	Sun Tong Garden	R	3
E4_SV	Sereno Verde	R	13
E4_SYTT	Sheung Yau Tin Tsuen	R	2

NSR[1]	NSR Description[2]	Use[3]	No. of Storey[5]
E4_SPT	Shek Po Tsuen	R	3
E4_TKL	Tai Kei Leng	R	1
E4_TKLT	Tai Kei Leng Tsuen	R	2-3
E4_TKT	Tan Kwai tsuen	R	3
E4_TLT	Tin Liu Tsuen	R	3
E4_TR	The Reach	R	24
E4_TTT	Tai Tao Tsuen	R	3
E4_TYST	Tong Yan San Tsuen	R	2
E4_UT	Uptown	R	3-19
E4_WNT	Wong Nai Tun Tsuen	R	1-3
E4_WY	Wa Yuen	R	3
E4_YH	Yoho Town	R	34-37
E4_YHM	Yoho Midtown	R	40
Committed and Planned NSRs			
P_E1.1	Planned primary school	E	8
P_E2.1	Planned primary school	E	8
P_E2.2	Planned primary school	E	8
P_E2.3	Planned primary school	E	8
P_E2.4	Planned primary school	E	8
P_E3.1	Planned primary school	E	8
P_E3.2	Planned primary school	E	8
P_E3.3	Planned primary school	E	8
P_G2.4	Married quarters	R	19
P_R1.1	Private housing	R	4
P_R1.2	Private housing	R	4
P_R1.3	Private housing	R	4-5
P_R1.4	Private housing	R	4-5
P_R1.5	Private housing	R	4
P_R1.6	Private housing	R	3-4
P_R1.7	Private housing	R	4-5
P_R1.8	Private housing	R	4
P_R1.9	Private housing	R	3

NSR[1]	NSR Description[2]	Use[3]	No. of Storey[5]
P_R1.10	Private housing	R	6
P_R1.11	Private housing	R	5
P_R1.12	Private housing	R	4
P_R1.13	Private housing	R	4-5
P_R1.14	Private housing	R	4
P_R1.15	Private housing	R	4
P_R1.16	Private housing	R	4
P_R2.1a	Private housing	R	30-32
P_R2.1b	Public housing	R	32-34
P_R2.2	Public housing	R	32-34
P_R2.3	Public housing	R	31-34
P_R2.4	Public housing	R	28-34
P_R2.5	Public housing	R	22-31
P_R2.6a	Private housing	R	20-26
P_R2.6b	Private housing	R	18-24
P_R2.6c	Private housing	R	20-22
P_R2.6d	Private housing	R	20-22
P_R2.7	Private housing	R	9-11
P_R2.8	Public housing	R	33
P_R3.1a	Private housing	R	17
P_R3.1b	Private housing	R	20-22
P_R3.1c	Private housing	R	11
P_R3.1d	Private housing	R	12
P_R3.1e	Private housing	R	11
P_R3.2	Private housing	R	22
P_R3.3	Private housing	R	20
P_R3.4a	Private housing	R	18
P_R3.4b	Private housing	R	14
P_R3.5a	Private housing	R	18-19
P_R3.5b	Private housing	R	21-22
P_R3.6a	Private housing	R	9
P_R3.6b	Private housing	R	8

NSR[1]	NSR Description[2]	Use[3]	No. of Storey[5]
P_OU1.1	Private housing	R	12
P_OU1.2a	Private housing	R	12
P_OU1.3	Private housing	R	10-11
P_OU2.3	Private housing	R	11
P_OU3.2a	Private housing	R	11
P_OU3.2b	Private housing	R	9
P_VR1.1	Village rehousing	R	3
P_VR2.1	Village rehousing	R	3
P_VR3.1	Village rehousing	R	3
P_HSK[4]	Planned residential development in HSK NDA	R	28-29
P_LBIH[4]	Planned public housing development in Long Bin	R	40

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within 300m from the PDA boundary and the associated infrastructure works is included.
[3] R – Residential Premises, E – Educational Institutions; W – Places of Public Worship; C – Clinics, Residential Care Home for the Elderly (RCHE). The land uses of the planned NSRs within the PDA are identified based on the RODP.
[4] Planned NSRs outside PDA.
[5] No. of storeys with noise-sensitive uses.

5.4.2 Construction Noise

5.4.2.1 For construction noise assessment, the assessment area includes those within 300m from the PDA boundary and the associated infrastructure works. The first layer of NSRs that would be located closest to the construction activities have been selected from **Table 5.4.1**. The locations of representative NSRs and Noise Assessment Points (NAPs) for various stages for construction noise assessment are shown in **Figure 5.3** and are summarised in **Table 5.4.2**.

Table 5.4.2 Representative NSRs and NAPs for construction noise assessment

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP
Existing NSRs (Within PDA)				
Stage 1	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R01
	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R04
	E2_TLY	Tin Lung Yuen	R	E2_TLY_R01
Stage 2	E1_RLS	Recours La Serre	R	E1_RLS_R02
	E1_ELD	Eldorado	R	E1_ELD_R02
	E1_PV	Park Villa	R	E1_PV_R02
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R07

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R13
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R32
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R36
	E1_MG	Marbella Gardens	R	E1_MG_R02
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R05
	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R01
	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R04
	E2_TLY	Tin Lung Yuen	R	E2_TLY_R01
	E2_MKTT	Muk Kiu Tau Tsuen	R	E2_MKTT_R01
Stage 3	E2_MKTT	Muk Kiu Tau Tsuen	R	E2_MKTT_R01
Stage 4	E1_RLS	Recours La Serre	R	E1_RLS_R02
	E1_ELD	Eldorado	R	E1_ELD_R02
	E1_PV	Park Villa	R	E1_PV_R02
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R07
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R13
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R36
	E1_MG	Marbella Gardens	R	E1_MG_R02
	E1_PH	The Parkhill	R	E1_PH_R01
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R05
	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R01
	E1_SHT	Shan Ha Tsuen	R	E1_SHT_R04
	E2_TLY	Tin Lung Yuen	R	E2_TLY_R01
Existing NSRs (Outside PDA)				
Stage 1	E4_SHT	Shan Ha Tsuen	R	E4_SHT_R07
	E4_LTT	Lung Tin Tsuen	R	E4_LTT_R01
	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R02
	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R12
Stage 2	E4_TTT	Tai Tao Tsuen	R	E4_TTT_R01
	E4_LTT	Lung Tin Tsuen	R	E4_LTT_R01
	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R02
	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R12
	E4_SHT	Shan Ha Tsuen	R	E4_SHT_R07

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP
	E4_PST	Pak Sha Tsuen	R	E4_PST_R01
	E4_MKT	Muk Kiu Tau Tsuen	R	E4_MKT_R05
	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R11
	E4_HTS	Ha Tsuen Shi	R	E4_HTS_R01
	E4_HTS	Ha Tsuen Shi	R	E4_HTS_R02
	E4_HTS	Ha Tsuen Shi	R	E4_HTS_R03
	E4_SPT	Shek Po Tsuen	R	E4_SPT_R01
	E4_UT	Uptown	R	E4_UT_R01
	E4_LA	Lutheran Academy	E	E4_LA_E01
	E4_KMYLS	Kwong Ming Ying Loi School	E	E4_KMYLS_E03
Stage 3	E4_PST	Pak Sha Tsuen	R	E4_PST_R01
	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R11
Stage 4	E4_TTT	Tai Tao Tsuen	R	E4_TTT_R01
	E4_LA	Lutheran Academy	E	E4_LA_E01
Planned NSRs				
Stage 1	P_R2.2	Public housing	R	P_R2.2_R01
	P_R2.2	Public housing	R	P_R2.2_R10
	P_R2.2	Public housing	R	P_R2.2_R17
	P_VR1.1	Village rehousing	R	P_VR1.1_R01
	P_VR2.1	Village rehousing	R	P_VR2.1_R01
Stage 2	P_VR3.1	Village rehousing	R	P_VR3.1_R02
	P_E2.1	Planned primary school	E	P_E2.1_E01
	P_R2.3	Public housing	R	P_R2.3_R08
	P_R2.4	Public housing	R	P_R2.4_R06
[4]	P_HSK	Planned residential development in HSKND	R	P_HSK_R01
	P_HSK	Planned residential development in HSK NDA	R	P_HSK_R02
	P_HSK	Planned residential development in HSK NDA	R	P_HSK_R03
[5]	P_LBIH	Planned public housing development in Long Bin	R	P_LBIH_R02

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within 300m from the PDA boundary and

the associated infrastructure works is included.
[3] R – Residential Premises, E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP.
[4] Planned NSRs outside PDA.
[5] Planned NSRs outside PDA. Since the layout of Long Bin development is yet to be developed, the location of NAPs have been assumed at 10m from TYST Interchange and 10m from Long Tin Road.

5.4.3 Road Traffic Noise

5.4.3.1 For operational road traffic noise assessment, the assessment area includes areas within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project. All levels of the first layer of existing and planned NSRs which are most affected by the project in the vicinity of project roads and committed and planned NSRs under the Project located in the vicinity of existing roads have been selected from **Table 5.4.1**. The locations of representative NSRs and NAPs for operational road traffic noise assessment are shown in **Figure 5.4** and are summarised in **Table 5.4.3**.

Table 5.4.3 Representative NSRs and NAPs for road traffic noise assessment

NSR[1],[2]	NSR Description	Use[3]	NAP
Existing NSRs (Within PDA)			
E1_CLMY	Chuk Lam Ming Yuen Monastery	W	E1_CLMY_T01
E1_ELD	Eldorado	R	E1_ELD_R01
E1_FLG	Fu Lai Garden	R	E1_FLG_R01
E1_GR	Greenville Residence	R	E1_GR_R01
E1_KLG	Kam Lai Garden	R	E1_KLG_R01
E1_KLK	Kam Lan Koon	W	E1_KLK_T01
E1_KV	Kisland Villa	R	E1_KV_R01 to E1_KV_R02
E1_LHT	Lam Hau Tsuen	R	E1_LHT_R01
E1_MG	Marbella Gardens	R	E1_MG_R01
E1_PH	The Parkhill	R	E1_PH_R01
E1_PV	Park Villa	R	E1_PV_R01 to E1_PV_R05
E1_RLS	Recours La Serre	R	E1_RLS_R01 to E1_RLS_R04
E1_SHT	Shan Ha Tsuen	R	E1_SHT_R01 to E1_SHT_R07
E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23, E1_TYST_R31 to E1_TYST_R38[5]
E1_WG	Winsor Garden	R	E1_WG_R01
E1_YHT	Yeung Hau Temple	W	E1_YHT_T01

NSR[1],[2]	NSR Description	Use[3]	NAP
E2_MKTT	Muk Kiu Tau Tsuen	R	E2_MKTT_R01
E2_TLT	Tin Lung Tsuen	R	E2_TLT_R01
E2_PST	Pak Sha Tsuen	R	E2_PST_R01
E2_TLY	Tin Lung Yuen	R	E2_TLY_R01 to E2_TLY_R02
Existing NSRs (Outside PDA)			
E4_CFG	Chun Fai Garden	R	E4_CFG_R01 to E4_CFG_R02
E4_CPG	Chung Pak Garden	R	E4_CPG_R01
E4_FSW	Fui Sha Wai	R	E4_FSW_R01 to E4_FSW_R02
E4_FWM	Fu Wa Mansion	R	E4_FWM_R01
E4_GM	Garden Maple	R	E4_GM_R01
E4_HCG	Hang Cheong Garden	R	E4_HCG_R01 to E4_HCG_R02
E4_HG	Hoover Garden	R	E4_HG_R01
E4_HYTT	Ha Yau Tin Tsuen	R	E4_HYTT_R01
E4_JC	Jasper Court	R	E4_JC_R01
E4_KMYLS	Kwong Ming Ying Loi School	E	E4_KMYLS_E01
E4_KTST	Kong Tau San Tsuen	R	E4_KTST_R01 to E4_KTST_R02
E4_LA	Lutheran Academy	E	E4_LA_E01
E4_LFM	Lung Fai Mansion	R	E4_LFM_R01
E4_LG	La Grove	R	E4_LG_R01 to E4_LG_R02
E4_LHT	Lam Hau Tsuen	R	E4_LHT_R01 to E4_LHT_R05
E4_LHWT	Lam Hau Wai Tsuen	R	E4_LHWT_R01 to E4_LHWT_R04
E4_LTG	Lung Teng Garden	R	E4_LTG_R01
E4_LTT	Lung Tin Tsuen	R	E4_LTT_R01 to E4_LTT_R28
E4_MKT	Muk Kiu Tau Tsuen	R	E4_MKT_R01 to E4_MKT_R05
E4_OHP	One Hyde Park	R	E4_OHP_R01 to E4_OHP_R02
E4_PCG	Pui Chak Garden	R	E4_PCG_R01 to E4_PCG_R02
E4_PHY	Pitt Hwa Yuen	R	E4_PHY_R01 to E4_PHY_R02
E4_PS	Park Signature	R	E4_PS_R01 to E4_PS_R03
E4_PST	Pak Sha Tsuen	R	E4_PST_R01 to E4_PST_R20
E4_PSV	Parkside Villa	R	E4_PSV_R01 to E4_PSV_R02

NSR[1],[2]	NSR Description	Use[3]	NAP
E4_RP	Regent's Park	R	E4_RP_R01 to E4_RP_R02
E4_SCST	Sham Chung San Tsuen	R	E4_SCST_R01 to E4_SCST_R02
E4_SCT	Sham Chung Tsuen	R	E4_SCT_R01 to E4_SCT_R03
E4_SG	Scenic Gardens	R	E4_SG_R01 to E4_SG_R02
E4_SHT	Shan Ha Tsuen	R	E4_SHT_R01 to E4_SHT_R07
E4_SMG	Sun Mei Garden	R	E4_SMG_R01 to E4_SMG_R04
E4_STG	Sun Tong Garden	R	E4_STG_R01
E4_SV	Sereno Verde	R	E4_SV_R01
E4_SYTT	Sheung Yau Tin Tsuen	R	E4_SYTT_R01
E4_TKL	Tai Kei Leng	R	E4_TKL_R01
E4_TKLT	Tai Kei Leng Tsuen	R	E4_TKLT_R01 to E4_TKLT_R06
E4_TKT	Tan Kwai Tsuen	R	E4_TKT_R01 to E4_TKT_R03
E4_TLT	Tin Liu Tsuen	R	E4_TLT_R01 to E4_TLT_R18
E4_TR	The Reach	R	E4_TR_R01
E4_TTT	Tai Tao Tsuen	R	E4_TTT_R01 to E4_TTT_R03
E4_TYST	Tong Yan San Tsuen	R	E4_TYST_R01
E4_UT	Uptown	R	E4_UT_R01 to E4_UT_R06
E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R01 to E4_WNT_R13
E4_WY	Wa Yuen	R	E4_WY_R01
E4_YH	Yoho Town	R	E4_YH_R01 to E4_YH_R02
E4_YHM	Yoho Midtown	R	E4_YHM_R01
Planned NSRs			
P_E1.1	Planned primary school	E	P_E1.1_E01 to P_E1.1_E02
P_E2.1	Planned primary school	E	P_E2.1_E01 to P_E2.1_E04
P_E2.2	Planned primary school	E	P_E2.2_E01 to P_E2.2_E02
P_E2.3	Planned primary school	E	P_E2.3_E01 to P_E2.3_E03
P_E2.4	Planned primary school	E	P_E2.4_E01 to P_E2.4_E04
P_E3.1	Planned primary school	E	P_E3.1_E01 to P_E3.1_E03
P_E3.2	Planned primary school	E	P_E3.2_E01 to P_E3.2_E03

NSR[1],[2]	NSR Description	Use[3]	NAP
P_E3.3	Planned primary school	E	P_E3.3_E01 to P_E3.3_E03
P_G2.4	Married quarters	R	P_G2.4_R01
P_R1.1	Private housing	R	P_R1.1_R01 to P_R1.1_R04
P_R1.2	Private housing	R	P_R1.2_R01 to P_R1.2_R05
P_R1.3	Private housing	R	P_R1.3_R01 to P_R1.3_R09
P_R1.4	Private housing	R	P_R1.4_R01 to P_R1.4_R09
P_R1.5	Private housing	R	P_R1.5_R01 to P_R1.5_R03
P_R1.6	Private housing	R	P_R1.6_R01 to P_R1.6_R02
P_R1.7	Private housing	R	P_R1.7_R01 to P_R1.7_R08
P_R1.8	Private housing	R	P_R1.8_R01 to P_R1.8_R08
P_R1.9	Private housing	R	P_R1.9_R01 to P_R1.9_R03
P_R1.10	Private housing	R	P_R1.10_R01 to P_R1.10_R09
P_R1.11	Private housing	R	P_R1.11_R01 to P_R1.11_R03
P_R1.12	Private housing	R	P_R1.12_R01 to P_R1.12_R06
P_R1.13	Private housing	R	P_R1.13_R01 to P_R1.13_R04
P_R1.14	Private housing	R	P_R1.14_R01 to P_R1.14_R05
P_R1.15	Private housing	R	P_R1.15_R01 to P_R1.15_R02
P_R1.16	Private housing	R	P_R1.16_R01 to P_R1.16_R07
P_R2.1a	Private housing	R	P_R2.1a_R01 to P_R2.1a_R05
P_R2.1b	Public housing	R	P_R2.1b_R01 to P_R2.1b_R03
P_R2.2	Public housing	R	P_R2.2_R01 to P_R2.2_R20
P_R2.3	Public housing	R	P_R2.3_R01 to P_R2.3_R08
P_R2.4	Public housing	R	P_R2.4_R01 to P_R2.4_R14
P_R2.5	Public housing	R	P_R2.5_R01 to P_R2.5_R10
P_R2.6a	Private housing	R	P_R2.6a_R01 to P_R2.6a_R07
P_R2.6b	Private housing	R	P_R2.6b_R01 to P_R2.6b_R03
P_R2.6c	Private housing	R	P_R2.6c_R01 to P_R2.6c_R04
P_R2.6d	Private housing	R	P_R2.6d_R01 to P_R2.6d_R03

NSR[1],[2]	NSR Description	Use[3]	NAP
P_R2.7	Private housing	R	P_R2.7_R01 to P_R2.7_R04
P_R2.8	Public housing	R	P_R2.8_R01 to P_R2.8_R06
P_R3.1a	Private housing	R	P_R3.1a_R01 to P_R3.1a_R02
P_R3.1b	Private housing	R	P_R3.1b_R01 to P_R3.1b_R03
P_R3.1c	Private housing	R	P_R3.1c_R01 to P_R3.1c_R02
P_R3.1d	Private housing	R	P_R3.1d_R01
P_R3.1e	Private housing	R	P_R3.1e_R01 to P_R3.1e_R02
P_R3.2	Private housing	R	P_R3.2_R01 to P_R3.2_R04
P_R3.3	Private housing	R	P_R3.3_R01 to P_R3.3_R03
P_R3.4a	Private housing	R	P_R3.4a_R01 to P_R3.4a_R04
P_R3.4b	Private housing	R	P_R3.4b_R01 to P_R3.4b_R04
P_R3.5a	Private housing	R	P_R3.5a_R01 to P_R3.5a_R03
P_R3.5b	Private housing	R	P_R3.5b_R01 to P_R3.5b_R03
P_R3.6a	Private housing	R	P_R3.6a_R01 to P_R3.6a_R02
P_R3.6b	Private housing	R	P_R3.6b_R01 to P_R3.6b_R02
P_OU1.1	Private housing	R	P_OU1.1_R01 to P_OU1.1_R09
P_OU1.2a	Private housing	R	P_OU1.2a_R01 to P_OU1.2a_R10
P_OU1.3	Private housing	R	P_OU1.3_R01 to P_OU1.3_R05
P_OU2.3	Private housing	R	P_OU2.3_R01 to P_OU2.3_R04
P_OU3.2a	Private housing	R	P_OU3.2a_R01 to P_OU3.2a_R04
P_OU3.2b	Private housing	R	P_OU3.2b_R01 to P_OU3.2b_R02
P_VR1.1	Village rehousing	R	P_VR1.1_R01 to P_VR1.1_R02
P_VR2.1	Village rehousing	R	P_VR2.1_R01 to P_VR2.1_R02
P_VR3.1	Village rehousing	R	P_VR3.1_R01
P_LBIH[4]	Planned Public Housing Development in Long Bin	R	P_LBIH_R01 to P_LBIH_R02

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises, E – Educational Institutions; W – Places of Public Worship; C – Clinics, Residential Care Home for the Elderly (RCHE). The land uses of the planned NSRs within the PDA are

identified based on the RODP.
[4] Planned NSRs outside PDA. Since the layout of Long Bin development is yet to be developed, the location of NAPs have been assumed at 10m from TYST Interchange and 10m from Long Tin Road.
[5] E1_TYST_R31 to E1_TYST_R38 would be demolished before commencement of construction works of Stage 3.

5.4.4 Fixed Noise Sources

5.4.4.1 For operational fixed noise sources impact assessment, the assessment area should be: i) 300m assessment area for committed and planned NSRs affected by planned and existing fixed noise sources; ii) 300m assessment area for planned fixed noise sources. The existing and planned NSRs affected by existing and planned fixed noise sources in the vicinity of project have been selected from **Table 5.4.1**. The representative NSRs and NAPs at various representative floor levels for operational fixed noise sources impact assessment are shown in **Figure 5.5** and are summarised in **Table 5.4.4**. Detailed information about the determination of their fixed noise criteria are provided in **Appendix 5.7**.

Table 5.4.4 Representative NSRs and NAPs for fixed noise sources impact assessment

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP	ASR
Existing NSRs (Within PDA)					
Stage 2	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R01	B
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R07	B
	E2_MKTT	Muk Kiu Tau Tsuen	R	E2_MKTT_R01	A
	E2_TLY	Tin Lung Yuen	R	E2_TLY_R02	B
Stage 3	E1_ELD	Eldorado	R	E1_ELD_R02	A
	E1_ELD	Eldorado	R	E1_ELD_R03	A
	E2_MKTT	Muk Kiu Tau Tsuen	R	E2_MKTT_R01	A
Existing NSRs (Outside PDA)					
Stage 2	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R01	B
	E4_TLT	Tin Liu Tsuen	R	E4_TLT_R17	A
	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R11	A
	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R12	A
	E4_KMYLS	Kwong Ming Ying Loi School	E	E4_KMYLS_E02	C
	E4_TTT	Tai Tao Tsuen	R	E4_TTT_R01	B
	E4_FSW	Fui Sha Wai	R	E4_FSW_R02	B
Stage 3	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R11	A
	E4_WNT	Wong Nai Tun Tsuen	R	E4_WNT_R12	A
	E4_SHT	Shan Ha Tsuen	R	E4_SHT_R04	A

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP	ASR
	E4_CPG	Chung Pak Garden	R	E4_CPG_R01	A
	E4_PST	Pak Sha Tsuen	R	E4_PST_R18	A
	E4_LHWT	Lam Hau Wai Tsuen	R	E4_LHWT_R01	B
Planned NSRs					
Stage 1	P_R2.2	Public housing	R	P_R2.2_R02	C
	P_R2.2	Public housing	R	P_R2.2_R19	B
Stage 2	P_R2.4	Public housing	R	P_R2.4_R04	B
	P_R2.5	Public housing	R	P_R2.5_R07	B
	R_VR3.1	Village rehousing	R	P_VR3.1_R03	B
Stage 3	P_R2.6a	Private housing	R	P_R2.6a_R03	B
	P_R2.6d	Private housing	R	P_R2.6d_R02	B
	P_R3.5b	Private housing	R	P_R3.5b_R02	B
	P_E3.1	Planned primary school	E	P_E3.1_E01	B
	P_OU2.3	Private housing	R	P_OU2.3_R02	B
	P_OU3.2a	Private housing	R	P_OU3.2a_R01	B
	P_OU3.2b	Private housing	R	P_OU3.2b_R03	B
Stage 4	P_OU1.1	Private housing	R	P_OU1.1_R02	B
	P_OU1.1	Private housing	R	P_OU1.1_R10	B
	P_OU1.3	Private housing	R	P_OU1.3_R01	B
	P_R1.2	Private housing	R	P_R1.2_R05	B

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within i) 300m assessment area for committed and planned NSRs affected by planned and existing fixed noise sources; ii) 300m assessment area for planned fixed noise sources is included.
[3] R – Residential Premises, E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP.

5.4.5 Rail Noise

5.4.5.1 For operational rail noise assessment, the assessment area includes those within 300m from the PDA boundary and EFTS alignment. Representative NSRs have been selected from **Table 5.4.1**. The locations of representative NSRs and NAPs for operational rail noise assessment are shown in **Figure 5.6** and are summarised in **Table 5.4.5**.

Table 5.4.5 Representative NSRs and NAPs for rail noise assessment

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP
Existing NSRs (Within PDA)				
N/A	E1_PV	Park Villa	R	E1_PV_R02
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R07
Existing NSRs (Outside PDA)				
N/A	E4_LHT	Lam Hau Tsuen	R	E4_LHT_R06
Planned NSRs				
Stage 1	P_R2.2	Public Housing	R	P_R2.2_R10
Stage 2	P_R2.1b	Private Housing	R	P_R2.1b_R02
	P_R2.4	Public Housing	R	P_R2.4_R12
Stage 3	P_R2.6d	Private Housing	R	P_R2.6d_R02
Stage 4	P_OU1.2a	Private Housing	R	P_OU1.2a_R10

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within 300m from the PDA boundary is included.
[3] R – Residential Premises. The land uses of the planned NSRs within the PDA are identified based on the RODP.

5.5 Construction Noise Impact Assessment

5.5.1 Construction Noise Impact Assessment Methodology

5.5.1.1 Construction noise assessment has been conducted based on the following procedures:

- Determine 300m from the PDA boundary and the associated infrastructure works;
- Identify and locate representative NSRs in each stage that may be affected by the works;
- Obtain the construction method and work sequence for the construction period;
- Obtain the construction plant inventory for each corresponding construction work sequence;
- Determine the sound power levels (SWLs) of the plant items according to the information stated in the TM-GW or other recognised sources of reference, where appropriate;
- Calculate the correction factors based on the distance between the NSRs and the notional noise source positions of the work sites;

- Apply corrections for façade, distance, barrier attenuation, acoustic reflection where applicable;
- Predict construction noise levels at the NSRs;
- Quantify the level of impact at the NSRs, in accordance with TM-GW;
- Predict the cumulative noise impacts for any concurrent construction works in the vicinity of the proposed work;
- For any exceedance of noise criteria, all practical mitigation measures such as alternative construction methodology, quiet plant, silencer, enclosure, etc, shall be examined to alleviate the predicted noise impacts as much as practicable; and
- Consideration of noise mitigation measures will follow Annex 13 of EIAO-TM and EIA Ordinance Guidance Note "Preparation of Construction Noise Impact Assessment under the Environmental Impact Assessment Ordinance" [GN 9/2010].

Construction Noise Assessment Tool

5.5.1.2 An in-house program has been used for construction noise calculations. Initially, program runs were conducted without any mitigation measures (i.e. the "Unmitigated Scenario"). Where noise level exceedance was identified, further runs would be made assuming different combinations of mitigation measures to be incorporated (i.e. the "Mitigated Scenario").

Utilisation Rates and SWLs of Powered Mechanical Equipment

5.5.1.3 Practically, the PMEs will not be operating for all times within a work site. The utilisation rates would depend on the construction sequences, work fronts scale and construction nature. In this assessment, the utilisation rates have been reviewed by engineer and have been concluded to be practical for the purpose of this EIA. **Appendix 5.3** tabulates the adopted utilisation rates. The construction plant inventory has been developed together with the construction programme to achieve the completion dates as given in **Appendix 5.4** of the EIA Report. For typical construction activities, it is realistic that there are always some idling time for various construction equipment. Take dump truck as an example, it is part of the normal logistics that they would need to await for other construction activities before loading and unloading. Hence, it is reasonable to assume a 70% on-time for dump trucks. Similar principle applies for other construction equipment.

5.5.2 Prediction and Evaluation of Construction Noise Impacts

5.5.2.1 According to the latest engineering design, construction would mainly comprise activities including site formation and road works as described in **Section 5.3.1**. The corresponding SWLs of these activities have been estimated according to the PME's SWLs and the assessment

methodology in the TM-GW. **Appendix 5.2** presents the SWLs for each PME. **Appendix 5.3** gives the plant inventory adopted for each workfront and **Appendix 5.5** shows the locations of workfronts adopted for this construction noise assessment.

5.5.2.2 **Appendix 5.9** presents the calculated construction noise impacts at selected representative NSRs under unmitigated scenario with consideration of cumulative impacts from concurrent projects. The predicted construction noise impacts at the NSRs under unmitigated scenario are summarised in **Table 5.5.1**.

Table 5.5.1 Predicted construction noise impact at NSRs under unmitigated scenario

Stage	NSR[1]	NAP[2][3]	Uses[4]	$L_{eq\ (30mins)}$, dB(A)		
				Criterion[5]	Unmitigated Noise Level[6]	Exceedance
Existing NSRs (Within PDA)						
Stage 1	E1_SHT	E1_SHT_R01	R	75	75	-
	E1_SHT	E1_SHT_R04	R	75	89	14
	E2_TLY	E2_TLY_R01	R	75	89	14
Stage 2	E1_RLS	E1_RLS_R02	R	75	66	-
	E1_ELD	E1_ELD_R02	R	75	71	-
	E1_PV	E1_PV_R02	R	75	76	1
	E1_TYST	E1_TYST_R07	R	75	77	2
	E1_TYST	E1_TYST_R13	R	75	70	-
	E1_TYST	E1_TYST_R32	R	75	80	5
	E1_TYST	E1_TYST_R36	R	75	68	-
	E1_MG	E1_MG_R02	R	75	64	-
	E1_TYST	E1_TYST_R05	R	75	69	-
	E1_SHT	E1_SHT_R01	R	75	75	-
	E1_SHT	E1_SHT_R04	R	75	89	14
	E2_TLY	E2_TLY_R01	R	75	89	14
	E2_MKTT	E2_MKTT_R01	R	75	81	6
Stage 3	E2_MKTT	E2_MKTT_R01	R	75	76	1
Stage 4	E1_RLS	E1_RLS_R02	R	75	86	11
	E1_ELD	E1_ELD_R02	R	75	85	10
	E1_PV	E1_PV_R02	R	75	88	13
	E1_TYST	E1_TYST_R07	R	75	84	9
	E1_TYST	E1_TYST_R13	R	75	76	1
	E1_TYST	E1_TYST_R36	R	75	68	-
	E1_MG	E1_MG_R02	R	75	86	11
	E1_PH	E1_PH_R01	R	75	80	5
	E1_TYST	E1_TYST_R05	R	75	80	5
	E1_SHT	E1_SHT_R01	R	75	80	5
	E1_SHT	E1_SHT_R04	R	75	73	-
	E2_TLY	E2_TLY_R01	R	75	70	-
Existing NSRs (Outside PDA)						
Stage 1	E4_SHT	E4_SHT_R07	R	75	85	10
	E4_LTT	E4_LTT_R01	R	75	90	15
	E4_TLT	E4_TLT_R02	R	75	91	16

Stage	NSR[1]	NAP[2][3]	Uses[4]	$L_{eq\ (30mins)}$, dB(A)		
				Criterion[5]	Unmitigated Noise Level[6]	Exceedance
	E4_TLT	E4_TLT_R12	R	75	89	14
Stage 2	E4_TTT	E4_TTT_R01[8]	R	75	73	-
	E4_LTT	E4_LTT_R01	R	75	90	15
	E4_TLT	E4_TLT_R02	R	75	91	16
	E4_TLT	E4_TLT_R12	R	75	89	14
	E4_SHT	E4_SHT_R07	R	75	85	10
	E4_PST	E4_PST_R01	R	75	79	4
	E4_MKT	E4_MKT_R05	R	75	81	6
	E4_WNT	E4_WNT_R11	R	75	75	-
	E4_HTS	E4_HTS_R01[8]	R	75	93	18
	E4_HTS	E4_HTS_R02[8]	R	75	77	2
	E4_HTS	E4_HTS_R03[8]	R	75	89	14
	E4_SPT	E4_SPT_R01[8]	R	75	77	2
	E4_UT	E4_UT_R01[8]	R	75	78	3
	E4_LA	E4_LA_E01	E	70 (65)	70	- (5)
	E4_KMYLS	E4_KMYLS_E03	E	70 (65)	78	8 (13)
Stage 3	E4_PST	E4_PST_R01	R	75	79	4
	E4_WNT	E4_WNT_R11	R	75	75	-
Stage 4	E4_TTT	E4_TTT_R01	R	75	71	-
	E4_LA	E4_LA_E01	E	70 (65)	74	4 (9)
Planned NSRs						
Stage 1	P_R2.2	P_R2.2_R01	R	75	77	2
	P_R2.2	P_R2.2_R10	R	75	76	1
	P_R2.2	P_R2.2_R17	R	75	76	1
	P_VR1.1	P_VR1.1_R01	R	75	80	5
	P_VR2.1	P_VR2.1_R01	R	75	78	3
Stage 2	P_VR3.1	P_VR3.1_R02	R	75	81	6
	P_E2.1	P_E2.1_E01	E	70 (65)	70	- (5)
	P_R2.3	P_R2.3_R08	R	75	74	-
	P_R2.4	P_R2.4_R06	R	75	74	-
[7]	P_HSK	P_HSK_R01[8]	R	75	89	14
	P_HSK	P_HSK_R02[8]	R	75	77	2
	P_HSK	P_HSK_R03[8]	R	75	77	2
	P_LBIH	P_LBIH_R02	R	75	80	5

Note:

[1] The assessment only includes NSRs which rely on opened windows for ventilation.

[2] NAP – Noise Assessment Point.

[3] Representative NSRs within 300m from the PDA boundary and the associated infrastructure works is included.

[4] R – Residential Premises; E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP.

[5] Values in parentheses indicate the noise criterion during examination period (Jan, Feb, May and June) of education institution.

[6] Noise levels presented are rounded to the nearest dB(A).

[7] Planned NSRs outside PDA.

[8] Cumulative construction noise impacts from the concurrent project (i.e. HSK NDA) are considered for E4_TTT_R01, E4_HTS_R01, E4_HTS_R02, E4_HTS_R03, E4_SPT_R01, E4_UT_R01, P_HSK_R01, P_HSK_R02 and P_HSK_R03 in the assessment. For other NSRs located outside 300m from HSK NDA, cumulative construction noise impacts are not anticipated.

5.5.2.3 It can be seen from the above **Table 5.5.1** that without any mitigation measures, exceedances of noise criteria range from 1 to 18 dB(A) for existing residential premises and from 4 to 13 dB(A) for existing educational institutions, while the exceedances of noise criteria range from 2 to 14 dB(A) for planned residential premises and 5 dB(A) for planned educational institutions. Mitigation measures are therefore required.

5.5.2.4 The existing NSRs at Ha Tsuen Shi along Tin Ha Road, which include E4_HTS (i.e. E4_HTS_R01), are anticipated to be the worst affected existing NSRs by the construction works mainly due to short buffer distance between the NSR and utility works on Tin Ha Road (i.e works area P2u).

5.5.2.5 The existing NSRs along Kung Um Road and Kiu Hing Road, which include E4_TLT (i.e. E4_TLT_R02) and E4_LTT (i.e. E4_LTT_R01), are also anticipated to be the worst affected existing NSRs by the construction works mainly due to short buffer distance between the NSR and road works on Kung Um Road and Kiu Hing Road (i.e. works area P1b and P2f).

5.5.2.6 The NSR P_VR_1.1 (i.e. P_VR1.1_R01) is anticipated to be the worst affected planned NSR in Stage 1 mainly due to short buffer distance between the NSR and road works on works area P2d in Stage 2.

5.5.2.7 The NSR P_VR3.1 (i.e. P_VR3.1_R02) is anticipated to be the worst affected planned NSR in Stage 2 mainly due to short buffer distance between the NSR and site formation and slope works on works area P3d in Stage 3.

5.5.2.8 The NSR P_HSK (i.e. P_HSK_R01) is anticipated to be the worst affected planned NSR mainly due to short buffer distance between the NSR and utility works on works area P2u in Stage 2. Please note that the planned NSR P_HSK belongs to the HSK NDA project. Details about the NSR should refer to the approved EIA report for HSK NDA (AEIAR-203/2016).

5.5.3 Mitigation Measures

5.5.3.1 The predicted noise levels show that the unmitigated construction noise impacts would exceed the daytime noise criteria. Mitigation measures are therefore required. The following mitigation measures have been considered:

- Good site practices to limit noise emissions at the source;
- Use of quality powered mechanical equipment (QPME);

- Use of temporary noise barriers and noise enclosure to screen noise from relatively static PMEs; and
- Alternative use of plant items within one worksite, wherever practicable.

5.5.3.2 The above mitigation measures would need to be implemented in work sites as good practices where appropriate. Detailed descriptions of these mitigation measures are given in the following sections.

Good Site Management Practices

5.5.3.3 Good site practice and noise management techniques could considerably reduce the noise impact from construction site activities on nearby NSRs. The following measures should be practised during each phase of construction:

- only well-maintained plant should be operated on-site and plant should be serviced regularly during the construction programme;
- machines and plant (such as trucks, cranes) that may be in intermittent use should be shut down between work periods or should be throttled down to a minimum;
- plant known to emit noise strongly in one direction, where possible, be orientated so that the noise is directed away from nearby NSRs;
- silencers or mufflers on construction equipment should be properly fitted and maintained during the construction works;
- mobile plant should be sited as far away from NSRs as possible and practicable; and
- material stockpiles, site office and other structures should be effectively utilised, where practicable, to screen noise from on-site construction activities.

5.5.3.4 The benefits of these techniques can vary according to specific site conditions and operations. The environmental noise climate would certainly be improved with these control practices, although the improvement can only be quantified during implementation when specific site parameters are known.

Use of Quality Powered Mechanical Equipment

5.5.3.5 The use of quiet plant associated with the construction works is made reference to the PME listed in the QPME and other commonly used PME listed in EPD web pages as far as possible. It is generally known (supported by field measurement) that particular models of construction equipment are quieter than standard types given in the TM-GW. Whilst it is generally considered too restrictive to specify that the Contractor has to use specific models or items of plant, it is reasonable and practicable to set plant noise performance specifications for specific PME so that some flexibility in selection of plant is allowed. A pragmatic approach would be to request that the Contractor independently verifies the noise level of the plant proposed to be used

and demonstrates through furnishing of these results, that the plant proposed to be used on the site meets the requirements.

Use of Movable Noise Barrier and Full Enclosure for Relatively Fixed Plant Source

5.5.3.6 Movable temporary noise barriers that can be located close to noisy plant and be moved concurrently with the plant along a worksite can be very effective for screening noise from NSRs. A typical design which has been used locally is a wooden framed barrier with a small-cantilevered upper portion of superficial density no less than 7kg/m^2 on a skid footing with 25mm thick internal sound absorptive lining. This measure is particularly effective for low level zone of NSRs. A cantilevered top cover would be required to achieve screening benefits at upper floors of NSRs.

5.5.3.7 Movable temporary noise barriers will be used for some PME (e.g. excavator). It is anticipated that suitably designed barriers could achieve at least 5dB(A) reduction for movable plant and 10dB(A) for stationary plant.

5.5.3.8 The use of standard enclosure has been considered in this assessment to shelter relatively fixed plant including air compressor, generator. These standard enclosures can provide at least 15dB(A) noise reduction.

5.5.3.9 A summary of the movable temporary noise barrier and standard enclosure adopted for various PMEs is given in **Appendix 5.2** and indicative drawing for barrier and standard enclosure is shown in **Appendix 5.10**, and the associated noise reduction is summarised in **Table 5.5.2**.

Table 5.5.2 Summary of barrier and standard enclosure adopted for PMEs

PME	Enclosure / Barrier	Attenuation, dB(A)
Asphalt Paver	Barrier	-5
Bar Bender and Cutter (Electric)	Barrier	-10
Hand Held Breaker	Barrier	-10
Bulldozer	Barrier	-5
Concrete Lorry Mixer	Barrier	-5
Concrete pump	Enclosure	-15
Mobile Crane	Barrier	-5
Compactor, vibratory	Barrier	-5
Dump Truck	Barrier	-5
Drill Rig, rotary type (Diesel)	Barrier	-10
Excavator	Barrier	-5
Backhoe	Barrier	-5
Generator	Enclosure	-15
Grout Mixer	Enclosure	-15
Grout Pump	Enclosure	-15
Hoist, passenger/ material (Petrol)	Barrier	-5
Lorry	Barrier	-5

PME	Enclosure / Barrier	Attenuation, dB(A)
Piling, large dia bored, reverse circulation drill	Barrier	-5
Vibratory Poker	Barrier	-10
Road Roller	Barrier	-5
Saw, Circular Wood	Barrier	-10
Water Pump (Electric)	Barrier	-10

5.5.3.10 In addition to the above-listed movable temporary noise barrier and standard enclosure, retractable noise barrier will be erected along the worksites at Kung Um Road, Kiu Hing Road and Tin Ha Road in front of the NSRs to further mitigate the construction noise. It is anticipated that these retractable noise barriers can provide about 10dB(A) noise reduction for the PME used at the worksites. Detail information about these PME are shown in **Appendix 5.2**.

Alternative Use of Plant Items within One Worksite

5.5.3.11 In practice, some plant items will operate sequentially within the same work site, and certain reduction of the predicted noise impacts could be achieved. However, any additional control on the sequencing of plant will impose a restrictive constraint to the Contractor on the operation and planning of plant items, and the implementation of the requirement would be difficult to be monitored. Hence, sequencing operation of PME has not been taken into consideration.

Assessment Results of Construction Noise Under Mitigated Scenario

5.5.3.12 **Appendix 5.12** presents the calculated construction noise impacts at selected representative NSRs with the adoption of the above mitigation measures. The predicted construction noise impacts at the NSRs under mitigated scenario are summarised in **Table 5.5.3**.

Table 5.5.3 Predicted construction noise impact at NSRs under mitigated scenario

Stage	NSR[1]	NAP[2][3]	Uses[4]	$L_{eq\ (30mins)}$, dB(A)		
				Criterion[5]	Mitigated Noise Level[6]	Exceedance
Existing NSRs (Within PDA)						
Stage 1	E1_SHT	E1_SHT_R01	R	75	61	-
	E1_SHT	E1_SHT_R04	R	75	75	-
	E2_TLY	E2_TLY_R01	R	75	75	-
Stage 2	E1_RLS	E1_RLS_R02	R	75	52	-
	E1_ELD	E1_ELD_R02	R	75	58	-
	E1_PV	E1_PV_R02	R	75	63	-
	E1_TYST	E1_TYST_R07	R	75	63	-
	E1_TYST	E1_TYST_R13	R	75	57	-
	E1_TYST	E1_TYST_R32	R	75	67	-
	E1_TYST	E1_TYST_R36	R	75	55	-
	E1_MG	E1_MG_R02	R	75	49	-
	E1_TYST	E1_TYST_R05	R	75	56	-
	E1_SHT	E1_SHT_R01	R	75	61	-

Stage	NSR[1]	NAP[2][3]	Uses[4]	$L_{eq\ (30mins)}$, dB(A)		
				Criterion[5]	Mitigated Noise Level[6]	Exceedance
	E1_SHT	E1_SHT_R04	R	75	75	-
	E2_TLY	E2_TLY_R01	R	75	75	-
	E2_MKTT	E2_MKTT_R01	R	75	67	-
Stage 3	E2_MKTT	E2_MKTT_R01	R	75	63	-
Stage 4	E1_RLS	E1_RLS_R02	R	75	73	-
	E1_ELD	E1_ELD_R02	R	75	71	-
	E1_PV	E1_PV_R02	R	75	75	-
	E1_TYST	E1_TYST_R07	R	75	71	-
	E1_TYST	E1_TYST_R13	R	75	63	-
	E1_TYST	E1_TYST_R36	R	75	55	-
	E1_MG	E1_MG_R02	R	75	73	-
	E1_PH	E1_PH_R01	R	75	67	-
	E1_TYST	E1_TYST_R05	R	75	67	-
	E1_SHT	E1_SHT_R01	R	75	67	-
	E1_SHT	E1_SHT_R04	R	75	60	-
	E2_TLY	E2_TLY_R01	R	75	57	-
Existing NSRs (Outside PDA)						
Stage 1	E4_SHT	E4_SHT_R07	R	75	72	-
	E4_LTT	E4_LTT_R01	R	75	75	-
	E4_TLT	E4_TLT_R02	R	75	72	-
	E4_TLT	E4_TLT_R12	R	75	70	-
Stage 2	E4_TTT	E4_TTT_R01[8]	R	75	60	-
	E4_LTT	E4_LTT_R01	R	75	75	-
	E4_TLT	E4_TLT_R02	R	75	72	-
	E4_TLT	E4_TLT_R12	R	75	70	-
	E4_SHT	E4_SHT_R07	R	75	72	-
	E4_PST	E4_PST_R01	R	75	66	-
	E4_MKT	E4_MKT_R05	R	75	68	-
	E4_WNT	E4_WNT_R11	R	75	62	-
	E4_HTS	E4_HTS_R01[8]	R	75	74	-
	E4_HTS	E4_HTS_R02[8]	R	75	71	-
	E4_HTS	E4_HTS_R03[8]	R	75	76	1
	E4_SPT	E4_SPT_R01[8]	R	75	71	-
	E4_UT	E4_UT_R01[8]	R	75	61	-
	E4_LA	E4_LA_E01	E	70 (65)	57	-
	E4_KMYLS	E4_KMYLS_E03	E	70 (65)	63	-
Stage 3	E4_PST	E4_PST_R01	R	75	66	-
	E4_WNT	E4_WNT_R11	R	75	62	-
Stage 4	E4_TTT	E4_TTT_R01	R	75	58	-
	E4_LA	E4_LA_E01	E	70 (65)	60	-

Stage	NSR[1]	NAP[2][3]	Uses[4]	$L_{eq\ (30mins)}$, dB(A)		
				Criterion[5]	Mitigated Noise Level[6]	Exceedance
Planned NSRs						
Stage 1	P_R2.2	P_R2.2_R01	R	75	64	-
	P_R2.2	P_R2.2_R10	R	75	63	-
	P_R2.2	P_R2.2_R17	R	75	63	-
	P_VR1.1	P_VR1.1_R01	R	75	66	-
	P_VR2.1	P_VR2.1_R01	R	75	65	-
Stage 2	P_VR3.1	P_VR3.1_R02	R	75	68	-
	P_E2.1	P_E2.1_E01	E	70 (65)	57	-
	P_R2.3	P_R2.3_R08	R	75	60	-
	P_R2.4	P_R2.4_R06	R	75	61	-
[7]	P_HSK	P_HSK_R01[8]	R	75	75	-
	P_HSK	P_HSK_R02[8]	R	75	74	-
	P_HSK	P_HSK_R03[8]	R	75	67	-
	P_LBIH	P_LBIH_R02	R	75	67	-

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] NAP – Noise Assessment Point.
[3] Representative NSRs within 300m from the PDA boundary and the associated infrastructure works is included.
[4] R – Residential Premises; E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP.
[5] Values in parentheses indicate the noise criterion during examination period (Jan, Feb, May and June) of education institution.
[6] Noise levels presented are rounded to the nearest dB(A).
[7] Planned NSRs outside PDA.
[8] Cumulative construction noise impacts from the concurrent project (i.e. HSK NDA) are considered for E4_TTT_R01, E4_HTS_R01, E4_HTS_R02, E4_HTS_R03, E4_SPT_R01, E4_UT_R01, P_HSK_R01, P_HSK_R02 and P_HSK_R03 in the assessment. For other NSRs located outside 300m from HSK NDA, cumulative construction noise impacts are not anticipated.

5.5.3.13 It can be seen from the above **Table 5.5.3** that construction noise impact arising from the project at all existing and planned NSRs, except for E4_HTS_R03 (Ha Tsuen Shi), will be mitigated after implementation of the recommended noise mitigation measures. It is anticipated that the noise level at this NSR at Ha Tsuen Shi will exceed the criterion by 1dB(A) for 10 months from August 2032 to May 2033 due to works of the HSK NDA development. According to the approved EIA report for HSK NDA (AEIAR-203/2016), mitigation measures such as use of quiet plant, temporary noise barrier, etc. are recommended to reduce the construction noise impacts.

5.5.4 Assessment of Side Effects and Constraints

5.5.4.1 As discussed in **Section 10.12**, measures such as provision of temporary landscape and screen hoarding at construction sites as well as controlling construction light are proposed to alleviate the induced impacts from the recommended construction noise mitigation measures.

5.5.4.2 As the construction of the Project may involve different parties, it is proposed to set up a liaison group among relevant government departments, contractors of the Works contracts, etc. during construction phase of the Project so as to ensure the proper implementation of the proposed noise mitigation measures.

5.5.5 Evaluation of Constraints on Planned Noise Sensitive Developments / Land Uses

5.5.5.1 With all practicable direct technical remedies in place, no additional measures within the planned noise sensitive uses would be required.

5.5.6 Evaluation of Residual of Construction Noise Impacts

5.5.6.1 Construction noise impact arising from the Project can be properly mitigated by implementing the proposed noise mitigation measures. Adverse residual noise impacts are thus not anticipated.

5.6 Road Traffic Noise Impact Assessment

5.6.1 Road Traffic Noise Impact Assessment Methodology

5.6.1.1 Road traffic noise calculation is based on the methods described in the U.K. Department of Transport's "Calculation of Road Traffic Noise (CRTN)". The predicted noise levels at the NSRs include a 2.5dB(A) façade reflection and correction factors of effects due to gradient, distance, view angle, road surface and barriers.

5.6.1.2 In preparation for noise prediction, the project roads and existing roads within the assessment area have been included in the model with parameters of road width and surface type. Computer-plots of the road traffic noise model with the extent of Project roads are shown in **Appendix 5.13**.

5.6.1.3 The computer programme, RoadNoise 2000, has been used to model road traffic noise from road networks. It complies with the Calculation of Road Traffic Noise (CRTN) developed by the UK Department of Transport. The road traffic noise will be presented in terms of noise levels exceeded for 10% of the one-hour period during peak traffic flow [i.e. $L_{10\ (1hr)}$ dB(A)].

5.6.1.4 According to the Revised Technical Feasibility Study (TFS) for Retrofitting of Noise Barriers on Long Tin Road, the tentative construction commencement and project completion dates for the retrofitting project are September 2019 and December 2022 respectively. As such, the retrofitting noise barriers at Long Tin Road has been included in the road traffic noise model.

5.6.1.5 As discussed in **Section 2.5**, based on the current tentative implementation programme, the proposed Project would be implemented in four stages and the expected operation years would be

in Years 2027, Year 2033 and Year 2038. Thus the assessment years for road traffic noise are taken as Year 2027, Year 2033 and Year 2053 (which is maximum traffic projection within 15 years upon operation of road networks or full occupation of planned NSRs). The traffic projection will take into account the induced traffic due to the operation of other planned projects including:

- HSK NDA Planning and Engineering Study;
- Engineering Works at Lin Cheung Road Site, Sham Shui Po and Wang Chau, Yuen Long;
- Site Formation and Infrastructural Works for the Development at Long Bin, Yuen Long, Feasibility Study;
- Site Formation and Infrastructural Works for the Development near Tan Kwai Tsuen, Yuen Long – Feasibility Study;
- Improvement to Pok Oi Interchange; and
- Property Development of Kam Sheung Road Station and its Adjoining Areas.

5.6.1.6 Hence, the following stages have been conducted:

- Stage 1 – Expected operation year of the proposed Project at Year 2027;
- Stage 2 – Expected operation year of the proposed Project at Year 2033; and
- Stage 3 and 4 - Expected operation year of the proposed Project at Year 2038 (Year 2053 for the maximum traffic projection within 15 year is choosen for road traffic noise assessment).

5.6.1.7 **Figure 5.2** shows the 300m from the boundary of the planned road works and the committed and planned NSRs of the Project for noise assessment.

5.6.1.8 Where the predicted noise impacts exceed the noise criteria, direct mitigation measures shall be considered to reduce the noise from the project road to a level that it (According to EIA Ordinance Guidance Note No. 12/2010):

- Is not higher than the standard; and
- Has no significant contribution to the overall noise from other existing roads, if the cumulative noise level, i.e. noise from the project road together with other existing roads exceeds the standard (i.e. less than 1.0 dB(A)).

5.6.1.9 In addition, a review for NSRs along existing roads will be conducted to assess whether the traffic noise impact due to Project is significant. Implementation of mitigation measures will be considered if necessary on those existing roads under this Project to alleviate the road traffic noise impact.

5.6.1.10 Details of traffic forecast for "With Project" and prevailing scenarios are given in **Appendix 5.14**. Transport Department (TD) has endorsed the traffic forecast as shown in **Appendix 5.14a**.

5.6.2 Prediction and Evaluation of Road Traffic Noise Impact

Scenarios

5.6.2.1 The road traffic noise assessment includes following scenarios:

- Prevailing scenario at assessment year 2019 (**Appendix 5.15**);
- Unmitigated scenario at assessment year 2027 for Stage 1 of the proposed Project (**Appendix 5.15a**);
- Mitigated scenario at assessment year 2027 for Stage 1 of the proposed Project (**Appendix 5.16a**);
- Unmitigated scenario at assessment year 2033 for Stage 2 of the proposed Project (**Appendix 5.15b**);
- Mitigated scenario at assessment year 2033 for Stage 2 of the proposed Project (**Appendix 5.16b**);
- Unmitigated scenario at assessment year 2053 for Stages 3 and 4 of the proposed Project (Maximum traffic project within 15 year after the full operation of the proposed Project at year 2038) (**Appendix 5.15c**); and
- Mitigated scenario at assessment year 2053 for Stages 3 and 4 of the proposed Project (Maximum traffic project within 15 year after the full operation of the proposed Project at year 2038) (**Appendix 5.16c**).

Prediction of Noise Impact

5.6.2.2 The predicted road traffic noise levels at each representative NSR and NAP under unmitigated scenario are presented in **Table 5.6.1**. **Appendix 5.15** shows the details of the noise impacts of the NSRs and NAPs at various floor levels in different stages.

Table 5.6.1a Summary of the predicted road traffic noise impact at existing and planned NSRs under unmitigated scenario (Year 2027)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
Existing NSR (within PDA)						
E1_CLMY	E1_CLMY_T01	Chuk Lam Ming Yuen Monastery	W	65	55 - 57	Y
E1_ELD	E1_ELD_R01	Eldorado	R	70	65	Y
E1_FLG	E1_FLG_R01	Fu Lai Garden	R	70	61 - 67	Y
E1_GR	E1_GR_R01	Greenville Residence	R	70	64 - 65	Y
E1_KLG	E1_KLG_R01	Kam Lai Garden	R	70	62 - 67	Y
E1_KLK	E1_KLK_T01	Kam Lan Koon	W	65	49	Y
E1_KV	E1_KV_R01 to E1_KV_R02	Kisland Villa	R	70	63 - 65	Y
E1_LHT	E1_LHT_R01	Lam Hau Tsuen	R	70	68 - 69	Y
E1_MG	E1_MG_R01	Marbella Gardens	R	70	63 - 64	Y
E1_PH	E1_PH_R01	The Parkhill	R	70	67	Y
E1_PV	E1_PV_R01 to E1_PV_R05	Park Villa	R	70	46 - 71	N
E1_RLS	E1_RLS_R01 to E1_RLS_R04	Recours La Serre	R	70	45 - 64	Y
E1_SHT	E1_SHT_R01 to E1_SHT_R07	Shan Ha Tsuen	R	70	54 - 64	Y
E1_TYST	E1_TYST_R01 to E1_TYST_ R14, E1_TYST_R20 to E1_TYST_ R23, E1_TYST_R31 to E1_TYST_R38	Tong Yan San Tsuen	R	70	46 - 71	N
E1_WG	E1_WG_R01	Winsor Garden	R	70	65	Y
E1_YHT	E1_YHT_T01	Yeung Hau Temple	W	65	48	Y
E2_MKTT	E2_MKTT_R01	Muk Kiu Tau Tsuen	R	70	71	N
E2_TLT	E2_TLT_R01	Tin Lung Tsuen	R	70	58	Y
E2_PST	E2_PST_R01	Pak Sha Tsuen	R	70	44	Y
E2_TLY	E2_TLY_R01 to E2_TLY_R02	Tin Lung Yuen	R	70	68 - 70	Y
Existing NSR (outside PDA)						
E4_CFG	E4_CFG_R01 to E4_CFG_R02	Chun Fai Garden	R	70	67 - 71	N
E4_CPG	E4_CPG_R01	Chung Pak Garden	R	70	69	Y
E4_FSW	E4_FSW_R01 to E4_FSW_R02	Fui Sha Wai	R	70	63 - 67	Y

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_FWM	E4_FWM_R01	Fu Wa Mansion	R	70	57	Y
E4_GM	E4_GM_R01	Garden Maple	R	70	71 - 76	N
E4_HCG	E4_HCG_R01 to E4_HCG_R02	Hang Cheong Garden	R	70	61 - 62	Y
E4_HG	E4_HG_R01	Hoover Garden	R	70	75	N
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	70 - 75	N
E4_JC	E4_JC_R01	Jasper Court	R	70	69 - 74	N
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	70 - 71	N
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	75	N
E4_LA	E4_LA_E01	Lutheran Academy	E	65	66 - 72	N
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	73	N
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	71 - 78	N
E4_LHT	E4_LHT_R01 to E4_LHT_R05	Lam Hau Tsuen	R	70	63 - 68	Y
E4_LHWT	E4_LHWT_R01 to E4_LHWT_R04	Lam Hau Wai Tsuen	R	70	63 - 66	Y
E4_LTG	E4_LTG_R01	Lung Teng Garden	R	70	61	Y
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	46 - 78	N
E4_MKT	E4_MKT_R01 to E4_MKT_R05	Muk Kiu Tau Tsuen	R	70	59 - 70	Y
E4_OHP	E4_OHP_R01 to E4_OHP_R02	One Hyde Park	R	70	73 - 74	N
E4_PCG	E4_PCG_R01 to E4_PCG_R02	Pui Chak Garden	R	70	68 - 70	Y
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	60 - 75	N
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	70 - 74	N
E4_PST	E4_PST_R01 to E4_PST_R20	Pak Sha Tsuen	R	70	59 - 73	N
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	68 - 75	N
E4_RP	E4_RP_R01 to E4_RP_R02	Regent's Park	R	70	59 - 72	N
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	67 - 72	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	63 - 73	N
E4_SG	E4_SG_ R01 to E4_SG_ R02	Scenic Gardens	R	70	61 – 75	N
E4_SHT	E4_SHT_R01 to E4_SHT_R07	Shan Ha Tsuen	R	70	32 - 52	Y
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	76 - 78	N
E4_STG	E4_STG_R01	Sun Tong Garden	R	70	66	Y
E4_SV	E4_SV_R01	Sereno Verde	R	70	68 - 73	N
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	78	N
E4_TKL	E4_TKL_R01	Tai Kei Leng	R	70	67	Y
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	70 - 78	N
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	66 - 73	N
E4_TLT	E4_TLT_R01 to E4_TKT_R18	Tin Liu Tsuen	R	70	58 - 78	N
E4_TR	E4_TR_R01	The Reach	R	70	72 - 75	N
E4_TTT	E4_TTT_R01 to E4_TTT_R03	Tai Tao Tsuen	R	70	67 - 69	Y
E4_TYST	E4_TYST_R01	Tong Yan San Tsuen	R	70	64 - 66	Y
E4_UT	E4_UT_ R01 to E4_UT_ R06	Uptown	R	70	74 - 80	N
E4_WNT	E4_WNT_R01 to E4_WNT_R13	Wong Nai Tun Tsuen	R	70	36 - 63	Y
E4_WY	E4_WY_R01	Wa Yuen	R	70	61 - 62	Y
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	71 - 74	N
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	75 - 77	N
Planned NSRs						
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	52 - 74	N
P_VR1.1	P_VR1.1_R01 to P_VR1.1_R02	Village rehousing	R	70	59 - 60	Y
P_VR2.1	P_VR2.1_R01 to P_VR2.1_R02	Village rehousing	R	70	52 - 53	Y
P_LBIH	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	75 - 81	N

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed/ planned NSRs of the Project is included.

[3] R – Residential Premises; E – Educational Institutions; W – Place of Worship. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).

Table 5.6.1b Summary of the predicted road traffic noise impact at existing and planned NSRs under unmitigated scenario (Year 2033)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
Existing NSR (within PDA)						
E1_CLMY	E1_CLMY_T01	Chuk Lam Ming Yuen Monastery	W	65	52 - 54	Y
E1_ELD	E1_ELD_R01	Eldorado	R	70	74 - 75	N
E1_FLG	E1_FLG_R01	Fu Lai Garden	R	70	67 - 73	N
E1_GR	E1_GR_R01	Greenville Residence	R	70	74 - 75	N
E1_KLG	E1_KLG_R01	Kam Lai Garden	R	70	69 - 73	N
E1_KLK	E1_KLK_T01	Kam Lan Koon	W	65	50	Y
E1_KV	E1_KV_R01 to E1_KV_R02	Kisland Villa	R	70	65 - 75	N
E1_LHT	E1_LHT_R01	Lam Hau Tsuen	R	70	67 - 68	Y
E1_MG	E1_MG_R01	Marbella Gardens	R	70	74 - 76	N
E1_PH	E1_PH_R01	The Parkhill	R	70	72 - 73	N
E1_PV	E1_PV_R01 to E1_PV_R05	Park Villa	R	70	47 - 73	N
E1_RLS	E1_RLS_R01 to E1_RLS_R04	Recours La Serre	R	70	50 - 67	Y
E1_SHT	E1_SHT_R01 to E1_SHT_R07	Shan Ha Tsuen	R	70	63 - 71	N
E1_TYST	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23, E1_TYST_R31 to E1_TYST_R38	Tong Yan San Tsuen	R	70	48 - 76	N
E1_WG	E1_WG_R01	Winsor Garden	R	70	74 - 75	N
E1_YHT	E1_YHT_T01	Yeung Hau Temple	W	65	55	Y
E2_MKTT	E2_MKTT_R01	Muk Kiu Tau Tsuen	R	70	69	Y
E2_TLT	E2_TLT_R01	Tin Lung Tsuen	R	70	62	Y
E2_TLY	E2_TLY_R01 to E2_TLY_R02	Tin Lung Yuen	R	70	72 - 73	N
Existing NSR (outside PDA)						
E4_CFG	E4_CFG_R01 to E4_CFG_R02	Chun Fai Garden	R	70	65 - 70	Y
E4_CPG	E4_CPG_R01	Chung Pak Garden	R	70	67	Y

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_FSW	E4_FSW_R01 to E4_FSW_R02	Fui Sha Wai	R	70	64 - 68	Y
E4_FWM	E4_FWM_R01	Fu Wa Mansion	R	70	55	Y
E4_GM	E4_GM_R01	Garden Maple	R	70	72 - 76	N
E4_HCG	E4_HCG_R01 to E4_HCG_R02	Hang Cheong Garden	R	70	66	Y
E4_HG	E4_HG_R01	Hoover Garden	R	70	74 - 75	N
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	69 - 73	N
E4_JC	E4_JC_R01	Jasper Court	R	70	74 - 76	N
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	69 - 70	N
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	75	N
E4_LA	E4_LA_E01	Lutheran Academy	E	65	65 - 72	N
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	73	N
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	72 - 78	N
E4_LHT	E4_LHT_R01 to E4_LHT_R05	Lam Hau Tsuen	R	70	64 - 68	Y
E4_LHWT	E4_LHWT_R01 to E4_LHWT_R04	Lam Hau Wai Tsuen	R	70	63 - 67	Y
E4_LTG	E4_LTG_R01	Lung Teng Garden	R	70	67	Y
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	48 - 79	N
E4_MKT	E4_MKT_R01 to E4_MKT_R05	Muk Kiu Tau Tsuen	R	70	62 - 67	Y
E4_OHP	E4_OHP_R01 to E4_OHP_R02	One Hyde Park	R	70	68	Y
E4_PCG	E4_PCG_R01 to E4_PCG_R02	Pui Chak Garden	R	70	64 - 66	Y
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	60 - 74	N
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	72 - 75	N
E4_PST	E4_PST_R01 to E4_PST_R20	Pak Sha Tsuen	R	70	62 - 69	Y
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	65 - 73	N
E4_RP	E4_RP_R01 to E4_RP_R02	Regent's Park	R	70	54 - 68	Y

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	68 - 72	N
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	65 - 74	N
E4_SG	E4_SG_R01 to E4_SG_R02	Scenic Gardens	R	70	54 - 73	N
E4_SHT	E4_SHT_R01 to E4_SHT_R07	Shan Ha Tsuen	R	70	55 - 71	N
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	71 - 73	N
E4_STG	E4_STG_R01	Sun Tong Garden	R	70	66 - 67	Y
E4_SV	E4_SV_R01	Sereno Verde	R	70	68 - 73	N
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	77	N
E4_TKL	E4_TKL_R01	Tai Kei Leng	R	70	68	Y
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	70 - 78	N
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	66 - 73	N
E4_TLT	E4_TLT_R01 to E4_TLT_R18	Tin Liu Tsuen	R	70	59 - 74	N
E4_TR	E4_TR_R01	The Reach	R	70	69 - 75	N
E4_TTT	E4_TTT_R01 to R03	Tai Tao Tsuen	R	70	65 - 68	Y
E4_TYST	E4_TYST_R01	Tong Yan San Tsuen	R	70	66 - 68	Y
E4_UT	E4_UT_R01 to E4_UT_R06	Uptown	R	70	73 - 79	N
E4_WNT	E4_WNT_R01 to E4_WNT_R13	Wong Nai Tun Tsuen	R	70	38 - 67	Y
E4_WY	E4_WY_R01	Wa Yuen	R	70	66 - 67	Y
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	70 - 74	N
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	76 - 77	N
Planned NSRs						
P_E2.1	P_E2.1_E01 to P_E2.1_E04	Planned primary school	E	65	60 - 67	N
P_E2.2	P_E2.2_E01 to P_E2.2_E02	Planned primary school	E	65	55 - 60	Y
P_E2.3	P_E2.3_E01 to P_E2.3_E03	Planned primary school	E	65	64 - 66	N
P_E2.4	P_E2.4_E01 to P_E2.4_E04	Planned primary school	E	65	59 - 63	Y
P_R2.1a	P_R2.1a_R01 to P_R2.1a_R05	Private housing	R	70	54 - 72	N
P_R2.1b	P_R2.1b_R01 to P_R2.1b_R03	Public housing	R	70	57 - 73	N
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	55 - 75	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
P_R2.3	P_R2.3_R01 to P_R2.3_R08	Public housing	R	70	62 - 73	N
P_R2.4	P_R2.4_R01 to P_R2.4_R14	Public housing	R	70	62 - 72	N
P_R2.5	P_R2.5_R01 to P_R2.5_R10	Public housing	R	70	62 - 70	Y
P_VR1.1	P_VR1.1_R01 to P_VR1.1_R02	Village rehousing	R	70	69 - 71	N
P_VR2.1	P_VR2.1_R01 to P_VR2.1_R02	Village rehousing	R	70	71 - 72	N
P_VR3.1	P_VR3.1_R01	Village rehousing	R	70	54	Y
P_LBIH	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	77 - 80	N

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions; W – Place of Worship. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).

Table 5.6.1c Summary of the predicted road traffic noise impact at existing and planned NSRs under unmitigated scenario (Year 2053)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
Existing NSR (within PDA)						
E1_CLMY	E1_CLMY_T01	Chuk Lam Ming Yuen Monastery	W	65	60 - 61	Y
E1_ELD	E1_ELD_R01	Eldorado	R	70	75	N
E1_FLG	E1_FLG_R01	Fu Lai Garden	R	70	69 - 74	N
E1_GR	E1_GR_R01	Greenville Residence	R	70	69 - 70	Y
E1_KLG	E1_KLG_R01	Kam Lai Garden	R	70	69 - 76	N
E1_KLK	E1_KLK_T01	Kam Lan Koon	W	65	63	Y
E1_KV	E1_KV_R01 to E1_KV_R02	Kisland Villa	R	70	72 - 73	N
E1_LHT	E1_LHT_R01	Lam Hau Tsuen	R	70	73	N
E1_MG	E1_MG_R01	Marbella Gardens	R	70	69 - 70	Y
E1_PH	E1_PH_R01	The Parkhill	R	70	70 - 71	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E1_PV	E1_PV_R01 to E1_PV_R05	Park Villa	R	70	53 - 76	N
E1_RLS	E1_RLS_R01 to E1_RLS_R04	Recours La Serre	R	70	63 - 72	N
E1_SHT	E1_SHT_R01 to E1_SHT_R07	Shan Ha Tsuen	R	70	66 - 76	N
E1_TYST	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23	Tong Yan San Tsuen	R	70	62 - 77	N
E1_WG	E1_WG_R01	Winsor Garden	R	70	71 - 72	N
E1_YHT	E1_YHT_T01	Yeung Hau Temple	W	65	55	Y
E2_MKTT	E2_MKTT_R01	Muk Kiu Tau Tsuen	R	70	73	N
E2_TLT	E2_TLT_R01	Tin Lung Tsuen	R	70	65	Y
E2_TLY	E2_TLY_R01 to E2_TLY_R02	Tin Lung Yuen	R	70	73 - 74	N
Existing NSR (outside PDA)						
E4_CFG	E4_CFG_R01 to E4_CFG_R02	Chun Fai Garden	R	70	68 - 72	N
E4_CPG	E4_CPG_R01	Chung Pak Garden	R	70	70 - 71	N
E4_FSW	E4_FSW_R01 to E4_FSW_R02	Fui Sha Wai	R	70	65 - 69	Y
E4_FWM	E4_FWM_R01	Fu Wa Mansion	R	70	55 - 56	Y
E4_GM	E4_GM_R01	Garden Maple	R	70	72 - 76	N
E4_HCG	E4_HCG_R01 to E4_HCG_R02	Hang Cheong Garden	R	70	66	Y
E4_HG	E4_HG_R01	Hoover Garden	R	70	75 - 76	N
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	70- 75	N
E4_JC	E4_JC_R01	Jasper Court	R	70	76 - 78	N
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	70 - 71	N
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	76	N
E4_LA	E4_LA_E01	Lutheran Academy	E	65	65 - 73	N
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	74	N
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	72 - 79	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_LHT	E4_LHT_R01 to E4_LHT_R05	Lam Hau Tsuen	R	70	65 - 73	N
E4_LHWT	E4_LHWT_R01 to E4_LHWT_R04	Lam Hau Wai Tsuen	R	70	64 - 67	Y
E4_LTG	E4_LTG_R01	Lung Teng Garden	R	70	67	Y
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	49 - 79	N
E4_MKT	E4_MKT_R01 to E4_MKT_R05	Muk Kiu Tau Tsuen	R	70	63 - 71	N
E4_OHP	E4_OHP_R01 to E4_OHP_R02	One Hyde Park	R	70	71	N
E4_PCG	E4_PCG_R01 to E4_PCG_R02	Pui Chak Garden	R	70	67 - 69	Y
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	62 - 76	N
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	72 - 75	N
E4_PST	E4_PST_R01 to E4_PST_R20	Pak Sha Tsuen	R	70	57 - 75	N
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	69 - 75	N
E4_RP	E4_RP_R01 to E4_RP_R02	Regent's Park	R	70	57 - 71	N
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	69 - 72	N
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	65 - 75	N
E4_SG	E4_SG_R01 to E4_SG_R02	Scenic Gardens	R	70	61 - 75	N
E4_SHT	E4_SHT_R01 to E4_SHT_R07	Shan Ha Tsuen	R	70	57 - 76	N
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	73 - 75	N
E4_STG	E4_STG_R01	Sun Tong Garden	R	70	69	Y
E4_SV	E4_SV_R01	Sereno Verde	R	70	69 - 74	N
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	78	N
E4_TKL	E4_TKL_R01	Tai Kei Leng	R	70	68	Y
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	72 - 80	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	67 - 74	N
E4_TLT	E4_TLT_R01 to E4_TLT_R18	Tin Liu Tsuen	R	70	61 - 77	N
E4_TR	E4_TR_R01	The Reach	R	70	70 - 76	N
E4_TTT	E4_TTT_R01 to R03	Tai Tao Tsuen	R	70	66 - 71	N
E4_TYST	E4_TYST_R01	Tong Yan San Tsuen	R	70	68 - 69	Y
E4_UT	E4_UT_R01 to E4_UT_R06	Uptown	R	70	74 - 81	N
E4_WNT	E4_WNT_R01 to E4_WNT_R13	Wong Nai Tun Tsuen	R	70	52 - 67	Y
E4_WY	E4_WY_R01	Wa Yuen	R	70	67	Y
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	71 - 75	N
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	76 - 78	N
Planned NSRs						
P_E1.1	P_E1.1_E01 to P_E1.1_E02	Planned primary school	E	65	60 - 62	Y
P_E2.1	P_E2.1_E01 to P_E2.1_E04	Planned primary school	E	65	62 - 70	N
P_E2.2	P_E2.2_E01 to P_E2.2_E02	Planned primary school	E	65	56 - 63	Y
P_E2.3	P_E2.3_E01 to P_E2.3_E03	Planned primary school	E	65	66 - 69	N
P_E2.4	P_E2.4_E01 to P_E2.4_E04	Planned primary school	E	65	62 - 66	N
P_E3.1	P_E3.1_E01 to P_E3.1_E03	Planned primary school	E	65	64 - 68	N
P_E3.2	P_E3.2_E01 to P_E3.2_E03	Planned primary school	E	65	62 - 66	N
P_E3.3	P_E3.3_E01 to P_E3.3_E03	Planned primary school	E	65	67 - 73	N
P_G2.4	P_G2.4_R01	Married quarters	R	70	71 - 74	N
P_R1.1	P_R1.1_R01 to P_R1.1_R04	Private housing	R	70	66 - 69	Y
P_R1.2	P_R1.2_R01 to P_R1.2_R05	Private housing	R	70	62 - 66	Y
P_R1.3	P_R1.3_R01 to P_R1.3_R09	Private housing	R	70	67 - 73	N
P_R1.4	P_R1.4_R01 to P_R1.4_R09	Private housing	R	70	67 - 71	N
P_R1.5	P_R1.5_R01 to P_R1.5_R03	Private housing	R	70	70 - 73	N
P_R1.6	P_R1.6_R01 to P_R1.6_R02	Private housing	R	70	71 - 73	N
P_R1.7	P_R1.7_R01 to P_R1.7_R08	Private housing	R	70	67 - 73	N
P_R1.8	P_R1.8_R01 to P_R1.8_R08	Private housing	R	70	70 - 74	N
P_R1.9	P_R1.9_R01 to P_R1.9_R03	Private housing	R	70	58 - 64	Y
P_R1.10	P_R1.10_R01 to P_R1.10_R09	Private housing	R	70	63 - 77	N
P_R1.11	P_R1.11_R01 to P_R1.11_R03	Private housing	R	70	64 - 70	Y

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
P_R1.12	P_R1.12_R01 to P_R1.12_R06	Private housing	R	70	69 - 73	N
P_R1.13	P_R1.13_R01 to P_R1.13_R04	Private housing	R	70	62 - 73	N
P_R1.14	P_R1.14_R01 to P_R1.14_R05	Private housing	R	70	71 - 74	N
P_R1.15	P_R1.15_R01 to P_R1.15_R02	Private housing	R	70	65 - 66	Y
P_R1.16	P_R1.16_R01 to P_R1.16_R07	Private housing	R	70	72 - 75	N
P_R2.1a	P_R2.1a_R01 to P_R2.1a_R05	Private housing	R	70	60 - 73	N
P_R2.1b	P_R2.1b_R01 to P_R2.1b_R03	Public housing	R	70	64 - 73	N
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	63 - 76	N
P_R2.3	P_R2.3_R01 to P_R2.3_R08	Public housing	R	70	67 - 74	N
P_R2.4	P_R2.4_R01 to P_R2.4_R14	Public housing	R	70	66 - 73	N
P_R2.5	P_R2.5_R01 to P_R2.5_R10	Public housing	R	70	64 - 69	Y
P_R2.6a	P_R2.6a_R01 to P_R2.6a_R07	Private housing	R	70	64 - 73	N
P_R2.6b	P_R2.6b_R01 to P_R2.6b_R03	Private housing	R	70	63 - 66	Y
P_R2.6c	P_R2.6c_R01 to P_R2.6c_R04	Private housing	R	70	49 - 73	N
P_R2.6d	P_R2.6d_R01 to P_R2.6d_R03	Private housing	R	70	55 - 74	N
P_R2.7	P_R2.7_R01 to P_R2.7_R04	Private housing	R	70	66 - 74	N
P_R2.8	P_R2.8_R01 to P_R2.8_R06	Public housing	R	70	67 - 75	N
P_R3.1a	P_R3.1a_R01 to P_R3.1a_R02	Private housing	R	70	65 - 70	Y
P_R3.1b	P_R3.1b_R01 to P_R3.1b_R03	Private housing	R	70	69 - 76	N
P_R3.1c	P_R3.1c_R01 to P_R3.1c_R02	Private housing	R	70	67 - 73	N
P_R3.1d	P_R3.1d_R01	Private housing	R	70	63 - 65	Y
P_R3.1e	P_R3.1e_R01 to P_R3.1e_R02	Private housing	R	70	69 - 72	N
P_R3.2	P_R3.2_R01 to P_R3.2_R04	Private housing	R	70	67 - 71	N
P_R3.3	P_R3.3_R01 to P_R3.3_R03	Private housing	R	70	64 - 68	Y
P_R3.4a	P_R3.4a_R01 to P_R3.4a_R04	Private housing	R	70	65 - 74	N
P_R3.4b	P_R3.4b_R01 to P_R3.4b_R04	Private housing	R	70	65 - 72	N
P_R3.5a	P_R3.5a_R01 to P_R3.5a_R03	Private housing	R	70	69 - 75	N
P_R3.5b	P_R3.5b_R01 to P_R3.5b_R03	Private housing	R	70	52 - 68	Y
P_R3.6a	P_R3.6a_R01 to P_R3.6a_R02	Private housing	R	70	67 - 70	Y
P_R3.6b	P_R3.6b_R01 to P_R3.6b_R02	Private housing	R	70	68 - 69	Y
P_OU1.1	P_OU1.1_R01 to P_OU1.1_R09	Private housing	R	70	67 - 76	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Unmitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Compliance of Noise Criteria (Y/N)
P_OU1.2a	P_OU1.2a_R01 to P_OU1.2a_R10	Private housing	R	70	70 - 77	N
P_OU1.3	P_OU1.3_R01 to P_OU1.3_R05	Private housing	R	70	64 - 74	N
P_OU2.3	P_OU2.3_R01 to P_OU2.3_R04	Private housing	R	70	64 - 69	Y
P_OU3.2a	P_OU3.2a_R01 to P_OU3.2a_R04	Private housing	R	70	67 - 69	Y
P_OU3.2b	P_OU3.2b_R01 to P_OU3.2b_R02	Private housing	R	70	65 - 68	Y
P_VR1.1	P_VR1.1_R01 to P_VR1.1_R02	Village rehousing	R	70	74 - 75	N
P_VR2.1	P_VR2.1_R01 to P_VR2.1_R02	Village rehousing	R	70	72	N
P_VR3.1	P_VR3.1_R01	Village rehousing	R	70	67	Y
P_LBIH	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	77 - 81	N

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions; W – Place of Worship. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).

Existing NSRs (Unmitigated Scenario)

5.6.2.3 Referring to **Table 5.6.1** and **Appendix 5.15**, the predicted noise levels at the representative existing NSRs within and outside the PDA under unmitigated scenario would be in the range of 45 – 77 dB(A) and 32 – 81 dB(A) respectively. Most of the representative existing NSRs would be exposed to noise levels exceeding the relevant noise criteria (i.e. 70dB(A) for residential and 65dB(A) for schools and places of public worship). Mitigation measures are proposed where appropriate to alleviate the potential road traffic noise impact.

Planned NSRs (Unmitigated Scenario)

5.6.2.4 From **Table 5.6.1** and **Appendix 5.15**, the predicted noise levels at Year 2053 at the planned NSRs will be in the range of 49 to 77 dB(A). Other than the environmental friendly design such as EFTS considered in the RODP, the RODP has also been proactively refined during the EIA process as far as practicable to maximise separation distance between road alignment and existing and planned NSRs, locating non-noise sensitive buildings at the site OU(OS), OU (SU)1, OU(SU)2 and OU(S+W) next to YLH to screen the road traffic noise for the planned NSRs and providing of depressed roads for some slip roads of TYST Interchange. All these initiatives would result in lower traffic flow in various roads including Kung Um Road. However, results indicate that most planned residential and educational NSRs would still exceed the noise criteria by up to 7 dB(A) and 8 dB(A) respectively. Hence, direct mitigation measures for these affected NSRs are required. For those non-compliant planned educational institutions, the detailed design shall consider suitable mitigation measures as recommended in the Class Assessment Document following ETWB Technical Circular (Works) No. 13/2003.

5.6.2.5 Hence, the total number of dwellings and classrooms that will be exposed to noise impact exceeding the criteria set out in Annex 5 of the EIAO-TM under unmitigated scenario has been summarised in **Table 5.6.1d**. All these affected NSRs would be significantly reduced after implementation of mitigation measures. (See **Sections 5.6.3**).

Table 5.6.1d Number of residential dwellings/classrooms exceeding the respective criteria under unmitigated scenario

Stages	No. of Existing Dwellings	No. of Existing Classrooms	No. of Planned Dwellings	No. of Planned Classrooms
Stage 1	4300	50	300	0
Stage 2	4050	50	1250	50
Stages 3 and 4	4950	50	5350	150

5.6.3 Mitigation of Road Traffic Noise Impact

Direct Mitigation Measures

5.6.3.1 According to the traffic forecast, there will be high volume of traffic flows on some major roads within or surrounding the PDA, namely

YLH, TYST Interchange, TSWW Interchange, Shap Pat Heung Interchange, Long Ho Road, Tai Kei Leng Road and Planned Road D1. With the existing noise barriers and LNRS on YLH and the retrofitting barriers and existing LNRS on Long Tin Road, exceedance of the noise criteria at some existing and planned NSRs are still predicted in the unmitigated scenario. Noise mitigation measures would therefore be required to alleviate the potential road traffic noise impact.

5.6.3.2 The following noise mitigation measures are proposed to alleviate adverse traffic noise impact on the affected NSRs:

- Absorptive vertical barriers and cantilevered noise barriers along some sections of Project roads and Kung Um Road;
- LNRS on some road sections;
- Semi-enclosures or full enclosures at primary distributor roads at TYST Interchange, local roads at sections of Kung Um Road;
- Nullah features / barriers along some sections of Yuen Long Nullah (Kiu Hing Road); and
- Alternative building orientation for some planned schools.

5.6.3.3 The details of proposed LNRS, noise barriers, enclosures and alternative building orientation are summarised in **Table 5.6.2**, **Table 5.6.3, Table 5.6.4** and **Table 5.6.5** respectively. The locations of the noise mitigation measures and LNRS for road traffic noise impacts are shown in **Figure 5.7**. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Table 5.6.2 Extents and locations of proposed low noise road surfacing

ID	Location	Approximate Length, m	Figure Reference
Stage 1			
LNRS28	Lam Tai East Road	465	**Figure 5.7b**
LNRS32	Kiu Hing Road (North of YLH)	260	**Figure 5.7b**
LNRS35	Proposed Road L1	410	**Figure 5.7b**
LNRS36	Proposed Road L1	410	**Figure 5.7b**
LNRS37	Proposed Road L1	375	**Figure 5.7b**
LNRS38	Proposed Road L1	375	**Figure 5.7b**
LNRS45	Kung Um Road (North of YLH)	40	**Figure 5.7b**
Stage 2			
LNRS02	Proposed Road L12	290	**Figure 5.7a**
LNRS03	Proposed Road L20	1050	**Figure 5.7a**
LNRS05	Proposed Road L12	130	**Figure 5.7a**
LNRS06	TYST Interchange	120	**Figure 5.7a**
LNRS07	TYST Interchange	265	**Figure 5.7a**
LNRS08	TYST Interchange (Loop)	625	**Figure 5.7a**
LNRS09	Slip Road near TYST Interchange	555	**Figure 5.7a**
LNRS10	TYST Interchange	125	**Figure 5.7a**
LNRS11	TYST Interchange	140	**Figure 5.7a**
LNRS12	TYST Interchange	115	**Figure 5.7a**
LNRS13	Long Tin Road	230	**Figure 5.7a**
LNRS14	TYST Interchange	80	**Figure 5.7a**
LNRS15	Slip Road near TYST Interchange	160	**Figure 5.7a**
LNRS16	TYST Interchange	95	**Figure 5.7a**

ID	Location	Approximate Length, m	Figure Reference
LNRS20	Proposed Road L14	50	**Figure 5.7a**
LNRS21	Proposed Road D1	935	**Figure 5.7a**
LNRS22	Proposed Road D1	930	**Figure 5.7a**
LNRS23	Proposed Road D2	510	**Figure 5.7c**
LNRS24	Proposed Road D2	510	**Figure 5.7c**
LNRS27	Lam Tai West Road	475	**Figure 5.7b**
LNRS29	Proposed Road L2	90	**Figure 5.7b**
LNRS30	Lam Tai West Road	385	**Figure 5.7b & c**
LNRS31	Lam Tai East Road	385	**Figure 5.7b & c**
LNRS33	Proposed Road L25	2165	**Figure 5.7b & c**
LNRS39	Tai Kei Leng Road	310	**Figure 5.7b**
LNRS40	Tai Kei Leng Road	320	**Figure 5.7b**
LNRS41	Tai Kei Leng Road	430	**Figure 5.7b**
LNRS43	Long Ho Road	215	**Figure 5.7b**
LNRS47	TYST Interchange	210	**Figure 5.7a**
LNRS48	Proposed Road L5	140	**Figure 5.7c**
Stage 3 and 4			
LNRS01	Slip road near TSWW Interchange	140	**Figure 5.7a**
LNRS04	Proposed Road L13	90	**Figure 5.7a**
LNRS17	Shan Ha Road	95	**Figure 5.7a**
LNRS18	Proposed Road L15	100	**Figure 5.7a**
LNRS19	Proposed Road L15	250	**Figure 5.7a**
LNRS25	Proposed Road D1	795	**Figure 5.7c**
LNRS26	Proposed Road D1	790	**Figure 5.7c**

ID	Location	Approximate Length, m	Figure Reference
LNRS34	Proposed Road L6	640	**Figure 5.7c**
LNRS42	Proposed Road L18	100	**Figure 5.7a**
LNRS46	Proposed Road L15	115	**Figure 5.7b**
LNRS49	Proposed Road L7	130	**Figure 5.7c**
LNRS50	TYST Road	315	**Figure 5.7a**

Note:
[1] The provision of LNRS should be constructed and maintained by responsible parties, i.e. CEDD and Highways Department respectively.

Table 5.6.3 Extents and locations of proposed noise barriers

Noise Barrier ID[1]	Location	Barrier Type	Height, m	Approximate Length, m	Figure Reference
Stage 1					
CB_E02	Proposed Road L1	Cantilevered Barrier	5+3m at 45º	265	**Figure 5.7b**
CB_E03	Proposed Road L25	Cantilevered Barrier	4+3m at 45º	15	**Figure 5.7b**
CB_E04	Proposed Road L25	Cantilevered Barrier	4+3m at 45º	80	**Figure 5.7b**
CB_E05	Proposed Road L25	Cantilevered Barrier	4+3m at 45º	90	**Figure 5.7b**
CB_E26	Proposed Road L25	Cantilevered Barrier	4+3m at 45º	30	**Figure 5.7b**
CB_E27	Proposed Road L25	Cantilevered Barrier	4+3m at 45º	20	**Figure 5.7b**
VB_E14	Proposed Road L1	Vertical Barrier	3	20	**Figure 5.7b**
VB_E25	Proposed Road L25	Vertical Barrier	2	11	**Figure 5.7b**
VB_E42	Proposed Road L1	Vertical Barrier	3	155	**Figure 5.7b**
VB_E47	Proposed Road L1	Vertical Barrier	3	25	**Figure 5.7b**
VB_P17	Lam Tai East Road	Vertical Barrier	6	50	**Figure 5.7b**
Stage 2					
CB_E01	Proposed Road L12	Cantilevered Barrier	5+3m at 45º	75	**Figure 5.7a**
CB_E16	Tai Kei Leng Road	Cantilevered Barrier	5+3m at 45º	95	**Figure 5.7b**
CB_E18	Tai Kei Leng Road	Cantilevered Barrier	5+3m at 45º	65	**Figure 5.7b**
CB_E19	Kung Um Road	Cantilevered Barrier	5+3m at 45º	50	**Figure 5.7b**

Noise Barrier ID[1]	Location	Barrier Type	Height, m	Approximate Length, m	Figure Reference
CB_E20	Kung Um Road	Cantilevered Barrier	5+3m at 45º	20	**Figure 5.7b**
CB_E21	TYST Interchange	Cantilevered Barrier	5+3m at 45º	65	**Figure 5.7a**
CB_E22	Kung Um Road	Cantilevered Barrier	5+3m at 45º	11	**Figure 5.7b**
CB_E25	TYST Interchange	Cantilevered Barrier	5+3m at 45º	25	**Figure 5.7a**
CB_P10	Lam Tai West Road	Cantilevered Barrier	5+3m at 45º	40	**Figure 5.7b**
NF_E01	Yuen Long Nullah	Nullah Feature / Barrier	5	80	**Figure 5.7b**
NF_E05	Yuen Long Nullah	Nullah Feature / Barrier	3	150	**Figure 5.7b**
NF_E06	Yuen Long Nullah	Nullah Feature / Barrier	3	115	**Figure 5.7b**
VB_E03	Proposed Road L20	Vertical Barrier	3	140	**Figure 5.7a**
VB_E04	Proposed Road L11	Vertical Barrier	5	70	**Figure 5.7a**
VB_E05	TYST Interchange	Vertical Barrier	6[2]	100	**Figure 5.7a**
VB_E06	Proposed Road L12	Vertical Barrier	1	20	**Figure 5.7a**
VB_E08	Proposed Road L12	Vertical Barrier	3	40	**Figure 5.7a**
VB_E10	TYST Interchange	Vertical Barrier	5	25	**Figure 5.7a**
VB_E11	TYST Interchange	Vertical Barrier	5	125	**Figure 5.7a**
VB_E12	TYST Interchange	Vertical Barrier	6	125	**Figure 5.7a**
VB_E13	TYST Interchange	Vertical Barrier	6	95	**Figure 5.7a**
VB_E15	Slip Road near TYST Interchange	Vertical Barrier	3	95	**Figure 5.7a**
VB_E16	Slip Road near TYST Interchange	Vertical Barrier	6	125	**Figure 5.7a**
VB_E17	Slip Road near TYST Interchange	Vertical Barrier	6	95	**Figure 5.7a**
VB_E18	Shan Ha Road	Vertical Barrier	5	75	**Figure 5.7a**
VB_E19	TYST Interchange	Vertical Barrier	6	125	**Figure 5.7a**
VB_E20	Proposed Road D1	Vertical Barrier	3	110	**Figure 5.7a**
VB_E23	Proposed Road L25	Vertical Barrier	1	9	**Figure 5.7c**
VB_E26	Proposed Road L25	Vertical Barrier	5	20	**Figure 5.7b**
VB_E27	Proposed Road L25	Vertical Barrier	4	25	**Figure 5.7b**
VB_E28	Proposed Road L25	Vertical Barrier	5	9	**Figure 5.7b**
VB_E29	Proposed Road L25	Vertical Barrier	5	9	**Figure 5.7b**

Noise Barrier ID[1]	Location	Barrier Type	Height, m	Approximate Length, m	Figure Reference
VB_E30	Proposed Road L25	Vertical Barrier	5	8	**Figure 5.7b**
VB_E31	Proposed Road L25	Vertical Barrier	3	6	**Figure 5.7b**
VB_E32	Proposed Road L25	Vertical Barrier	3	25	**Figure 5.7b**
VB_E33	Proposed Road L25	Vertical Barrier	3	30	**Figure 5.7b**
VB_E34	Proposed Road L25	Vertical Barrier	1	16	**Figure 5.7c**
VB_E35	Proposed Road L25	Vertical Barrier	4	50	**Figure 5.7c**
VB_E36	Proposed Road L25	Vertical Barrier	5	7	**Figure 5.7c**
VB_E37	Proposed Road L25	Vertical Barrier	5	7	**Figure 5.7c**
VB_E38	Proposed Road L25	Vertical Barrier	5	13	**Figure 5.7c**
VB_E39	Proposed Road L25	Vertical Barrier	5	30	**Figure 5.7c**
VB_E40	Proposed Road L25	Vertical Barrier	3	12	**Figure 5.7c**
VB_E41	Proposed Road L25	Vertical Barrier	3	13	**Figure 5.7c**
VB_E43	Proposed Road L1	Vertical Barrier	3	20	**Figure 5.7b**
VB_E45	Tai Kei Leng Road	Vertical Barrier	3	285	**Figure 5.7b**
VB_E48	Tai Kei Leng Road	Vertical Barrier	3	50	**Figure 5.7b**
VB_E49	Tai Kei Leng Road	Vertical Barrier	3	70	**Figure 5.7b**
VB_E53	Proposed Road L12	Vertical Barrier	4	19	**Figure 5.7a**
VB_E54	Proposed Road D1	Vertical Barrier	4	25	**Figure 5.7a**
VB_E55	Proposed Road D1	Vertical Barrier	5	20	**Figure 5.7a**
VB_E56	Kung Um Road	Vertical Barrier	1	8	**Figure 5.7c**
VB_P18	Lam Tai West Road	Vertical Barrier	4	75	**Figure 5.7b**
VB_P19	Lam Tai East Road	Vertical Barrier	3	30	**Figure 5.7b**
VB_P20	Lam Tai East Road	Vertical Barrier	3	25	**Figure 5.7b**
Stage 3 and 4					
VB_E01	TSWW Interchange	Vertical Barrier	5	145	**Figure 5.7a**
VB_E02	TSWW Interchange	Vertical Barrier	4	75	**Figure 5.7a**
VB_E07	Proposed Road L13	Vertical Barrier	4	40	**Figure 5.7a**
VB_E09	Proposed Road D1	Vertical Barrier	3	100	**Figure 5.7a**

Noise Barrier ID[1]	Location	Barrier Type	Height, m	Approximate Length, m	Figure Reference
VB_E21	Proposed Road L15	Vertical Barrier	3	35	**Figure 5.7a**
VB_E22	Proposed Road L15	Vertical Barrier	2	100	**Figure 5.7a**
VB_E46	Proposed Road D1	Vertical Barrier	4	25	**Figure 5.7a**
VB_E51	Proposed Road L15	Vertical Barrier	2	25	**Figure 5.7b**
VB_E52	TSWW Interchange	Vertical Barrier	3	80	**Figure 5.7a**
VB_P01	Proposed Road L15	Vertical Barrier	3	25	**Figure 5.7a**
VB_P02	Proposed Road D1	Vertical Barrier	5	220	**Figure 5.7a**
VB_P05	Proposed Road D1	Vertical Barrier	4	100	**Figure 5.7a**
VB_P06	Proposed Road L15	Vertical Barrier	4	40	**Figure 5.7a**
VB_P07	Proposed Road D2	Vertical Barrier	3	45	**Figure 5.7c**
VB_P08	Proposed Road D2	Vertical Barrier	3	45	**Figure 5.7c**
VB_P09	Proposed Road D1	Vertical Barrier	5	75	**Figure 5.7c**
VB_P10	Proposed Road D1	Vertical Barrier	5	195	**Figure 5.7c**
VB_P11	Proposed Road D1	Vertical Barrier	3	80	**Figure 5.7c**
VB_P12	Proposed Road L6	Vertical Barrier	3	25	**Figure 5.7c**
VB_P23	Proposed Road L20	Vertical Barrier	3	30	**Figure 5.7a**
VB_P26	Proposed Road L20	Vertical Barrier	2	35	**Figure 5.7a**
VB_P29	Proposed Road D1	Vertical Barrier	6	90	**Figure 5.7c**
CB_P05	Proposed Road L20	Cantilevered Barrier	5+3m at 45º	40	**Figure 5.7a**
CB_P06	Proposed Road D1	Cantilevered Barrier	5+3m at 45º	95	**Figure 5.7c**
CB_P07	Proposed Road D1	Cantilevered Barrier	5+3m at 45º	55	**Figure 5.7c**
CB_P08	Proposed Road D1	Cantilevered Barrier	5+3m at 45º	90	**Figure 5.7c**
CB_P09	Proposed Road D1	Cantilevered Barrier	5+3m at 45º	50	**Figure 5.7c**
CB_P11	Proposed Road L20	Cantilevered Barrier	5+3m at 45º	40	**Figure 5.7a**
CB_P18	Proposed Road L6	Cantilevered Barrier	5+3m at 45º	45	**Figure 5.7c**
CB_P19	Proposed Road L6	Cantilevered Barrier	5+3m at 45º	40	**Figure 5.7c**
CB_P20	Proposed Road L6	Cantilevered Barrier	5+3m at 45º	50	**Figure 5.7c**

Note:

[1] Noise barrier ID with prefix E indicates barrier required to protect existing NSRs and prefix P indicated barrier required to protect planned NSRs.

[2] VB_E05 is located on the retained wall of the depressed road. The height of this vertical barrier shall be incorporated with the retained wall such that the overall height of the retained wall and the vertical barrier is 6m from the road surface.

[3] The provision of noise barrier should be constructed and maintained by responsible parties, i.e. CEDD and Highways Department respectively.

Table 5.6.4 Extents and locations of proposed enclosures

Enclosure ID	Location	Enclosure Type[1]	Approximate Length, m	Figure Reference
Stage 1				
SE04	Yuen Long Nullah - Kung Um Road	Semi-enclosure	70	**Figure 5.7b**
SE05	Yuen Long Nullah - Kung Um Road	Semi-enclosure	30	**Figure 5.7b**
Stage 2				
SE01	Slip Road near TYST Interchange	Semi-enclosure	310	**Figure 5.7a**
SE02	Slip Road near TYST Interchange	Semi-enclosure	160	**Figure 5.7a**
SE03	Slip Road near TYST Interchange	Semi-enclosure	255	**Figure 5.7a**
FE01	TYST Interchange	Full enclosure	180	**Figure 5.7a**
FE02	TYST Interchange	Full enclosure	175	**Figure 5.7a**
FE03	TYST Interchange	Full enclosure	195	**Figure 5.7a**
FE04	TYST Interchange	Full enclosure	180	**Figure 5.7a**

Note:

[1] 6m tall enclosure are assumed in the road traffic noise assessment for conservative purpose.

[2] The provision of enclosures should be constructed and maintained by responsible parties, i.e. CEDD and Highways Department respectively.

[3] All full enclosures and semi-enclosures are proposed to protect existing NSRs.

Table 5.6.5 Locations of proposed alternative building orientation

Site ID	NSR Description	Figure Reference
Stage 2		
E2.1	Planned Primary School	**Figure 5.7c**
E2.3	Planned Primary School	**Figure 5.7c**
Stage 3 and 4		
E3.1	Planned Primary School	**Figure 5.7c**
E3.3	Planned Primary School	**Figure 5.7c**

Mitigated Noise Levels for Existing NSRs

5.6.3.4 With the implementation of proposed mitigation measures in **Table 5.6.2**, **Table 5.6.3, Table 5.6.4** and **Table 5.6.5**, the predicted mitigated road traffic noise levels at the representative existing NSRs within and outside the PDA, and the major noise source causing exceedance on these existing NSRs are presented in **Table 5.6.6**.

Table 5.6.6a Summary of the predicted road traffic noise impact at existing NSRs under mitigated scenario (Year 2027)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
Existing NSR (within PDA)						
E1_PV	E1_PV_R01 to E1_PV_R05	Park Villa	R	70	46 - 71	YLH
E1_TYST	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23, E1_TYST_R31 to E1_TYST_R38	Tong Yan San Tsuen	R	70	46 - 71	Sha Tseng Road and TYST Road (existing section)
E2_MKTT	E2_MKTT_R01	Muk Kiu Tau Tsuen	R	70	71	Kung Um Road (existing section)
Existing NSR (outside PDA)						
E4_CFG	E4_CFG_R01 to E4_CFG_R02	Chun Fai Garden	R	70	67 - 71	Kung Um Road (existing section)
E4_GM	E4_GM_R01	Garden Maple	R	70	71 - 75	Shap Pat Heung Road
E4_HG	E4_HG_R01	Hoover Garden	R	70	75	YLH
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	70 - 75	YLH
E4_JC	E4_JC_R01	Jasper Court	R	70	69 - 74	YLH
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	70 - 71	YLH
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	75	YLH
E4_LA	E4_LA_E01	Lutheran Academy	E	65	66 - 72	YLH
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	72	Shap Pat Heung Road
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	68 - 78	Shap Pat Heung Road
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	46 - 77	Shap Pat Heung Road
E4_OHP	E4_OHP_R01 to E4_OHP_R02	One Hyde Park	R	70	73 - 74	Kung Um Road (existing section)
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	60 - 75	Kung Um Road (existing section)
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	66 - 73	Shap Pat Heung Road

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
E4_PST	E4_PST_R01 to E4_PST_R20	Pak Sha Tsuen	R	70	59 - 73	Kung Um Road (existing section)
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	68 - 74	Long Tin Road
E4_RP	E4_RP_R01 to E4_RP_R02	Regent's Park	R	70	59 - 72	Kung Um Road (existing section)
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	67 - 72	YLH
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	63 - 72	YLH
E4_SG	E4_SG_R01 to E4_SG_R02	Scenic Gardens	R	70	61 - 75	Long Tin Road
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	76 - 78	Kung Um Road (existing section)
E4_SV	E4_SV_R01	Sereno Verde	R	70	68 - 73	YLH
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	78	YLH
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	70 - 78	Tai Shu Ha Rd
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	66 - 73	YLH
E4_TLT	E4_TLT_R01 to E4_TLT_R18	Tin Liu Tsuen	R	70	57 - 78	Kung Um Road (existing section)
E4_TR	E4_TR_R01	The Reach	R	70	72 - 75	YLH
E4_UT	E4_UT_R01 to E4_UT_R06	Uptown	R	70	74 - 80	Hung Tin Road
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	71 - 74	YLH and Castle Peak Road
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	75 - 77	YLH and Castle Peak Road

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed/ planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions.
[4] Noise levels presented are rounded to the nearest dB(A).

Table 5.6.6b Summary of the predicted road traffic noise impact at existing NSRs under mitigated scenario (Year 2033)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
Existing NSR (within PDA)						
E1_ELD	E1_ELD_R01	Eldorado	R	70	74 - 75	Sha Tseng Road and TYST Road (existing section)
E1_FLG	E1_FLG_R01	Fu Lai Garden	R	70	67 - 73	Sha Tseng Road and TYST Road (existing section)
E1_GR	E1_GR_R01	Greenville Residence	R	70	74 - 75	Sha Tseng Road and TYST Road (existing section)
E1_KLG	E1_KLG_R01	Kam Lai Garden	R	70	69 - 73	Sha Tseng Road and TYST Road (existing section)
E1_KV	E1_KV_R01 to E1_KV_R02	Kisland Villa	R	70	64 - 75	Sha Tseng Road and TYST Road (existing section)
E1_MG	E1_MG_R01	Marbella Gardens	R	70	74 - 76	Sha Tseng Road and TYST Road (existing section)
E1_PH	E1_PH_R01	The Parkhill	R	70	72 - 73	Sha Tseng Road and TYST Road (existing section)
E1_RLS	E1_RLS_R01 to E1_RLS_R04	Recours La Serre	R	70	47 - 68	-
E1_SHT	E1_SHT_R01 to E1_SHT_R07	Shan Ha Tsuen	R	70	60 - 67	-
E1_TYST	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23, E1_TYST_R31 to E1_TYST_R38	Tong Yan San Tsuen	R	70	47 - 76	Sha Tseng Road and TYST Road (existing section)
E1_WG	E1_WG_R01	Winsor Garden	R	70	74 - 75	Sha Tseng Road and TYST Road (existing section)
E2_TLY	E2_TLY_R01 to E2_TLY_R02	Tin Lung Yuen	R	70	64 - 69	-
Existing NSR (outside PDA)						
E4_GM	E4_GM_R01	Garden Maple	R	70	71 - 75	Shap Pat Heung Road

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
E4_HG	E4_HG_R01	Hoover Garden	R	70	74 - 75	YLH
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	69 - 73	YLH
E4_JC	E4_JC_R01	Jasper Court	R	70	67 - 73	YLH
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	69 - 70	YLH
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	75	YLH
E4_LA	E4_LA_E01	Lutheran Academy	E	65	65 - 72	YLH
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	72	Shap Pat Heung Road
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	67 - 78	Shap Pat Heung Road
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	48 - 77	Shap Pat Heung Road
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	56 - 68	-
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	66 - 74	Shap Pat Heung Road
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	63 - 69	-
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	68 - 71	YLH
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	63 - 73	YLH
E4_SG	E4_SG_R01 to E4_SG_R02	Scenic Gardens	R	70	52 - 72	Long Tin Road
E4_SHT	E4_SHT_R01 to E4_SHT_R07	Shan Ha Tsuen	R	70	54 - 70	-
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	69 - 70	-
E4_SV	E4_SV_R01	Sereno Verde	R	70	64 - 72	YLH
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	77	YLH
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	62 - 72	Tai Shu Ha Rd
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	66 - 73	YLH

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
E4_TLT	E4_TLT_R01 to E4_TLT_R18	Tin Liu Tsuen	R	70	52 - 71	Kung Um Road (existing section)
E4_TR	E4_TR_R01	The Reach	R	70	64 - 75	YLH
E4_UT	E4_UT_R01 to E4_UT_R06	Uptown	R	70	73 – 79	Hung Tin Road
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	70 - 74	YLH and Castle Peak Road
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	76 - 77	YLH and Castle Peak Road

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed/ planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions.
[4] Noise levels presented are rounded to the nearest dB(A).

Table 5.6.6c Summary of the predicted road traffic noise impact at existing NSRs under mitigated scenario (Year 2053)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
Existing NSR (within PDA)						
E1_ELD	E1_ELD_R01	Eldorado	R	70	75	Sha Tseng Road and TYST Road (existing section)
E1_FLG	E1_FLG_R01	Fu Lai Garden	R	70	69 - 74	Sha Tseng Road and TYST Road (existing section)
E1_KLG	E1_KLG_R01	Kam Lai Garden	R	70	69 - 76	Sha Tseng Road and TYST Road (existing section)
E1_KV	E1_KV_R01 to E1_KV_R02	Kisland Villa	R	70	72 - 73	Sha Tseng Road and TYST Road (existing section)
E1_LHT	E1_LHT_R01	Lam Hau Tsuen	R	70	70	-
E1_PH	E1_PH_R01	The Parkhill	R	70	68 - 69	-
E1_PV	E1_PV_R01 to E1_PV_R05	Park Villa	R	70	52 - 70	-
E1_RLS	E1_RLS_R01 to E1_RLS_R04	Recours La Serre	R	70	62 - 71	Sha Tseng Road and TYST Road

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
E1_SHT	E1_SHT_R01 to E1_SHT_R07	Shan Ha Tsuen	R	70	56 - 70	-
E1_TYST	E1_TYST_R01 to E1_TYST_R14, E1_TYST_R20 to E1_TYST_R23	Tong Yan San Tsuen	R	70	56 - 76	Sha Tseng Road and TYST Road (existing section)
E1_WG	E1_WG_R01	Winsor Garden	R	70	70 - 72	Sha Tseng Road and TYST Road (existing section)
E2_MKTT	E2_MKTT_R01	Muk Kiu Tau Tsuen	R	70	69	-
E2_TLY	E2_TLY_R01 to E2_TLY_R02	Tin Lung Yuen	R	70	65 - 69	-
Existing NSR (outside PDA)						
E4_CFG	E4_CFG_R01 to E4_CFG_R02	Chun Fai Garden	R	70	66 - 70	-
E4_CPG	E4_CPG_R01	Chung Pak Garden	R	70	68	-
E4_GM	E4_GM_R01	Garden Maple	R	70	72 - 76	Shap Pat Heung Road
E4_HG	E4_HG_R01	Hoover Garden	R	70	75 - 76	YLH
E4_HYTT	E4_HYTT_R01	Ha Yau Tin Tsuen	R	70	70 - 75	YLH
E4_JC	E4_JC_R01	Jasper Court	R	70	66 - 73	YLH
E4_KMYLS	E4_KMYLS_E01	Kwong Ming Ying Loi School	E	65	70 - 71	YLH
E4_KTST	E4_KTST_R01 to E4_KTST_R02	Kong Tau San Tsuen	R	70	76	YLH
E4_LA	E4_LA_E01	Lutheran Academy	E	65	65 - 73	YLH
E4_LFM	E4_LFM_R01	Lung Fai Mansion	R	70	73	Shap Pat Heung Road
E4_LG	E4_LG_R01 to E4_LG_R02	La Grove	R	70	67 - 78	Shap Pat Heung Road
E4_LHT	E4_LHT_R01 to E4_LHT_R05	Lam Hau Tsuen	R	70	63 - 70	-
E4_LTT	E4_LTT_R01 to E4_LTT_R28	Lung Tin Tsuen	R	70	49 - 78	Shap Pat Heung Road
E4_MKT	E4_MKT_R01 to E4_MKT_R05	Muk Kiu Tau Tsuen	R	70	63 - 69	-
E4_OHP	E4_OHP_R01 to E4_OHP_R02	One Hyde Park	R	70	70	-
E4_PHY	E4_PHY_R01 to E4_PHY_R02	Pitt Hwa Yuen	R	70	57 - 69	-

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Major Noise Source causing exceedance
E4_PS	E4_PS_R01 to E4_PS_R03	Park Signature	R	70	67 - 75	Shap Pat Heung Road
E4_PST	E4_PST_R01 to E4_PST_R20	Pak Sha Tsuen	R	70	56 - 70	-
E4_PSV	E4_PSV_R01 to E4_PSV_R02	Parkside Villa	R	70	66 - 70	-
E4_RP	E4_RP_R01 to E4_RP_R02	Regent's Park	R	70	56 - 69	-
E4_SCST	E4_SCST_R01 to E4_SCST_R02	Sham Chung San Tsuen	R	70	68 - 72	YLH
E4_SCT	E4_SCT_R01 to E4_SCT_R03	Sham Chung Tsuen	R	70	64 - 74	YLH
E4_SG	E4_SG_R01 to E4_SG_R02	Scenic Gardens	R	70	61 - 74	Long Tin Road
E4_SHT	E4_SHT_R01 to E4_SHT_R07	Shan Ha Tsuen	R	70	55 - 70	-
E4_SMG	E4_SMG_R01 to E4_SMG_R04	Sun Mei Garden	R	70	70 - 71	Kung Um Road (existing section)
E4_SV	E4_SV_R01	Sereno Verde	R	70	65 - 73	YLH
E4_SYTT	E4_SYTT_R01	Sheung Yau Tin Tsuen	R	70	78	YLH
E4_TKLT	E4_TKLT_R01 to E4_TKLT_R06	Tai Kei Leng Tsuen	R	70	63 - 72	Tai Shu Ha Road
E4_TKT	E4_TKT_R01 to E4_TKT_R03	Tan Kwai Tsuen	R	70	67 - 74	YLH
E4_TLT	E4_TLT_R01 to E4_TLT_R18	Tin Liu Tsuen	R	70	54 - 71	Kung Um Road (existing section)
E4_TR	E4_TR_R01	The Reach	R	70	65 - 76	YLH
E4_TTT	E4_TTT_R01 to E4_TTT_R03	Tai Tao Tsuen	R	70	63 - 67	-
E4_UT	E4_UT_R01 to E4_UT_R06	Uptown	R	70	72 - 81	Hung Tin Road
E4_YH	E4_YH_R01 to E4_YH_R02	Yoho Town	R	70	71 - 75	YLH and Castle Peak Road
E4_YHM	E4_YHM_R01	Yoho Midtown	R	70	76 - 78	YLH and Castle Peak Road

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions.
[4] Noise levels presented are rounded to the nearest dB(A).

5.6.3.5 As shown in **Table 5.6.6**, the exceedance at the representative existing NSRs would be due to existing roads. All practicable direct mitigation measures on project roads have been considered such that the noise contribution due to project roads is insignificant. As discussed in **Section 5.6.1**, a review for NSRs along existing roads is conducted to assess whether the traffic noise impact due to the Project would be significant. **Appendix 5.16e** shows that significant road traffic noise impact from the Project would only be predicted on TYST Road (existing section) and Sha Tseng Road. Implementation of mitigation measures on these existing roads under this Project are proposed to alleviate the road traffic noise impact.

Sha Tseng Road and TYST Road (existing section)

5.6.3.6 Construction of noise barrier along Sha Tseng Road and TYST Road (existing section) is considered not feasible due to insufficient space and a number of existing ingress and egress. LNRS is therefore proposed to alleviate the road traffic noise impacts at these NSRs. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials.

Shap Pat Heung Road

5.6.3.7 With the existing vertical noise barrier (2m high) and cantilevered noise barrier (2m high vertical with 1m cantilevered arm) along east bound of Shap Pat Heung Road, the road traffic noise levels of some existing NSRs would exceed the noise criterion of 70dB(A). However, construction of noise barrier along this section of the existing Shap Pat Heung Road is considered not feasible due to insufficient space. Resurfacing with LNRS material may be considered to be implemented where practicable by the Government.

Kung Um Road (existing section)

5.6.3.8 Currently, along the Kung Um Road (existing section), there are low-rise villages such as Tin Liu Tsuen, etc., separated at about 5m with ingress and egress. In view of such short separation to the nearby NSRs, the road traffic noise impacts at these existing NSRs will still exceed the noise criterion of 70dB(A). In view of the existing ingress, egress and limited space for various villages, all at-source mitigation measures have been provided as far as practicable to reduce the road traffic noise impacts for NSRs along Kung Um Road.

YLH and Castle Peak Road

5.6.3.9 The road traffic noise levels of some existing NSRs would exceed the noise criterion of 70dB(A). LNRS has been already paved along YLH and with existing 4m noise barriers near Yoho Town and Yoho Midtown. However, construction of noise barrier along this section of the existing YLH and Castle Peak Road is considered not feasible due

to insufficient structural capacity and insufficient space. Resurfacing with LNRS material may be considered to be implemented where practicable by the Government.

Hung Tin Road

5.6.3.10 The road traffic noise levels of some existing NSRs would exceed the noise criterion of 70dB(A). Further refinement of noise barrier along the concerned road section may be considered to be implemented where practicable by the Government.

Long Tin Road

5.6.3.11 The road traffic noise levels of some existing NSRs would exceed the noise criterion of 70dB(A). The existing TYST Interchange will be re-constructed and noise mitigation measures, such as semi-enclosures, roadside barriers and LNRS, have been proposed as far as practicable.

Tai Shu Ha Road

5.6.3.12 The road traffic noise levels of some existing NSRs would exceed the noise criterion of 70dB(A). In addition, construction of noise barrier along this section of the existing Tai Shu Ha Road is considered not feasible due to insufficient space. Resurfacing with LNRS material may be considered to be implemented where practicable by the Government.

Mitigated Noise Levels for Planned NSRs

5.6.3.13 With the implementation of proposed mitigation measures in **Table 5.6.2**, **Table 5.6.3, Table 5.6.4** and **Table 5.6.5**, the predicted mitigated road traffic noise levels at the representative planned NSRs within the PDA are presented in **Table 5.6.7**.

Table 5.6.7a Summary of the predicted road traffic noise impact at planned NSRs under mitigated scenario (Year 2027)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	51 - 74	Y
P_LBIH	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	75 - 81	Y

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).

Table 5.6.7b Summary of the predicted road traffic noise impact at planned NSRs under mitigated scenario (Year 2033)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_E2.1[5]	P_E2.1_E01 to P_E2.1_E07	Planned primary school	E	65	55 - 63	N
P_E2.3[5]	P_E2.3_E01 to P_E2.3_E06	Planned primary school	E	65	58 - 64	N
P_R2.1a	P_R2.1a_R01 to P_R2.1a_R05	Private housing	R	70	52 - 72	Y
P_R2.1b	P_R2.1b_R01 to P_R2.1b_R03	Public housing	R	70	53 - 70	N
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	52 - 74	Y
P_R2.3	P_R2.3_R01 to P_R2.3_R08	Public housing	R	70	59 - 70	N
P_R2.4	P_R2.4_R01 to P_R2.4_R14	Public housing	R	70	60 - 69	N
P_VR1.1	P_VR1.1_R01 to P_VR1.1_R02	Village rehousing	R	70	64 - 67	N
P_VR2.1	P_VR2.1_R01 to P_VR2.1_R02	Village rehousing	R	70	69 - 70	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_LBIH	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	62 - 77	Y

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).
[5] The NAPs are assigned based on alternative building orientation as shown in **Figure 5.7c**.

Table 5.6.7c Summary of the predicted road traffic noise impact at planned NSRs under mitigated scenario (Year 2053)

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_E2.1[6]	P_E2.1_E01 to P_E2.1_E07	Planned primary school	E	65	56 - 65	N
P_E2.3[6]	P_E2.3_E01 to P_E2.3_E06	Planned primary school	E	65	61 - 64	N
P_E2.4	P_E2.4_E01 to P_E2.4_E04	Planned primary school	E	65	61 - 64	N
P_E3.1[6]	P_E3.1_E01 to P_E3.1_E04	Planned primary school	E	65	53 - 61	N
P_E3.2	P_E3.2_E01 to P_E3.2_E03	Planned primary school	E	65	61 - 64	N
P_E3.3	P_E3.3_E01 to P_E3.3_E03	Planned primary school	E	65	61 - 64	N
P_G2.4	P_G2.4_R01	Married quarters	R	70	69 - 70	N
P_R1.3	P_R1.3_R01 to P_R1.3_R09	Private housing	R	70	65 - 70	N
P_R1.4	P_R1.4_R01 to P_R1.4_R09	Private housing	R	70	64 - 70	N
P_R1.5	P_R1.5_R01 to P_R1.5_R03	Private housing	R	70	68 - 70	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_R1.6	P_R1.6_R01 to P_R1.6_R02	Private housing	R	70	69 - 70	N
P_R1.7	P_R1.7_R01 to P_R1.7_R08	Private housing	R	70	62 - 70	N
P_R1.8	P_R1.8_R01 to P_R1.8_R08	Private housing	R	70	69 - 70	N
P_R1.10	P_R1.10_R01 to P_R1.10_R09	Private housing	R	70	54 - 70	N
P_R1.12	P_R1.12_R01 to P_R1.12_R06	Private housing	R	70	66 - 70	N
P_R1.13	P_R1.13_R01 to P_R1.13_R04	Private housing	R	70	62 - 70	N
P_R1.14	P_R1.14_R01 to P_R1.14_R05	Private housing	R	70	60 - 70	N
P_R1.16	P_R1.16_R01 to P_R1.16_R07	Private housing	R	70	58 - 70	N
P_R2.1a	P_R2.1a_R01 to P_R2.1a_R05	Private housing	R	70	58 - 73	Y
P_R2.1b	P_R2.1b_R01 to P_R2.1b_R03	Public housing	R	70	60 - 70	N
P_R2.2	P_R2.2_R01 to P_R2.2_R20	Public housing	R	70	60 - 75	Y
P_R2.3	P_R2.3_R01 to P_R2.3_R08	Public housing	R	70	64 - 70	N
P_R2.4	P_R2.4_R01 to P_R2.4_R14	Public housing	R	70	64 - 70	N
P_R2.6a	P_R2.6a_R01 to P_R2.6a_R07	Private housing	R	70	64 - 70	N
P_R2.6c	P_R2.6c_R01 to P_R2.6c_R04	Private housing	R	70	47 - 70	N
P_R2.6d	P_R2.6d_R01 to P_R2.6d_R03	Private housing	R	70	52 - 70	N
P_R2.7	P_R2.7_R01 to P_R2.7_R04	Private housing	R	70	61 - 69	N
P_R2.8	P_R2.8_R01 to P_R2.8_R06	Public housing	R	70	65 - 70	N

NSR	NAP ID[1,2]	Description	Uses[3]	Noise Criterion, $L_{10\ (1\ hr)}$, dB(A)	Predicted Mitigated Overall Noise Level, ($L_{10\ (1\ hr)}$, dB(A))[4]	Require Additional Mitigation Measures
P_R3.1b	P_R3.1b_R01 to P_R3.1b_R03	Private housing	R	70	64 - 70	N
P_R3.1c	P_R3.1c_R01 to P_R3.1c_R02	Private housing	R	70	66 - 70	N
P_R3.1e	P_R3.1e_R01 to P_R3.1e_R02	Private housing	R	70	67 - 70	N
P_R3.2	P_R3.2_R01 to P_R3.2_R04	Private housing	R	70	64 - 69	N
P_R3.4a	P_R3.4a_R01 to P_R3.4a_R04	Private housing	R	70	60 - 70	N
P_R3.4b	P_R3.4b_R01 to P_R3.4b_R04	Private housing	R	70	63 - 70	N
P_R3.5a	P_R3.5a_R01 to P_R3.5a_R03	Private housing	R	70	63 - 70	N
P_OU1.1	P_OU1.1_R01 to P_OU1.1_R09	Private housing	R	70	63 - 72	Y
P_OU1.2a	P_OU1.2a_R01 to P_OU1.2a_R10	Private housing	R	70	67 - 75	Y
P_OU1.3	P_OU1.3_R01 to P_OU1.3_R05	Private housing	R	70	64 - 70	N
P_VR1.1	P_VR1.1_R01 to P_VR1.1_R02	Village rehousing	R	70	64 - 70	N
P_VR2.1	P_VR2.1_R01 to P_VR2.1_R02	Village rehousing	R	70	70	N
P_LBIH[5]	P_LBIH_R01 to P_LBIH_R02	Planned Public Housing Development in Long Bin	R	70	62 - 77	Y

Note:
[1] The assessment will only include NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs within 300m from the boundary of the planned road works and the committed and planned NSRs of the Project is included.
[3] R – Residential Premises; E – Educational Institutions. The land uses of the planned NSRs within the PDA are identified based on the RODP. The land uses of P_LBIH is identified based on DC Discussion Paper "Consultation on the proposed amendments to the approved TYST Outline Zoning Plan No. S/YL-TYST/10".
[4] Noise levels presented are rounded to the nearest dB(A).
[5] A separate Environmental Assessment Study (EAS) will be conducted in the detailed design stage by HKHA to address environmental impacts according to HKPSG and to comply with relevant criteria.
[6] The NAPs are assigned based on alternative building orientation as shown in **Figure 5.7c**.

5.6.3.14 Combination of proposed mitigation measures are formulated to protect both planned and existing NSRs from road traffic noise impact. The noise level reduction by these mitigation measures are illustrated in **Table 5.6.8**.

Table 5.6.8 Noise level reduction by combination of proposed mitigation measures

Intended NSRs to be Protected	Combination of Proposed Mitigation Measures	Noise Level Reduction, $L_{10\ (1\ hr)}$, dB(A)
Existing NSRs		
E1_LHT	FE01, FE02, FE03, FE04, VB_E21, LNRS17, LNRS18, LNRS19	3
E1_PV	CB_E01, VB_E03, VB_E04, LNRS02, LNRS03, LNRS04	3 - 11
E1_RLS	VB_E07, LNRS04, LNRS 50	2 - 9
E1_SHT	VB_E20, VB_E46, VB_E54, VB_E55, LNRS21, LNRS22	5 - 6
E1_TYST	SE03, CB_E25, VB_E05, VB_E08, VB_E06, VB_E10, VB_E11, VB_E12, VB_E53, LNRS02, LNRS05, LNRS06, LNRS08, LNRS10, LNRS47, LNRS50	1 - 18
E2_MKTT	VB_E23, VB_E34, LNRS04, LNRS33	4
E2_TLY	CB_E02, CB_E19, LNRS37, LNRS38	5 - 8
E4_CFG	VB_E32, VB_E33, LNRS33	3
E4_CPG	VB_E41, LNRS33	3
E4_GM	SE04, SE05, CB_E04, LNRS32, LNRS45	1
E4_JC	SE01, VB_E16, LNRS06, LNRS07, LNRS08, LNRS09, LNRS10, LNRS12, LNRS13, LNRS14, LNRS15, LNRS16, LNRS47	4 - 10
E4_LFM	SE04, SE05, CB_E04, LNRS32, LNRS45	1
E4_LG	SE04, SE05, CB_E04, CB_E05, LNRS32, LNRS45	1 - 7
E4_LHT	VB_E13, VB_E15, VB_E22, VB_E51	4 - 8
E4_LTT	CB_E03, CB_E04, CB_E05, CB_E26, CB_E27, VB_E25, LNRS 32, LNRS45	1 - 12
E4_MKT	LNRS33	2
E4_OHP	LNRS33	1
E4_PHY	VB_E26, VB_E28, VB_E30, LNRS33	4 - 9
E4_PS	SE04, SE05, CB_E04, CB_E05, LNRS32, LNRS45	1 - 6
E4_PST	VB_E35, VB_E36, VB_E37, VB_E38, VB_E39, VB_E40, VB_E41, VB_E56, LNRS33	2 - 11
E4_PSV	SE02, VB_E17, VB_E19, CE_E21, LNRS08, LNRS09, LNRS13, LNRS14, LNRS16	2 - 5
E4_RP	LNRS33	2
E4_SCST	LNRS43	1

Intended NSRs to be Protected	Combination of Proposed Mitigation Measures	Noise Level Reduction, $L_{10\ (1\ hr)}$, dB(A)
E4_SCT	VB_E14, VB_E42, VB_E47, LNRS35, LNRS36, LNRS43	1 - 4
E4_SG	SE02, VB_E17, LNRS09, LNRS13	1
E4_SHT	VB_E09, LNRS21, LNRS22, LNRS27, LNRS28, LNRS29, LNRS30, LNRS31	3 - 8
E4_SMG	LNRS33	3 - 4
E4_SV	CB_E16, CB_E18, VB_E45, VB_E49, LNRS39, LNRS40, LNRS41	2 - 3
E4_TKLT	CB_E16, CB_E18, VB_E48, VB_E45, VB_E49, LNRS39, LNRS40, LNRS41	3 - 16
E4_TLT	CB_E02, CB_E19, CB_E20, CB_E22, VB_E43, VB_E26, VB_E27, VB_E28, VB_E29, VB_E30, VB_E31, NF_E01, NF_E05, NF_E06, LNRS33, LNRS35, LNRS36, LNRS37, LNRS38	4 - 16
E4_TR	VB_E45, LNRS39, LNRS40	1 - 3
E4_TTT	VB_E01, VB_E02, VB_E52, LNRS01	5
E4_UT	VB_E01, VB_E02, VB_E52, LNRS01	1 - 2
Planned NSRs		
P_E2.1	Alternative building orientation, LNRS33, LNRS48	3 - 8
P_E2.3	Alternative building orientation, LNRS33, LNRS48	3 - 7
P_E2.4	LNRS33	2
P_E3.1	Alternative building orientation, LNRS33	9 - 13
P_E3.2	LNRS33, LNRS34, LNRS49	2
P_E3.3	Alternative building orientation, LNRS33, LNRS34, LNRS49	5 - 10
P_G2.4	VB_P09, VB_P10, LNRS25, LNRS26	3 - 4
P_OU1.1	FE01, FE02, FE03, FE04, SE03, VB_E11, VB_E12, VB_E13, VB_E18, LNRS08, LNRS10, LNRS11, LNRS12, LNRS20, LNRS47	3 - 10
P_OU1.2a	FE01, FE02, FE03, FE04, VB_P01, VB_E13, VB_E15, LNRS08, LNRS10, LNRS11, LNRS12, LNRS17, LNRS18, LNRS19, LNRS20, LNRS47	1 - 9
P_OU1.3	FE01, FE02, FE03, FE04, VB_P02, LNRS20, LNRS21, LNRS22	4
P_R1.3	VB_P23, LNRS03, LNRS04, LNRS42	2 - 5
P_R1.4	LNRS50	3
P_R1.5	LNRS50	2 - 3
P_R1.6	VB_P05, LNRS03, LNRS42	3
P_R1.7	VB_P26, LNRS03	3
P_R1.8	VB_P02, LNRS03, LNRS21, LNRS22	2 - 5
P_R1.10	VB_P02, LNRS20, LNRS21, LNRS22	7 - 18

Intended NSRs to be Protected	Combination of Proposed Mitigation Measures	Noise Level Reduction, $L_{10\ (1\ hr)}$, dB(A)
P_R1.12	VB_P23, LNRS03, LNRS42	3
P_R1.13	LNRS50	3
P_R1.14	CB_P05, CB_P11, LNRS03, LNRS42	4 - 11
P_R1.16	FE01, FE02, FE03, FE04, VB_E46, VB_P06, VB_E54, LNRS19, LNRS20, LNRS21, LNRS22	4 - 14
P_R2.1a	LNRS27, LNRS28	1 - 3
P_R2.1b	CB_P10, LNRS27, LNRS28, LNRS29	3 - 4
P_R2.2	CB_E02, VB_P17, LNRS27, LNRS28, LNRS29, LNRS37, LNRS38,	1 - 4
P_R2.3	VB_P18, VB_P08, LNRS23, LNRS24, LNRS30, LNRS31	3 - 5
P_R2.4	VB_P19, VB_P20, LNRS23, LNRS24, LNRS30, LNRS31	3 - 4
P_R2.6a	LNRS23, LNRS24, LNRS30, LNRS31	3
P_R2.6c	VB_P07, VB_P08, LNRS21, LNRS22, LNRS25, LNRS26, LNRS30, LNRS31	3
P_R2.6d	VB_P09, LNRS25, LNRS26	3 - 4
P_R2.7	VB_P10, LNRS25, LNRS26	5
P_R2.8	CB_P07, CB_P09, VB_P11, LNRS25, LNRS26	3 - 8
P_R3.1b	CB_P06, CB_P08, VB_P29, LNRS25, LNRS26, LNRS34	3 - 10
P_R3.1c	CB_P08, LNRS34	3
P_R3.1e	LNRS34	2 - 3
P_R3.2	VB_P29, LNRS25, LNRS26	5
P_R3.4a	CB_P20, VB_P12, LNRS34, LNRS49	3 - 14
P_R3.4b	LNRS34, LNRS49	3
P_R3.5a	CB_P18, CB_P19, LNRS25, LNRS26, LNRS34	3 - 12
P_VR1.1	VB_E46, VB_E20, VB_E54, VB_E55, LNRS21, LNRS22	5 - 10
P_VR2.1	LNRS27, LNRS28, LNRS29, LNRS30, LNRS31	2 - 3
P_LBIH	SE01, VB_E16, VB_E17, LNRS07, LNRS08, LNRS09, LNRS10, LNRS11, LNRS12, LNRS13, LNRS14, LNRS15, LNRS16, LNRS47	2 - 19

5.6.3.15 As shown in **Table 5.6.8**, with the recommended at-source mitigation measures in place, some planned residential premises would still exceed the noise criterion by 1 – 5dB(A).

5.6.3.16 After the exhaustion of at-source mitigation measures, at-receiver mitigation measures are proposed to alleviate the road traffic noise impacts:

- Provision of acoustic windows for some planned public and private housing.

5.6.3.17 Details of the at-receiver mitigation measures proposed at the affected NSRs are summarised in **Table 5.6.9.**

Table 5.6.9 Proposed at-receiver mitigation measures at planned sites

Site ID	Proposed Additional Mitigation Measures	Uses[1]	Applied to	Figure Reference
R2.1a	Acoustic windows	R	11/F – 30/F	**Figure 5.7b**
R2.2	Acoustic windows	R	4/F – 34/F	**Figure 5.7b**
OU1.1	Acoustic windows	R	5/F – 12/F	**Figure 5.7a**
OU1.2a	Acoustic windows	R	1/F – 12/F	**Figure 5.7a**

Note:
[1] R – Residential Premises.

5.6.3.18 **Appendix 5.16** shows the details of the noise impacts of the NSRs and NAPs at various floor levels under mitigated scenario. It has been demonstrated that the noise impact can be mitigated to an acceptable level by employing a package of mitigation measures including: 1) absorptive vertical barriers and cantilevered noise barriers along some sections of Project Roads and Kung Um Road; 2) LNRS on some road sections; 3) semi-enclosures or full enclosures at primary distributor roads at TYST Interchange and Kung Um Road; 4) nullah features / barriers along some sections of Yuen Long Nullah (Kiu Hing Road); 5) alternative building orientation for some planned schools to alleviate adverse traffic noise impact on the affected NSRs.

5.6.3.19 The proposed at-receiver mitigation measures i.e. the provision of acoustic windows for some planned public and private housing are subject to further study by Hong Kong Housing Authority (HKHA) and future private developers respectively. HKHA and future private developers can further explore alternative options which can achieve corresponding traffic noise reduction during the detailed design stage.

5.6.3.20 There is scope for adoption of alternative noise mitigation measures, for example, by means of alternative layout and design of individual developments at the detailed design stage. Noise impact assessment at the planned residential sites is proposed to be conducted by future developers at the detailed design stage to study whether the future development layout would avoid exposing to excessive road traffic noise levels so as to minimise the scale and extent of the proposed noise mitigation measures such as noise barriers and special building design. The requirement of noise impact assessment would be included in the land lease condition or planning briefs of the residential sites.

5.6.3.21 In addition, Environmental Assessment Study (EAS) for the planned public housings will be conducted by HKHA in the detailed design stage according to HKPSG to address environment impacts and to comply with relevant criteria.

5.6.3.22 With implementation of the proposed mitigation measures, the total number of dwellings and classrooms that will be benefited is summarized in **Table 5.6.9a**.

Table 5.6.9a Number of residential dwellings/classrooms benefited under mitigated scenario

Stages	No. of Existing Dwellings	No. of Existing Classrooms	No. of Planned Dwellings	No. of Planned Classrooms
Stage 1	250	0	300	0
Stage 2	1000	0	1250	50
Stages 3 and 4	1500	0	5350	150

5.6.3.23 The total number of dwellings and classrooms that will still be exposed to noise impact exceeding the criteria set out in Annex 5 of the EIAO-TM has been summarized in **Table 5.6.9b**. All direct mitigation measures have been proposed, such that the contribution from the Project Road is not significant (i.e. less than 1.0dB(A)). All of the exceedance are due to the existing roads.

Table 5.6.9b Number of residential dwellings/classrooms exposed the respective criteria under mitigated scenario

Stages	No. of Existing Dwellings	No. of Existing Classrooms	No. of Planned Dwellings	No. of Planned Classrooms
Stage 1	4100	50	0	0
Stage 2	3350	50	0	0
Stages 3 and 4	3850	50	0	0

5.6.4 Assessment of Side Effects and Constraints

5.6.4.1 As discussed in **Section 10.12**, the visual impact of noise mitigation measures will be mitigated by appropriate detailed design to reduce visual bulkiness and incorporate aesthetically pleasing surface treatments to promote visual amenity.

5.6.5 Evaluation of Constraints on Planned Noise Sensitive Developments / Land Uses

5.6.5.1 The planned use with proposed acoustic windows or alternative building orientation will comply with noise criteria. These at-receiver mitigation measures will not affect the development parameters Therefore, constraints on the planned noise sensitive development/land uses are not anticipated.

5.6.6 Evaluation of Residual of Road Traffic Noise Impacts

5.6.6.1 Road traffic noise impact from the Project can be properly mitigated by implementing the proposed noise mitigation measures such as noise barriers and enclosures on some sections of project roads, LNRS on some road sections and acoustic windows on some planned residential developments. Adverse residual road traffic noise impact is not anticipated.

5.7 Fixed Noise Sources Impact Assessment

5.7.1 Fixed Noise Sources Impact Assessment Methodology

5.7.1.1 Fixed noise sources impact assessment will be conducted based on the following procedures:

- Determine the assessment area;
- Identify and locate representative NSRs that may be affected by the noise sources;
- Determine the noise criteria for both daytime and night-time;
- Use standard acoustic principle for attenuation and directivity;
- Adopt correction of tonality, impulsiveness and intermittency as stipulated in TM-Places;
- Calculate the noise impacts using assumed plant inventories and utilisation schedule for worst case scenario; and
- Cumulative impacts will be included.

5.7.2 Prediction and Evaluation of Fixed Noise Sources Impact

Scenarios

5.7.2.1 The fixed noise sources impact assessment of unmitigated scenario and mitigated scenario of worst operation mode representing the maximum noise emission is conducted with respect to the criteria set out in Annex 5 of the EIAO-TM. For planned fixed noise sources, the worst operation mode with 100% operation will be predicted based on backward calculation of separation distance. Hence, the maximum SWL will be predicted regardless on the percentage usage. Validity of the operation modes have been confirmed with the project proponent. Project proponent would further liaise and agree with the relevant departments on the responsibility of implementation and maintenance of the predicted maximum allowable SWLs for the planned fixed plant noise sources in the detailed design stage. For existing fixed noise sources, their SWLs have been estimated based on measured sound pressure levels (SPLs) for cumulative assessment.

Prediction of Noise Impact

<u>Existing Industrial Zone (Outside PDA near Fui Sha Wai)</u>

5.7.2.2 An "I" zone located to the north of YLH and to the east of Fui Sha Wai. The nearest existing NSR E4_TYST_R01 is encroached within the "I" zone, and the nearest planned NSR P_R1.2_R05 is located at about 330m away from the "I" zone.

5.7.2.3 According to the prevailing noise measurements, the noise levels at PNM-04 (refer to **Appendix 5.1**), which is located at about 70m away from the "I" zone, are 60-62dB(A), 60-61dB(A) and 59-60dB(A) during daytime, evening time and night-time periods respectively, and have included the effects of all the noise sources in the vicinity such as road traffic noise, community noise, fixed noise sources impacts from the "I" zone. Since the total noise from all these various sources are within the criteria, adverse fixed noise sources impacts from the existing "I" zone near Fui Sha Wai alone on surrounding NSRs are not anticipated.

5.7.2.4 In addition, non-noise sensitive buildings of about 40m high will be developed on the OU zone (i.e. OU1.7, OU1.8, OU1.9 and OU1.10) in between the planned residential sites and the "I" zone. These developments can provide screening effect for fixed plant at the industrial and residential interface. Therefore, adverse fixed noise sources impacts from the existing "I" zone near Fui Sha Wai on NSRs within the PDA are not anticipated. Since there are no other major existing and planned fixed noise sources in the vicinity of the "I" zone, cumulative fixed noise sources impacts are not anticipated.

Existing Livestock Farm (AGR3.1)

5.7.2.5 A livestock farm located within the PDA (AGR3.1) and will be retained during the development. It is located in Area 3 at the southern end of Kung Um Road. The nearest existing NSR E4_WNT_R12 is located at about 90m away from the livestock farm, and the nearest planned NSR P_VR3.1_R03 is located at about 185m away from the livestock farm. The farm will be located in the vicinity of planned government reserve (intended for government depot) (G3.1 and G3.3) and STW (OU3.3) and therefore, cumulative fixed noise sources impacts are considered in the assessment.

5.7.2.6 The major noise sources of the livestock farm would mainly consist of ventilation system with louver and generator etc. and there would have intermittent nuisance of cock's crowing around the clock. Noise measurement has been conducted at a distance of 10m away from the chicken farm and the SPL was found to be about 55dB(A). Since the nearest existing NSR E4_WNT_R12 is located at about 90m away from the livestock farm, and the nearest planned NSR P_VR3.1_R03 is located at about 185m away from the livestock farm, adverse fixed noise sources impacts from the existing livestock farm alone on surrounding NSRs are not anticipated.

Existing Lung Tin Sewage Pumping Station (OU2.2)

5.7.2.7 The existing LTSPS is located at about 20m to the south of YLH and about 20m to the west of Kung Um Road. Noise measurement has been conducted at a distance of 10m away from the LTSPS and the SPL was found to be about 57dB(A). Since the nearest existing NSR E4_TLT_R01 is located at about 75m away from the LTSPS, and the nearest planned NSR P_R2.2_R02 is located at about 190m away from

the LTSPS, adverse fixed noise sources impacts from the LTSPS alone on surrounding NSRs are not anticipated.

Planned Electricity Substation (OU2.1)

5.7.2.8 An ESS (OU2.1) is planned to be located in Area 2 at the middle of the PDA at about 3m to the west of Kung Um Road. The major noise sources of ESS are from the operation of transformers and ventilation systems. Since there is no design information on the proposed ESS, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.1** below shows the maximum allowable SWL of the ESS. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.1 Summary of maximum allowable SWL of ESS (OU2.1)

Plant Item	Maximum Allowable SWL, dB(A)
ESS (OU2.1)	79

5.7.2.9 The nearest existing NSR E4_TLT_R17 is located at about 40m from the ESS, and the distance attenuation is about 40dB(A) according to standard point source correction. The nearest planned NSR P_R2.2_R19 is located at about 95m from the ESS, and the distance attenuation is about 48dB(A) according to standard point source correction. The maximum predicted fixed noise level is about 45dB(A). Since there are no other major existing and planned fixed noise sources in the vicinity of the ESS affecting the surrounding NSRs, cumulative fixed noise sources impacts are not anticipated.

5.7.2.10 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.1** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Refuse Collection Points and Sewage Pumping Stations (OU1.12, OU2.2)

5.7.2.11 Two RCP and two SPS (OU1.12, OU2.2) are planned within the PDA. One RCP and one SPS (OU1.12) will be located at the northern part of the PDA to south of YLH and East of TYST Interchange. One RCP and one SPS (OU1.12) will be located in Area 1 at about 200m from the planned PTI (OU1.1) and retention tank (OU1.11) and hence cumulative fixed noise sources impacts are considered in the assessment. The major noise sources of RCP and SPS are from the operation of pumps, mechanically racked screens, transformer and ventilation fan of deodorization unit. Since there is no design information on the above proposed RCP and SPS, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.2** below shows the maximum allowable

SWL of the RCP and SPS. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.2 Summary of maximum allowable SWL of RCP and SPS (OU1.12, OU2.2)

Plant Item	Maximum Allowable SWL, dB(A)
RCP and SPS (OU1.12)	89[1]
RCP and SPS (OU2.2)	86[2]

Note:
[1] Cumulative impacts from the nearby planned PTI (OU1.1) and retention tank (OU1.11) is included.
[2] Cumulative impacts from the nearby existing LTSPS is included.

5.7.2.12 The nearest existing NSR E4_LHWT_R01 is located at about 70m from the RCP and SPS (OU1.12) and the distance attenuation is about 45dB(A) according to standard point source correction. The nearest planned NSR P_OU1.2a_R02 is located at about 100m from the RCP and SPS (OU1.12) and the distance attenuation is about 48dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 50dB(A).

5.7.2.13 Another RCP and SPS (OU2.2) will be located in Area 2 at the northern part of the PDA to south of YLH and in the vicinity of the existing LTSPS, and hence cumulative fixed noise sources impacts are considered in the assessment. The nearest existing NSR E2_TLY_R02 is located at about 70m from the RCP and SPS (OU2.2) and the distance attenuation is about 45dB(A) according to standard point source correction. The nearest planned NSR P_R2.2_R02 is located at about 165m from the RCP and SPS (OU2.2) and the distance attenuation is about 52dB(A) according to standard point source correction. The maximum predicted fixed noise level is about 49dB(A).

5.7.2.14 The Contractor shall enclose pumps, noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.2** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Sewage Treatment Works (OU3.3)

5.7.2.15 A STW (OU3.3) is planned to be located in Area 3 at the southern end of the PDA to the southwest of the existing Wong Nai Tun Tsuen and in the vicinity of an existing livestock farm (AGR3.1) and planned government reserve (intended for government depot) (G3.1 and G3.3), and hence cumulative fixed noise sources impacts are considered in the assessment. The major noise sources of the STW are from the operation of sewage facilities and ventilation system. Since there is no design information on the proposed STW, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.3** below shows the maximum allowable SWL of the STW. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.3 Summary of maximum allowable SWL of STW (OU3.3)

Plant Item	Maximum Allowable SWL, dB(A)
STW (OU3.3)	89[1]

Note:
[1] Cumulative impacts from the nearby existing livestock farm (AGR3.1) and planned government reserve (intended for government depot) (G3.1 and G3.3) is included.

5.7.2.16 The nearest existing NSR E4_WNT_R11 is located at about 260m from the STW, and the distance attenuation is about 56dB(A) according to standard point source correction. The nearest planned NSR P_VR3.1_R03 is located at about 180m from the STW, and the distance attenuation is about 53dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is 45dB(A).

5.7.2.17 The Contractor shall enclose pumps, noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.3** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Divisional Fire Station and Ambulance Depot (G2.3)

5.7.2.18 A divisional fire station and ambulance depot (G2.3) is planned to be located in Area 2 at the middle of the PDA to the south of the existing Shan Ha Tsuen. The major fixed noise sources of the divisional fire station and ambulance depot include chiller, condenser, transformer, etc. Other noise sources associated with the divisional fire station and ambulance depot include station loudspeakers, siren and vehicle sirens. Since there is no design information on the proposed divisional fire station and ambulance depot, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.4** below shows the maximum allowable SWL of the divisional fire station and ambulance depot. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.4 Summary of maximum allowable SWL of fire station and ambulance depot (G2.3)

Plant Item	Maximum Allowable SWL, dB(A)
Divisional Fire Station and Ambulance Depot (G2.3)	88

5.7.2.19 The nearest existing NSR E4_SHT_R04 is located at about 270m from the divisional fire station and ambulance depot, and the distance attenuation is about 57dB(A) according to standard point source correction. The nearest planned NSR P_R2.6d_R02 is located at about 60m from the divisional fire station and ambulance depot, and the distance attenuation is about 44dB(A) according to standard point source correction. The maximum predicted fixed noise level is about 50dB(A). Since there are no other major existing and planned fixed noise sources in the vicinity of the divisional fire station and ambulance

depot affecting the surrounding NSRs, cumulative fixed noise sources impacts are not anticipated.

5.7.2.20 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.4** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned PTI (OU1.1, OU2.3, OU3.2a, in the vicinity of Yoho Midtown)

5.7.2.21 Three PTIs (OU1.1, OU2.3, OU3.2a) are planned within the PDA and one PTI outside the PDA near Yoho Midtown. The PTIs would be decked under proposed building structures and designed with no line-of-sight at the noise sensitive use. In addition, the PTI would be oriented with ingress and egress facing away from NSRs in close vicinity. The major fixed noise sources of PTIs include the operation of ventilation fans and bus idling. Since there is no design information on the above proposed PTIs, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.5** below shows the maximum allowable SWL of the PTIs. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.5 Summary of maximum allowable SWL for PTI (OU1.1, OU2.3, OU3.2a, near Yoho Midtown outside the PDA)

Plant Item	Maximum Allowable SWL, dB(A)
PTI (OU1.1)	83[1]
PTI (OU2.3)	83[2]
PTI (OU3.2a)	78[3]
PTI (near Yoho Midtown outside the PDA)	77

Note:
[1] Cumulative impacts from the nearby planned retention tank (OU1.11) as well as RCP and SPS (OU1.12) is included.
[2] Cumulative impacts from the nearby planned sports centre and market (G2.1) is included.
[3] Cumulative impacts from the nearby planned district police station (G3.2) is included.

5.7.2.22 One of the PTIs (OU1.1) will be located in Area 1 at the northern part of the PDA to south of TYST Interchange adjacent to a planned retention tank (OU1.11) and about 200m from a planned RCP and SPS (OU1.12), and hence cumulative fixed noise sources impacts are considered in the assessment. The nearest existing NSR E1_ELD_R03 is located at about 110m from the PTI and the distance attenuation is about 49dB(A) according to standard point source correction. The nearest planned NSR is P_OU1.1_R10. However, since the planned NSR P_OU1.1_R10 is not facing the PTI and directivity correction applied, the worst affected planned NSR would be P_OU1.3_R01 which is located at about 62m from the PTI and the distance attenuation is about 44dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 50dB(A).

5.7.2.23 Another PTI (OU2.3) will be located in Area 2 at the middle of the PDA at about 35m to west of Kung Um Road adjacent to a planned sports centre and market (G2.1), and hence cumulative fixed noise sources impacts are considered in the assessment. The nearest existing NSR E2_MKTT_R01 is located at about 80m from the PTI and the distance attenuation is about 46dB(A) according to standard point source correction. The nearest planned is NSR P_OU2.3_R02. However, since the planned NSR P_OU2.3_R02 is not facing the PTI and directivity correction applied, the worst affected planned NSR would be P_R2.6a_R03 which is located at about 50m from the PTI and the distance attenuation is about 42dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 48dB(A).

5.7.2.24 Another PTI (OU3.2a) will be located in Area 3 to the west of Pak Sha Tsuen and at about 25m to the west of Kung Um Road adjacent to a planned district police station (G3.2), and hence cumulative fixed noise sources impacts are considered in the assessment. The nearest existing NSR E4_PST_R18 is located at about 145m from the PTI and the distance attenuation is about 51dB(A) according to standard point source correction. The nearest planned NSR is P_OU3.2a_R01. However, since the planned NSR P_OU3.2a_R01 is not facing the PTI and directivity correction applied, the worst affected planned NSR would be P_OU3.2b_R03 which is located at about 29m from the PTI and the distance attenuation is about 37dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 48dB(A).

5.7.2.25 The fourth PTI will be located outside the PDA to the east of Yoho Midtown and north of Kwong Ming Ying Loi School. Since there are no other major existing and planned fixed noise sources in the vicinity of the PTI affecting the surrounding NSRs, cumulative fixed noise sources impacts are not anticipated. The nearest existing NSR E4_KMYLS_E02 is located at about 10m from the PTI and the distance attenuation is about 28dB(A) according to standard point source correction. There are no planned NSR in the vicinity of the PTI. The maximum predicted fixed noise level is about 55dB(A).

5.7.2.26 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.5** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Sports Centre and Market (G2.1)

5.7.2.27 A sports centre and a market (G2.1) are planned to be co-located in Area 2 at the middle of the PDA at about 25m to west of Kung Um Road and in the vicinity of a planned PTI (OU2.3), and hence

cumulative fixed noise sources impacts are considered in the assessment. The major fixed noise sources of the sports centre and market include chiller, ventilation louver, etc. Since there is no design information on the proposed sports centre and market, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. Since the nearest NSR (i.e. P_R2.5_R07) is located in close vicinity to the sports centre and market, the fixed noise sources should be located away from the NSR as far as possible and oriented with opening facing away from the NSR (i.e. opening facing south to the road). Therefore, the maximum allowable SWL has included directivity correction factor. **Table 5.7.6** below shows the maximum allowable SWL of the sports centre and market. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.6 Summary of maximum allowable SWL of sports centre and market (G2.1)

Plant Item	Maximum Allowable SWL, dB(A)
Sports Centre and Market (G2.1)	79[1]

Note:
[1] Cumulative impacts from the nearby planned PTI (OU2.3) is included.

5.7.2.28 The nearest existing NSR E2_MKTT_R01 is located at about 85m from the sports centre and market, and the distance attenuation is about 47dB(A) according to standard point source correction. The nearest planned NSR P_R2.5_R07 is located at about 10m from the sports centre and market, and the distance attenuation is about 28dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is 48dB(A).

5.7.2.29 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.6** will not be exceeded. The Contractor shall also ensure the fixed plants are located away from concerned NSRs located to the north and west of the centre and market as well as their openings facing south to the road. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned District Police Station (G3.2)

5.7.2.30 A district police station (G3.2) is planned to be located in Area 3 to the west of Pak Sha Tsuen and at about 150m to the west of Kung Um Road and in the vicinity of a planned PTI (OU3.2a), and hence cumulative fixed noise sources impacts are considered in the assessment. The major fixed noise sources of the district police station include chiller, ventilation louver, etc. Since there is no design information on the proposed station, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. Since the nearest NSR (i.e. P_R3.5b_R02) is located in close vicinity to the district police station, the fixed noise sources should be located away

from the NSR as far as possible and oriented with opening facing away from the NSR (i.e. opening facing north east to the road). Therefore, the maximum allowable SWL has included directivity correction factor. **Table 5.7.7** below shows the maximum allowable SWL of the district police station. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.7 Summary of maximum allowable SWL of district police station (G3.2)

Plant Item	Maximum Allowable SWL, dB(A)
District Police Station (G3.2)	77[1]

Note:
[1] Cumulative impacts from the nearby planned PTI (OU3.2a) is included.

5.7.2.31 The nearest existing NSR E4_CPG_R01 is located at about 240m from the district police station and the distance attenuation is about 56dB(A) according to standard point source correction. The nearest planned NSR P_R3.5b_R02 is located at about 8m from the district police station and the distance attenuation is about 26dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 48dB(A).

5.7.2.32 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.7** will not be exceeded. The Contractor shall also ensure the fixed plants are located away from concerned NSRs located to the south west of the station as well as their openings facing north east to the road. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Sites for Storage Use, Open Storage as well as Storage and Workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10)

5.7.2.33 Several sites in Area 1 at the north western part of the PDA to the south of YLH are planned for storage use, open storage as well as storage and workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10), and hence cumulative fixed noise sources impacts are considered in the assessment. The major fixed noise sources of the sites for storage use, open storage as well as storage and workshop include loading and unloading, forklifting, chiller, ventilation louver, etc.. Since there is no design information on the proposed sites for storage use, open storage as well as storage and workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10), analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.8** below shows the maximum allowable SWL of the sites for storage use, open storage as well as storage and workshop. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.8 Summary of maximum allowable SWL of sites for storage use, open storage as well as storage and workshop (OU1.5, OU1.6, OU1.7, OU1.8, OU1.9, OU1.10)

Plant Item	Maximum Allowable SWL, dB(A)
Storage Use Site (OU1.5)	87[1]
Open Storage Site (OU1.6)	83[2]
Storage Use Site (OU1.7)	80[3]
Storage and Workshop Site (OU1.8)	88[4]
Storage Use Site (OU1.9)	81[5]
Storage Use Site (OU1.10)	78[6]

Note:
[1] Cumulative impacts from the nearby planned open storage site (OU1.6), storage use sites (OU1.7, 1.9 and 1.10) as well as storage and workshop site (OU1.8) is included.
[2] Cumulative impacts from the nearby planned storage use sites (OU1.5, 1.7,1.9 and 1.10) as well as storage and workshop site (OU1.8) is included.
[3] Cumulative impacts from the nearby planned open storage site (OU1.6), storage use sites (OU1.5, 1.9 and 1.10) as well as storage and workshop site (OU1.8) is included.
[4] Cumulative impacts from the nearby planned open storage site (OU1.6) and storage use sites (OU1.5, 1.7, 1.9 and 1.10) is included.
[5] Cumulative impacts from the nearby planned open storage site (OU1.6), storage use sites (OU1.5, 1.7 and 1.10) as well as storage and workshop site (OU1.8) is included.
[6] Cumulative impacts from the nearby planned open storage site (OU1.6), storage use sites (OU1.5, 1.7 and 1.9) as well as storage and workshop site (OU1.8) is included.

5.7.2.34 The existing NSR located closest to these OU sites is E1_TYST_R07. It is located at about 45m from the storage use site (OU1.7) and the distance attenuation is about 41dB(A) according to standard point source correction. The planned NSR located closest to these OU sites is P_R1.2_R05. It is located at about 35m from the storage use site (OU1.10) and the distance attenuation is about 39dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 48dB(A).

5.7.2.35 The Contractor shall install acoustic silencers, noise barrier or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.8** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Retention Tank (OU1.11)

5.7.2.36 A retention tank (OU1.11) is planned to be located in a district open space in Area 1 near TYST Interchange and in the vicinity of a planned PTI (OU1.1), RCP and SPS (OU1.12), and hence cumulative fixed noise sources impacts are considered in the assessment. The retention tank is planned to be installed underground. The major noise sources would mainly be its associated pumping and ventilation system which is located above ground. Since there is no design information on the proposed retention tank, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.9** below shows the maximum allowable SWL of the retention tank. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.9 Summary of maximum allowable SWL of retention tank (OU1.11)

Plant Item	Maximum Allowable SWL, dB(A)
Retention Tank (OU1.11)	77[1]

Note:
[1] Cumulative impacts from the nearby planned PTI (OU1.1) as well as RCP and SPS (OU1.12) is included.

5.7.2.37 The nearest existing NSR E1_ELD_R02 is located at about 105m from the retention tank and the distance attenuation is about 48dB(A) according to standard point source correction. The nearest planned NSR P_OU1.1_R10 is located at about 32m from the retention tank and the distance attenuation is about 38dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 50dB(A).

5.7.2.38 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.9** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Planned Government Reserve (Intended for Government Depot) (G3.1, G3.3)

5.7.2.39 Two sites in Area 3 at the southern end of the PDA to the southwest of the existing Wong Nai Tun Tsuen are planned for government depot use (G3.1 and G3.3) and will be located in the vicinity of an existing livestock farm (AGR3.1) and planned STW (OU3.3), and hence cumulative fixed noise sources impacts are considered in the assessment. The major noise sources of the government depot include the operation of chiller, ventilation louver, etc.. Since there is no design information on the proposed government depot, analysis has been conducted to determine the maximum allowable SWL based on the separation distance. **Table 5.7.10** below shows the maximum allowable SWL of the government depot. Detailed calculation is shown in **Appendix 5.17**.

Table 5.7.10 Summary of maximum allowable SWL of government depot (G3.1 and G3.3)

Plant Item	Maximum Allowable SWL, dB(A)
Government Depot (G3.1 and G3.3)	88[1]

Note:
[1] Cumulative impacts from the nearby existing livestock farm (AGR3.1) and planned STW (OU3.3) is included.

5.7.2.40 The nearest existing NSR E4_WNT_R12 is located at about 155m from the government depot and the distance attenuation is about 52dB(A) according to standard point source correction. The nearest planned NSR P_VR3.1_R03 is located at about 280m from the government depot and the distance attenuation is about 57dB(A) according to standard point source correction. The maximum predicted cumulative fixed noise level is about 45dB(A).

5.7.2.41 The Contractor shall enclose noisy plants and install acoustic silencers, acoustic louvre or acoustic enclosure as appropriate to ensure the specified maximum SWLs as shown in **Table 5.7.10** will not be exceeded. The future design and selection of the equipment shall also aim to reduce the effect of tonality at the NSRs as much as practicable. However, the Contractor shall also take into account the latest available information at time of detailed design to review and update the maximum allowable SWL as appropriate.

Cumulative Fixed Noise Sources Impact

5.7.2.42 The predicted fixed noise sources impacts at existing and planned NSRs are given in **Table 5.7.11**. Detailed cumulative fixed noise calculation is shown in **Appendix 5.18**.

Table 5.7.11 Predicted fixed noise sources impacts at existing and planned NSRs

ID [1]	NAP [2]	ASR	Planned Noise Source Only			Existing + Planned Noise Source		
			Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]	Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]
Existing NSRs (Within PDA)								
E1_ELD	E1_ELD_R02	A	43	45	Y	43	50	Y
E1_ELD	E1_ELD_R03	A	42	45	Y	42	50	Y
E1_TYST	E1_TYST_R01	B	46	50	Y	46	55	Y
E1_TYST	E1_TYST_R07	B	48	50	Y	48	55	Y
E2_MKTT	E2_MKTT_R01	A	44	45	Y	44	50	Y
E2_TLY	E2_TLY_R02	B	47	50	Y	49	55	Y
Existing NSRs (Outside PDA)								
E4_TTT	E4_TTT_R01	B	46	50	Y	46	55	Y
E4_TLT	E4_TLT_R01	B	43	50	Y	47	55	Y
E4_TLT	E4_TLT_R17	A	45	45	Y	45	50	Y

ID [1]	NAP [2]	ASR	Planned Noise Source Only			Existing + Planned Noise Source		
			Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]	Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]
E4_WNT	E4_WNT_R11	A	41	45	Y	42	50	Y
E4_WNT	E4_WNT_R12	A	43	45	Y	45	50	Y
E4_SHT	E4_SHT_R04	A	37	45	Y	37	50	Y
E4_CPG	E4_CPG_R01	A	34	45	Y	34	50	Y
E4_PST	E4_PST_R18	A	34	45	Y	34	50	Y
E4_KMYLS	E4_KMYLS_E02	C	55	55	Y	55	60	Y
E4_FSW	E4_FSW_R02	B	46	50	Y	46	55	Y
E4_LHWT	E4_LHWT_R01	B	50	50	Y	50	55	Y
Planned NSRs								
P_R2.2	P_R2.2_R02	C	40	55	Y	42	60	Y
P_R2.2	P_R2.2_R19	B	37	50	Y	37	55	Y
P_R2.4	P_R2.4_R04	B	44	50	Y	44	55	Y
P_R2.5	P_R2.5_R07	B	48	50	Y	48	55	Y
P_VR3.1	P_VR3.1_R03	B	43	50	Y	44	55	Y
P_R2.6a	P_R2.6a_R03	B	48	50	Y	48	55	Y
P_R2.6d	P_R2.6d_R02	B	50	50	Y	50	55	Y
P_R3.5b	P_R3.5b_R02	B	48	50	Y	48	55	Y

ID [1]	NAP [2]	ASR	Planned Noise Source Only			Existing + Planned Noise Source		
			Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]	Max. Predicted SPL, dB(A) [3]	Night-time Criteria $L_{eq\ (30\ min)}$, dB(A) [4]	Compliance [Y/N]
P_E3.1	P_E3.1_E01	B	40	50	Y	41	55	Y
P_OU1.3	P_OU1.3_R01	B	46	50	Y	46	55	Y
P_OU2.3	P_OU2.3_R02	B	47	50	Y	47	55	Y
P_OU3.2a	P_OU3.2a_R01	B	45	50	Y	45	55	Y
P_OU3.2b	P_OU3.2b_R03	B	48	50	Y	48	55	Y
P_OU1.1	P_OU1.1_R02	B	48	50	Y	48	55	Y
P_OU1.1	P_OU1.1_R10	B	48	50	Y	48	55	Y
P_R1.2	P_R1.2_R05	B	46	50	Y	46	55	Y

Notes:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] NAP – Noise Assessment Point.
[3] Shortest distances between NSRs and the noise sources have been used to calculate the maximum predicted SPL.
[4] Only night-time criteria are assessed as worst case scenario. Noise criterion is proposed based on ANL-5 for planned fixed noise sources impact and ANL for existing and planned fixed noise sources impact.

5.7.3 Mitigation of Fixed Noise Sources Impact

5.7.3.1 For the proposed noise sources which are located near to existing and planned NSRs, the following tentative noise mitigation measures are considered:

- All the pumps and noisy plants should be enclosed inside a building structure;
- Proper selection of quiet plant aiming to reduce the tonality at NSRs;
- Installation of silencer / acoustic enclosure / acoustic louvre for the exhaust of ventilation system; and
- Openings of ventilation systems should be located away from NSRs as far as practicable and oriented away from the NSRs.

5.7.3.2 The feasibility, practicability, programming and effectiveness of the above mitigation measures have been reviewed by engineer. As the fixed noise sources impacts for planned and existing NSRs will be within respective noise criteria, the number of dwellings, classrooms and other sensitive receivers that will be exposed to noise impact exceeding the criteria set out in Annex 5 of the EIAO-TM is nil.

5.7.3.3 This EIA has demonstrated that the practicability of the noise mitigation measures to be implemented. In order to allow flexibility of the future developer of the proposed OU(SU) sites (i.e. OU1.5, 1.7, 1.9 and 1.10) and OU(S+W) site (i.e. OU1.8), it is proposed to state in the land lease condition that the future developer could demonstrate the noise impact of the proposed OU(SU) sites (i.e. OU1.5, 1.7, 1.9 and 1.10) and OU(S+W) site (i.e. OU1.8) to the satisfactory of the relevant authority (i.e. EPD). The detailed design should also incorporate the following good practice in order to minimize to the neighbouring NSRs.

- Louvres should be orientated away from nearby NSRs, preferably onto main roads which are less sensitive.
- Direct noise mitigation measures including silencers, acoustic louvers and acoustic enclosures should be allowed for the design of the noise sources of the industrial block.
- Changes to industrial noise sources in the future should only be made with reference to the noise criteria. For example, careful consideration is required for relocation of fixed plant etc.

5.7.3.4 **Table 5.7.12** below summarizes the planned fixed noise sources as well as mitigation measures proposed.

Table 5.7.12 Summary of planned fixed noise sources and key mitigation measure proposed

Fixed Noise Source	Key Mitigation Measure Proposed
Retention Tank, STW, SPS & RCP	i) All the pumps and noisy plants should be enclosed inside a building structure; ii) Proper selection of quiet plant aiming to reduce tonality at NSRs; iii) Installation of silencer / acoustic enclosure / acoustic louvre for the exhaust of ventilation system; iv) Openings of ventilation systems should be located away from NSRs as far as practicable.
Sports Centre and Market, District Police Station, Divisional Fire Station and Ambulance Depot, Government Reserve (Intended for Government Depot), PTI, ESS	i) All the noisy plants should be enclosed inside a building structure; ii) Proper selection of quiet plant aiming to reduce tonality at NSRs; iii) Installation of silencer / acoustic enclosure / acoustic louvre for the exhaust of ventilation system; iv) Openings of ventilation systems should be located and oriented away from NSRs as far as practicable.

Fixed Noise Source	Key Mitigation Measure Proposed
Sites for Storage Use, Open Storage as well as Storage and Workshop	i) All the noisy plants should be enclosed inside a building structure; ii) Proper selection of quiet plant aiming to reduce tonality at NSRs; iii) Installation of silencer / acoustic enclosure / acoustic louvre for the exhaust of ventilation system and fixed plants such as chillers / processing plant; iv) Openings of ventilation systems and fixed plants such as chillers / processing plant should be located and oriented away from NSRs as far as practicable; v) Changes to industrial noise sources in the future should only be made with reference to the noise criteria. For example, careful consideration is required for relocation of fixed plant etc.

5.7.4 Assessment of Side Effects and Constraints

5.7.4.1 Side effects and constraints are not anticipated through early design to allow implementation of the proposed noise mitigation measures including enclosure of pumps and noisy plants inside a building structure, proper selection of quiet plant, installation of silencer / acoustic enclosure / acoustic louvre for the exhaust of ventilation system as well as location and orientation of ventilation system openings away from NSRs as far as practicable.

5.7.5 Evaluation of Constraints on Planned Noise Sensitive Developments / Land Uses

5.7.5.1 With all practicable direct technical remedies in place, no additional measures within the planned noise sensitive uses would be required.

5.7.6 Evaluation of Residual Fixed Noise Sources Impact

5.7.6.1 Fixed noise sources impact arising from the Project can be properly mitigated by implementing the proposed noise mitigation measures. Adverse residual noise impacts are thus not anticipated.

5.8 Rail Noise Impact Assessment

5.8.1 Environmentally Friendly Transport Services

5.8.1.1 An EFTS is proposed to be developed as one of the major modes of transportation within the PDA with possible connection to West Rail Tin Shui Wai Station.

5.8.1.2 The EFTS system within the PDA will be an extension from that in the approved EIA report for HSK NDA (AEIAR-203/2016). Various modes of EFTS, which include Light Rail Transit (LRT), trolley bus, automatic people mover (APM), monorails, modern tramway and

electric / LPG bus, will be investigated in later stage by the respective project proponent. The assessment is based on the best and latest information available during the course of EIA study. For conservative rail noise impact assessment, the rail based EFTS has been selected in this study.

5.8.2 Rail Noise Assessment Assumption and Methodology

5.8.2.1 A reference scheme was developed with the assumptions and assessed level shown in **Table 5.8.1** below. The reference scheme has followed those in the approved EIA report for HSK NDA.

Table 5.8.1 Rail based EFTS reference scheme

Items	Description
Type of source	Automated guideway transit, steel wheel, welded rail
Noise source level	SEL 80 dB(A) at 50 feet and 50 mph Ref: Table 5-1 Transit Noise and Vibration Impact Assessment FTA-VA-90-1003-06 May 2006 by Office of Planning and Environment Federal Transit Administration
Operating speed	Assumed 40 kph
Frequency	Assumed 5 trains per headway
Track enhancement	• Embedded rail: 3 dB(A) reduction • Green track with vegetation: 2 dB(A) reduction Reference material as shown in **Appendix 5.19**

5.8.3 Prediction and Evaluation of Rail Noise Impact

5.8.3.1 Representative NSRs within 300m from the PDA and the associated infrastructure works which are closest to the EFTS alignment are identified and shown in **Figure 5.6**. Their buffer distances from the proposed EFTS alignment based on the indicative layout are summarized in **Table 5.8.2**. Information about the determination of noise criteria for representative NSRs are provided in **Appendix 5.20**.

Table 5.8.2 Buffer distances from representative NSRs to the proposed EFTS alignment

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP	Distance from Track
Existing NSRs (Within PDA)					
N/A	E1_PV	Park Villa	R	E1_PV_R02	36m
	E1_TYST	Tong Yan San Tsuen	R	E1_TYST_R07	23m
Existing NSRs (Outside PDA)					
N/A	E4_LHT	Lam Hau Tsuen	R	E4_LHT_R06	13m
Planned NSRs					

Stage	NSR[1]	NSR Description[2]	Use[3]	NAP	Distance from Track
Stage 1	P_R2.2	Public Housing	R	P_R2.2_R10	17m
Stage 2	P_R2.1b	Private Housing	R	P_R2.1b_R02	56m
	P_R2.4	Public Housing	R	P_R2.4_R12	14m
Stage 3	P_R2.6d	Private Housing	R	P_R2.6d_R02	21m
Stage 4	P_OU1.2a	Private Housing	R	P_OU1.2a_R10	33m[4]

Note:
[1] The assessment only includes NSRs which rely on opened windows for ventilation.
[2] The first layer of NSRs which are most affected by the project within 300m from the PDA and the associated infrastructure works is included.
[3] R – Residential Premises.
[4] Based on the indicate layout for assessment, the first level of the NSR is about 32m and 9m away from the EFTS track vertically and horizontally, which gives a slant distance of about 33m from the EFTS track.

5.8.3.2 As shown in **Table 5.8.2**, Lam Hau Tsuen (i.e. E4_LHT) is the closest existing NSR to the EFTS track, while the public housing P_R2.4 and the private housing P_R2.6d are the closest planned public housing NSR and planned private housing NSR to the EFTS track respectively. The representative existing and planned NSRs are located at least 13m and 14m respectively away from the proposed alignment for the EFTS.

5.8.3.3 According to the approved EIA report for HSK NDA, the predicted rail noise level for the NSRs with at least 10m buffer distance from the proposed alignment of EFTS would be lower than 55dB(A) based on the EFTS reference scheme as shown in **Table 5.8.1**. As shown in **Appendix 5.20** and **Table 5.8.2**, all representative NSRs along the EFTS in the PDA have ASR rating of B and will have buffer distance of at least 13m. The rail noise level would be about 54 dB(A) for NSR at Lam Hau Tsuen (i.e. E4_LHT), which is the closest existing NSR to the EFTS track alignment, and comply with the night-time noise criterion of 55 dB(A). Therefore, adverse rail noise impacts from EFTS are not anticipated for this study.

5.8.3.4 In case a rail cross or joint is designed for the EFTS, such rail cross or joint location should be located at the section with largest possible distance between the track and the NSR. Alternatively, erect a noise barrier or enclosure at the rail cross or joint section to control the noise impact to the nearby NSRs. Detailed design of the rail cross or joint along the track of EFTS and the necessity of associate mitigation measures will be carried out at later stage by the EFTS project proponent.

5.8.3.5 There is no design and operation detail of the EFTS during this Schedule 3 EIA study. In any case there are deviations from the assumption in **Table 5.8.1** and **Table 5.8.3** above and causing an exceedance of the noise level compared with the criteria, the EFTS project proponent should implement further noise mitigation measures such as noise barrier and enclosure to control the noise impact by the

EFTS to comply with the noise criteria. Noise barrier and enclosure are effective and traditional mitigation measures to reduce rail noise by 10 dB(A) and 15 dB(A) respectively.

5.8.3.6 As discussed in **Section 1.5.1** of this report, the proposed EFTS is a Schedule 2 DP under the EIA Ordinance. The proposed EFTS will be designed, constructed and operated by the others. The associated environmental impacts will be adequately addressed in a separate EIA study to be prepared and submitted under the EIA Ordinance by the future operator.

5.8.4 Assessment of Side Effects and Constraints

5.8.4.1 Side effects and constraints are not anticipated through early design to allow sufficient space between EFTS track alignment and NSRs.

5.8.5 Evaluation of Constraints on Planned Noise Sensitive Developments / Land Uses

5.8.5.1 With all practicable direct technical remedies in place, no additional measures within the planned noise sensitive uses would be required.

5.8.6 Evaluation of Residual Rail Noise Impact

5.8.6.1 With the sufficient buffer distance and implementation of suitable track enhancement measures, rail noise impacts are not expected. Adverse residual noise impacts are therefore not anticipated. There would be no existing dwellings, classrooms and other noise sensitive elements that will be exposed to adverse residual noise impact exceeding the criteria set out in Annex 5 of the EIAO-TM.

5.9 Conclusion

5.9.1.1 Construction noise associated with the use of PME for different stages of construction has been conducted. With the implementation of practical mitigation measures including good site management practices, use of movable noise barrier, full enclosure and retractable barrier, use of “quiet” plant and working method, construction noise impacts at all of the neighboring residential noise sensitive uses would be controlled to acceptable levels. For educational institutions, the construction noise impacts during school normal and examination periods would also comply with the relevant criteria.

5.9.1.2 Operational road traffic noise impact on planned and existing noise sensitive uses within the PDA have been predicted for the assessment year. Results indicate that the noise impacts can be mitigated by a combination of noise mitigation measures including 1) absorptive vertical barriers and cantilevered noise barriers along some sections of Project Roads and Kung Um Road; 2) LNRS on some road sections; 3) semi-enclosures or full enclosures at primary distributor roads at TYST

Interchange and Kung Um Road; 4) nullah features / barriers along some sections of Yuen Long Nullah (Kiu Hing Road); 5) provision of acoustic windows for some planned public and private housing and 6) alternative building orientation for some planned schools to alleviate adverse traffic noise impact on the affected NSRs. These mitigation measures will also ensure that the noise levels caused by project roads are within the respective noise criteria.

5.9.1.3 For the existing road sections such as Kung Um Road, Sha Tseng Road and TYST Road, LNRS would be considered to address the existing noise impacts from existing roads. Different materials for LNRS are being reviewed and tested by the Government for application in Hong Kong. Environmental reviews will be conducted at the later design stage to review the proposed noise mitigation measures taking into account the latest design standard at that time for the application of the LNRS materials. Similar considerations are also applied for planned roads.

5.9.1.4 The proposed at-receiver mitigation measures i.e. the provision of acoustic windows for some planned public and private housing are subject to further study by HKHA and future private developers respectively. HKHA and future developers can further explore alternative options which can achieve corresponding traffic noise reduction during the detailed design stage.

5.9.1.5 There is scope for adoption of alternative noise mitigation measures, for example, by means of alternative layout and design of individual developments at the detailed design stage. Noise impact assessment at the planned residential sites is proposed to be conducted by future developers at the detailed design stage to study whether the future development layout would avoid exposing to excessive road traffic noise levels so as to minimise the scale and extent of the proposed noise mitigation measures such as noise barriers and special building design. The requirement of noise impact assessment would be included in the land lease condition or planning briefs of the residential sites.

5.9.1.6 In addition, Environmental Assessment Study (EAS) for the planned public housings will be conducted by Hong Kong Housing Authority (HKHA) in the detailed design stage according to HKPSG to address environmental impacts and to comply with relevant criteria.

5.9.1.7 Fixed noise sources impact assessment has been conducted. Noise impact from planned fixed plant could be effectively mitigated by implementing noise mitigation measure at source during the detailed design stage. With the adoption of the proposed maximum permissible SWLs for the proposed fixed plant, the impact noise levels at representative NSRs complies with the relevant noise criteria for both existing and planned fixed noise sources. Therefore, adverse fixed noise sources impact on the NSRs is not anticipated.

5.9.1.8 An EFTS is proposed to be introduced to operate within the Project, which may be in the form of rail based mode of transport. For

conservative noise assessment, the rail based EFTS was assumed for rail noise impact assessment. Results indicate that the noise impacts on NSRs would comply with the statutory requirement after considering some track enhancement measures such as embedded rail, green track with vegetation, etc. With the proposed of 10m buffer distance, the operation of EFTS would not pose adverse noise impact on the nearby NSRs.

5.10 References

[5-1] EPD (2010), EIA Ordinance Guidance Note No. 9/2010 on "Preparation of Construction Noise Impact Assessment Under the Environmental Impact Assessment Ordinance".

[5-2] EPD (2010), EIA Ordinance Guidance Note No. 12/2010 on "Road Traffic Noise Impact Under the Environmental Impact Assessment Ordinance".